

Basics of electronics and components

5. Transistor Devices and Transistor Circuits

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Greek letters

Lower case greek letters & commonly used capitals

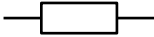
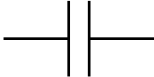
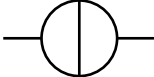
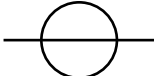
α	alpha
β	beta
γ Γ	gamma
δ Δ	delta
ϵ (ε)	epsilon ('varepsilon')
ζ	zeta
η	eta
θ Θ	theta

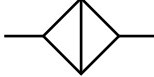
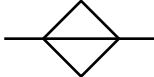
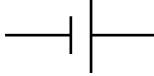
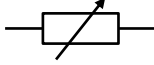
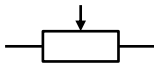
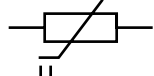
ι	iota
κ	kappa
λ Λ	lambda
μ	mu
ν	nu
ξ Ξ	xi
$\omicron$	omicron
π Π	pi

ρ	rho
σ Σ	sigma
τ	tau
υ Υ	upsilon
ϕ (φ) Φ	phi ('varphi')
χ	chi
ψ Ψ	psi
ω Ω	omega

Circuit elements

Basic passive and active elements

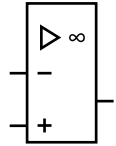
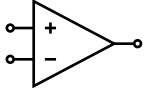
Type	EN-60617 symbol no.	
Resistor	04-01-01	
Capacitor (generic)	04-02-01	
Inductance (generic)	-	
Ground (generic)	02-15-01	
Current source	02-16-01	
Voltage source	02-16-02	

Type	EN-60617 symbol no.	
Dependent current source	-	
Dependent voltage source	-	
Battery cell	05-03-06	
Resistor (changeable)	04-01-03	
Potentiometer	04-01-07	
Varistor	04-01-04	

Circuit elements

Active and passive semiconductor devices

Type	EN-60617 symbol no.	
Diode (generic)	05-03-01	
Schottky diode	05-02-04 + 05-03-01	
Zener diode	05-03-06	
LED	05-03-02	
Photo-diode	05-06-02	

Type	EN-60617 symbol no.	
Operational amplifier	Part 13	
Operational amplifier	(old symbol)	

5.1 Bipolar junction transistor and BJT circuits

*Replica of the first transistor developed at the
Bell Telephone Laboratories, USA, in 1947*

Chapter outline

5.1.1 About transistors	5.1.2 Basics of BJT transistors	5.1.3 BJT large-signal model	5.1.4 BJT basic circuits	5.1.5 Transistor as switch
5.1.6 Transistor characteristic field	5.1.7 Transistor as amplifier	5.1.8 BJT small-signal model	5.1.9 Frequency response of transistor amplifiers	

5.1 Bipolar junction transistor and BJT circuits

Learning objective

- Learn the basic properties, circuit symbols and applications of bipolar junction transistors.
- Design basic circuits and for the correct biasing of transistors
- Describe bipolar junction transistors in terms of their large-signal model by applying the method of assumed states.
- Design circuits for the amplification of signals.
- Design circuits utilizing transistors as switch.
- Construct and describe the four-quadrant characteristic field of transistors.
- Calculate the small-signal properties of transistor circuits.
- Distinguish large-signal, small-signal, and frequency-domain descriptions

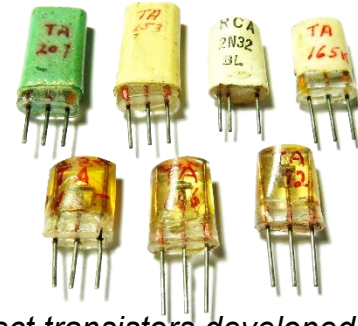
5.1 Bipolar junction transistor and BJT circuits

5.1.1 About transistors

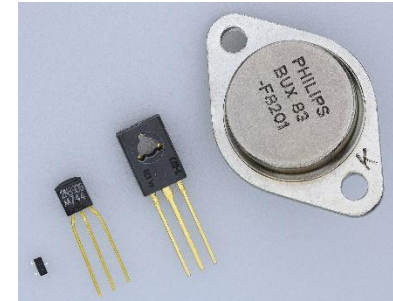
5.1.1 About transistors

About transistors

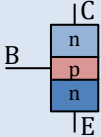
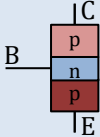
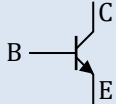
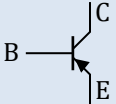
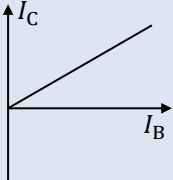
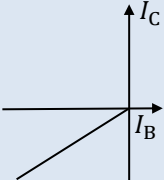
- Semiconductor device used to **switch** or **amplify** electrical signals and power.
- One of the **most important active devices** in electronics and a fundamental device class in **analogue**, **digital**, and **power electronics**.
- Operating principle: A **third terminal controls** the current between two other terminals.
- Two main transistor families:
 - ↳ bipolar junction transistor (BJT): current-controlled device
 - ↳ field-effect transistor (FET): voltage-controlled device
- Important FET subtypes: JFET and MOSFET.
- Transistors are available as **discrete components** or as **integrated devices** on chips.
- Historically, the **first practical transistor** devices were developed at Bell Labs in the **late 1950ies**.



Joint-contact transistors developed in 1948 at Bell Labs.



Modern BJT transistor packages, from left to right: SOT-23, TO-92, TO-126, TO-3.

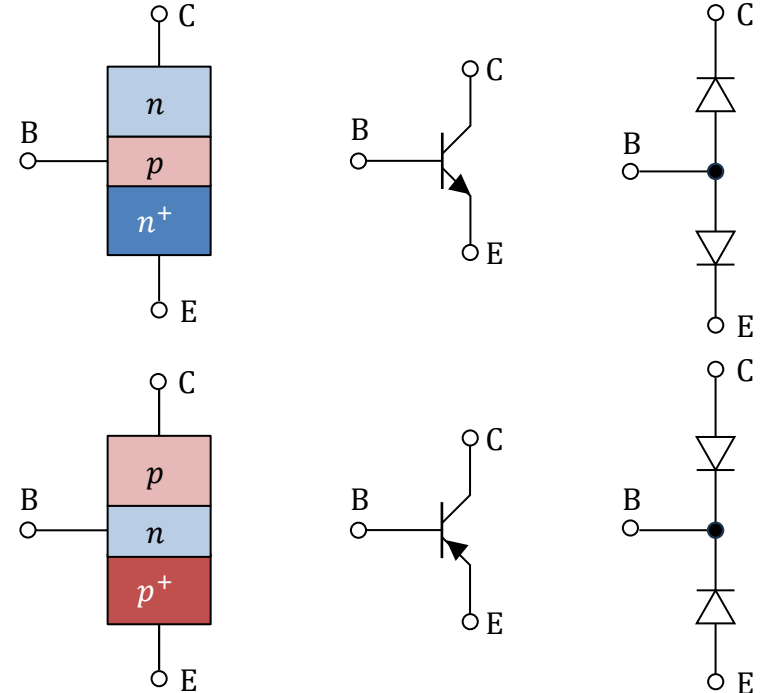
Type	Bipolar junction transistors	
	npn transistor	pnp transistor
Schematic		
Symbol		
Typical transfer characteristics		
Charac-teristics	pos. U_{CE}	neg. U_{CE}
	Normally-off	
	Current controlled	
Use cases	High-freq., analogue & power electronics	

5.1.2 Basics of BJT transistors

5.1.2 Basics of BJT transistors

Transistor design

- Three semiconductors forming **two pn-junctions**.
- Two types: **npn** and **pnp**
- Components of a BJT transistor
 - Collector **C**: Lightly doped → small depletion zone
 - Base **B**: lightly doped and very thin → very few majority charge carriers
 - Emitter **E**: heavily doped → generates many charge carriers
- Behaves similarly to opposing diodes:
 - BE-path:
 - Normal diode, conductive for $U_{BE} \geq 0.7 \text{ V}$.
 - Generates a lot of free charge carriers.
 - BC-path:
 - Typically operated in reverse mode.
 - “weak” diode due to the low doping.
 - Easily “overrun”** by the charges from the BE-diode.



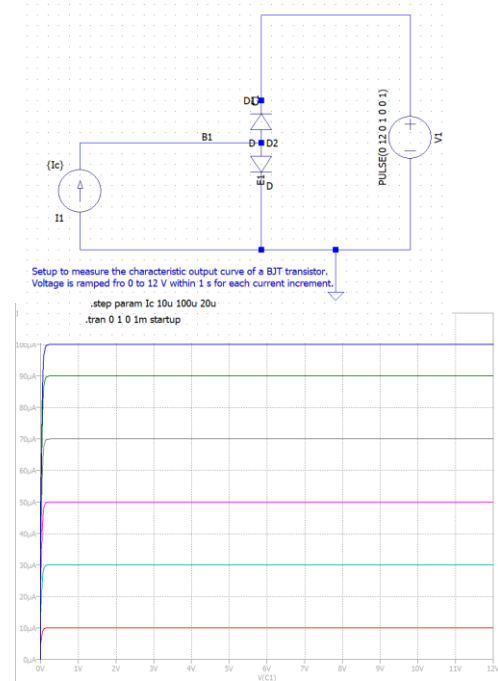
BJT transistor schematic, symbol (**middle**) and diode-equivalent circuit (**left**) for npn (**top**) and pnp-type (**bottom**).

5.1.2 Basics of BJT transistors

LTspice exercise: Transistor vs. two opposing diodes 2/2

The following circuit shows a simplified circuit to measure the characteristic field of **two diodes in opposite direction**. Please analyse the collector and emitter current as function of the CE-voltage and the base current.

- As both diodes are parametrized identically, the current from the ideal current source is forced through both diodes equally when no external voltage is applied.
- As soon as the voltage ramps up, diode D1 is under reverse bias and off $I_{D0} \rightarrow 0$.
- The current I_{D2} is equal to I_{B1}

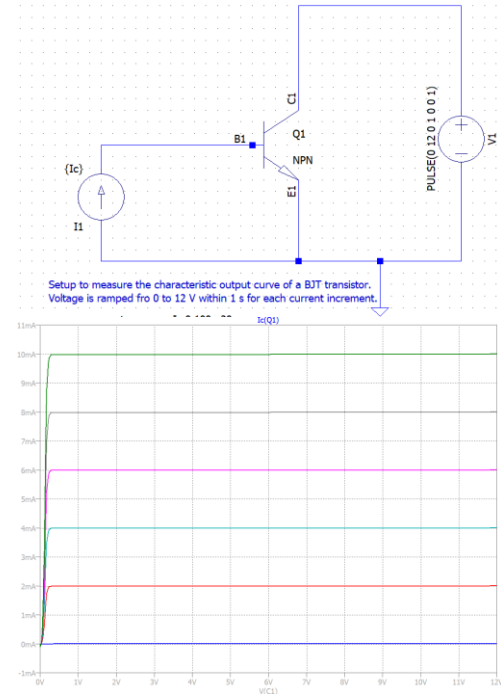


5.1.2 Basics of BJT transistors

LTspice exercise: Transistor vs. two opposing diodes 1/2

The following circuit shows a simplified circuit to measure the characteristic field of a **nnp transistor**. Please analyse the collector and emitter current as function of the CE-voltage and the base current.

- The characteristic field shows the expected behaviour
 - ┐ Cut-off: For $I_B = 0 \rightarrow I_C = 0$
 - ┐ Saturation: I_C no longer determined by βI_B but limited by the external circuit.
 - ┐ Forward active: I_C is a function of I_B and independent of U_{CE}
 - ┐ The current I_C is substantially larger than I_B



5.1.2 Basics of BJT transistors

Operation state: Cut-off

- **Terminal voltages:**

- BE is reverse biased: $U_{BE} < 0 \text{ V}$

- BC is reverse biased to avoid inverse operation: $U_{BC} < 0 \text{ V}$

- Any voltage across CE

- BE and BC diode under reverse bias:

- Majority **charge carriers** are **pulled from the depletion zone**, depletion zone growth & **electrical field built-up**.

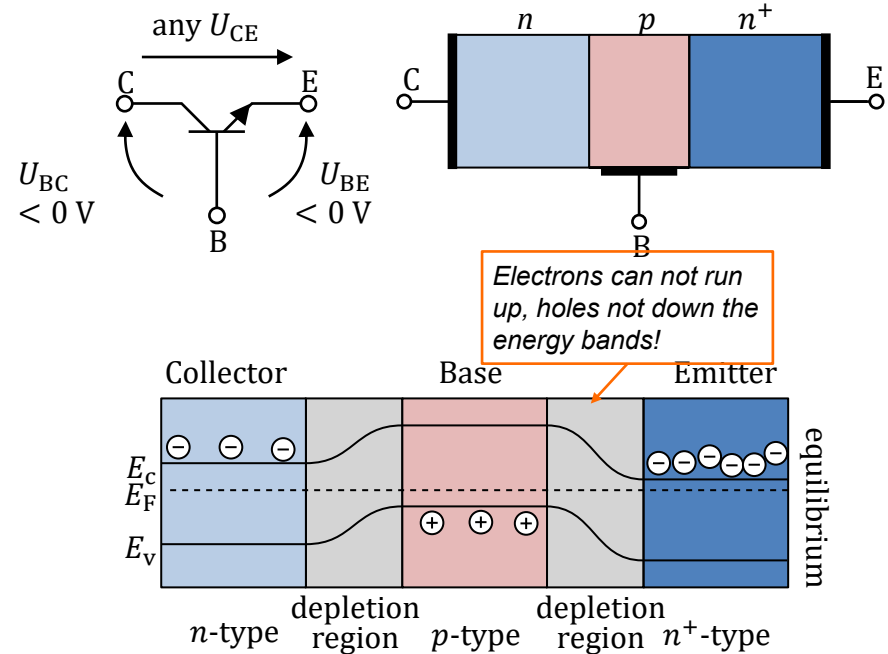
- No base current I_B

- No collector current I_C

- No emitter current I_E

- Corresponds to **logical 0** or “off”

- Equivalent circuit model: **open circuit**.



NPN in cut-off mode: Top left: Terminal voltages; Top right: schematic of npn junction; Bottom: band diagram.

5.1.2 Basics of BJT transistors

Operation state: Forward active (qualitative description)

- The **BE diode** is in forward direction; the **BC diode** is in reverse direction:

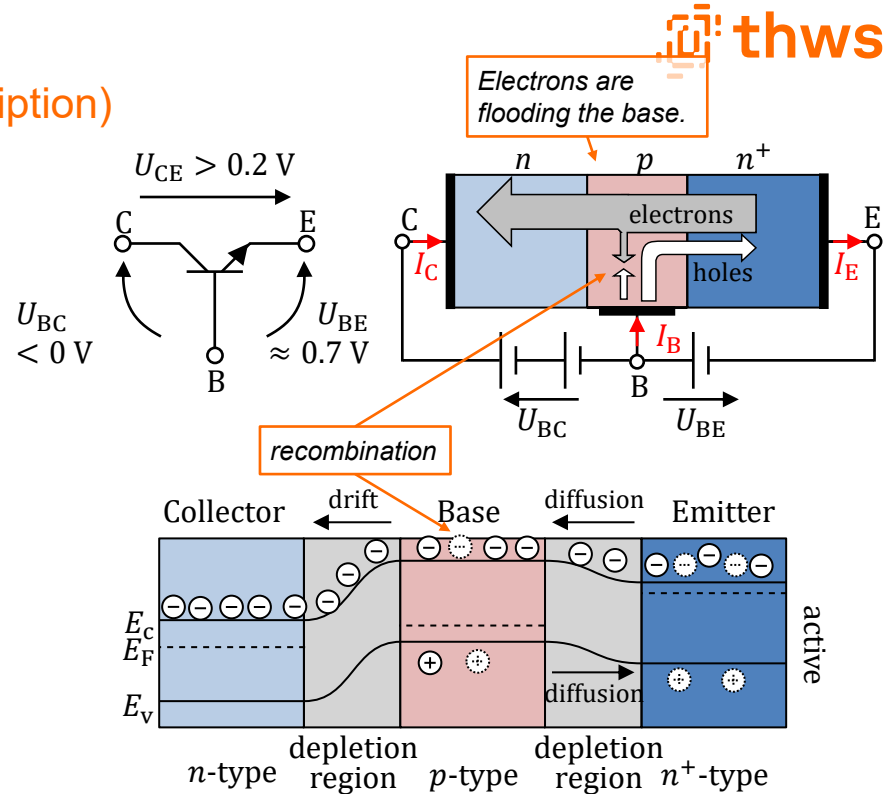
$$U_{BE} \approx 0.7 \text{ V}, U_{BC} < 0 \text{ V and } U_{CE} > 0.2 \text{ V}$$

Step-by-step mechanism:

1. Holes diffuse from the base to the emitter
 2. Due to the **n^+ doping** many **electrons** diffuse from the emitter to the base.
 3. Because of the **light doping** and **small diameter**, only a **fraction** of the electrons **recombine** in the base.
 4. The **vast majority** drifts **towards the collector** to close the circuit: flooding the base
- Control of **collector current** via base current.
 - Equivalent circuit model: **current controlled current source**.



A BJT operates in reversed active mode w/ reduced gain!



NPN in forward active mode: Top left: Terminal voltages; **Top right:** electron and hole current; **Bottom:** band diagram.

5.1.2 Basics of BJT transistors

Operation state: Forward active (quantitative description)

Terminal voltages

$$U_{BE} \approx 0.7 \text{ V}, U_{BC} < 0 \text{ V} \text{ and } U_{CE} > 0.2 \text{ V}$$

Base

- ⌈ Hole injection into emitter: $I_{B1} = I_{Ep} \ll I_{En}$
- ⌈ Recombination of p_B with e_E : $I_{B2} \ll I_{B1} \ll I_{En}$
- ⌈ Base current: $I_B = I_{B1} + I_{B2} \approx I_{B1}$

Emitter

- ⌈ Electron injection into base: I_{En}
- ⌈ Recombination with holes from the base.
- ⌈ Emitter current: $I_E = I_{Ep} + I_{En} \approx I_{En}$

Collector

- ⌈ Electron drift from emitter via base to collector.

Collector current:

$$I_C = I_E - I_B \approx I_E = \beta_{DC} I_B$$

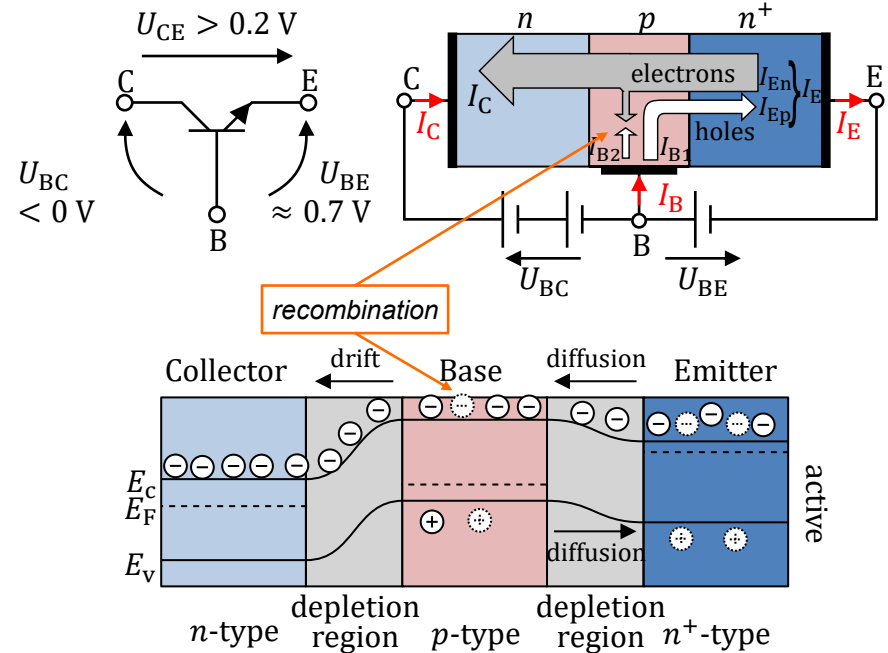
DC current gain:

Other symbols:
 B, β or h_{FE} .

$$\beta_{DC} = I_C / I_B \gg 1$$

Basics of electronics and components

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NPN in **forward active mode**: **Top left**: Terminal voltages; **Top right**: electron and hole current; **Bottom**: band diagram.

5.1.2 Basics of BJT transistors

Operation state: Saturation

Terminal voltages and collector current

- Both **BE** and **BC** are **forward biased**:

$$U_{BE} \approx 0.7 \text{ V and } U_{BC} \approx 0.5 \text{ V}$$

- Small collector-emitter voltage to drive the collector current:

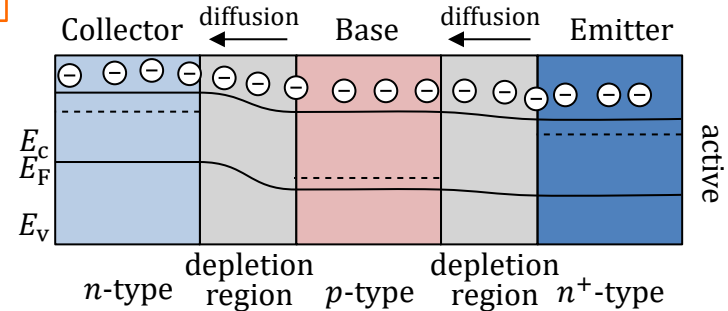
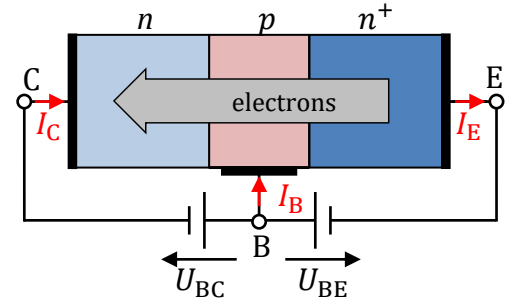
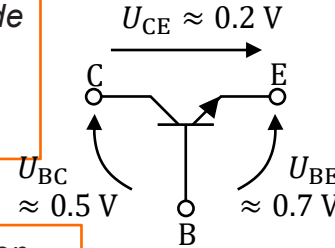
$$U_{CE} \approx 0.2 \text{ V}$$

- Collector current below amplification: U_{CE} be larger, as well.

$$I_C < \beta_{DC} I_B$$

- Electrons **diffuse** against the **electric field** from **emitter to collector** without control via base.
- Collector current I_C **independent** of current I_B
- Corresponds to **logical 1** or “on”
- Equivalent circuit model: **Short circuit** or small resistance
- Note: For high base currents, the collector-emitter voltages may be larger than 0.2 V.*

The BC diode is just at the threshold of conduction!



NPN in **saturation** mode: **Top left**: Terminal voltages; **Top right**: electron and hole current; **Bottom**: band diagram.

5.1.3 BJT large-signal model

5.1.3 BJT large-signal model

Transistor states

- **Forward active** ($U_{BE} \approx 0.7 \text{ V}$, $U_{CE} > 0.2 \text{ V}$, $U_{BC} < 0 \text{ V}$)

- BE path modelled by 0.7 V voltage source
 - CE path modelled by a controlled current source

$$I_C = \beta I_B$$

- U_{CE} across current source adjusts automatically

- **Saturation** ($U_{BE} \approx 0.7 \text{ V}$, $U_{CE} \approx 0.2 \text{ V}$, $U_{BC} \approx 0.5 \text{ V}$)

- CE path modelled by a **short circuit** or small ohmic resistance R_{CE} .

- I_C is controlled by the external circuit.

$$I_C < \beta I_B$$

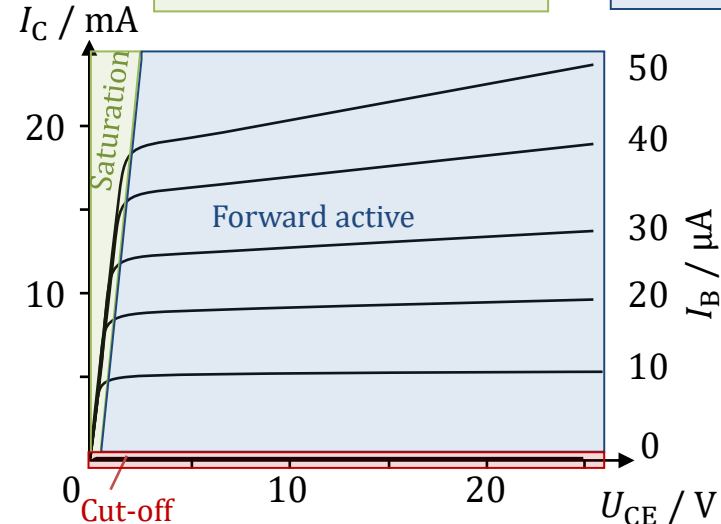
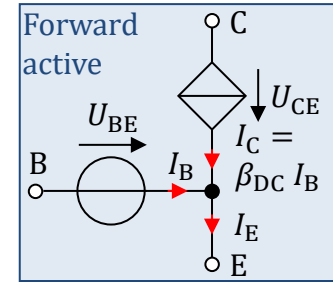
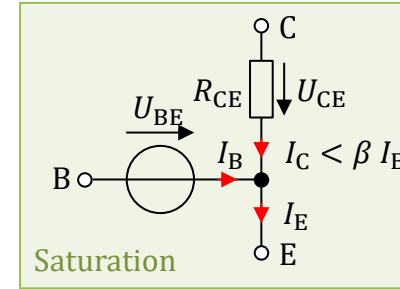
- BE path analogue to forward active

- **Cut-off** ($U_{BE} < 0 \text{ V}$, $U_{BC} < 0 \text{ V}$)

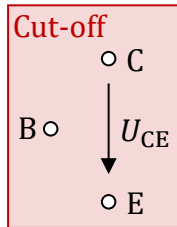
- No current independently of U_{CE}

- Modelled by an **open circuit**

- *Note: In reverse active collector and emitter are swapped. Analogue to forward active but less gain.*



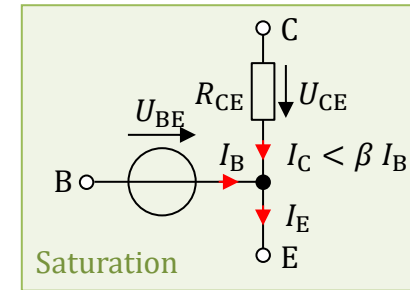
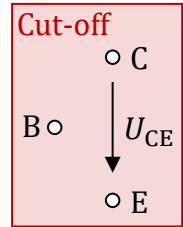
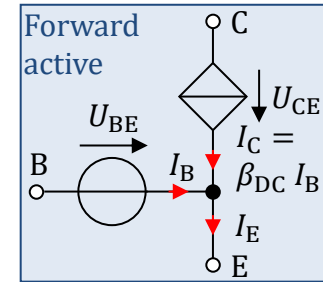
Characteristic output curve of a BJT transistor.



5.1.3 BJT large-signal model

Model of assumed states

- The **model of assumed states** is a **large-signal model** and used for **parametrizing** different transistor states.
- **Procedure** for finding and parametrizing the **right state**:
 1. **Assume** the most probable transistor state, e. g. saturation when $U_{BE} \approx 0.7 \text{ V}$, $U_{CE} \approx 0.2 \text{ V}$ or $I_C < \beta I_B$
 2. **Replace the transistor** by the corresponding large-signal **DC circuit model**.
 - ⌈ **Cut-off**: Open circuit
 - ⌈ **Saturation**: CE: short; BE: voltage source.
 - ⌈ **Active**: CE: controlled current source; BE: voltage source.
 3. **Solve** the resulting circuit. Not all model conditions are necessary for this purpose.
 4. **Check** if the remaining model conditions are satisfied.
 5. **If not**: select and test **another state model**.
- **Note**: The large-signal DC circuit model **always provides a unique solution**.



Large-signal equivalent circuits for the three main states: Cut-off, saturation, forward active.

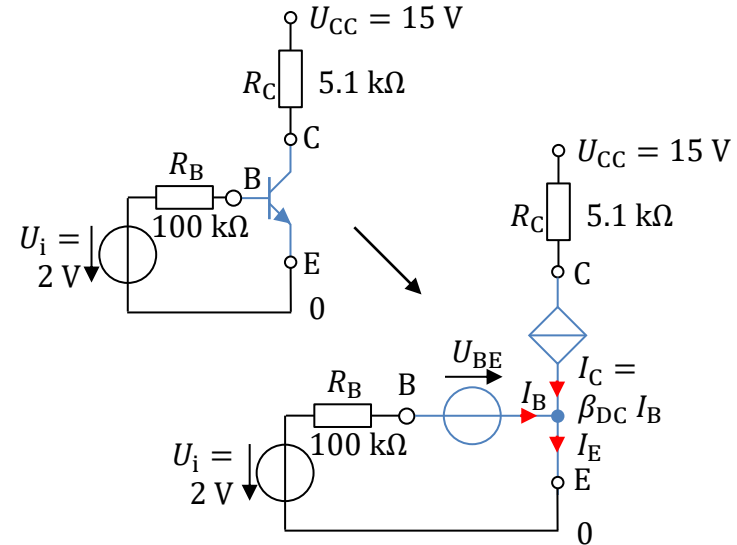
5.1.3 BJT large-signal model

Example: Model of assumed states

EXAM	Typical
•	Exam
•	Question

Solve the transistor circuit by determining unknown voltages U_{BE} and U_{CE} using the large-signal DC BJT model and the method of assumed states. Assume $\beta_{DC} = 100$.

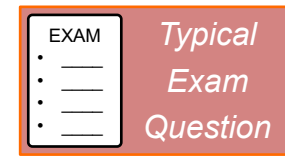
- First assume a state
 - Cut-off:** Can't be due to the BE-voltage ($0 < U_{BE} < 2 \text{ V}$)
 - Saturation** ($U_{BE} \approx 0.7 \text{ V}$; $U_{CE} \approx 0.2 \text{ V}$):
 - Base current: $I_B = \frac{U_i - U_{BE}}{R_B} = \frac{2 \text{ V} - 0.7 \text{ V}}{100 \text{ k}\Omega} = 13 \mu\text{A}$
 - Collector current: $I_{CS} = \frac{U_{CC} - U_{CE}}{R_C} = \frac{15 \text{ V} - 0.2 \text{ V}}{5.1 \text{ k}\Omega} = 2.9 \text{ mA}$
 - Check if condition $I_{CS} < \beta_{DC} I_B$ holds:
 $I_{CS} = 2.9 \text{ mA} > \beta_{DC} I_B = 1.3 \text{ mA} \rightarrow$ no saturation
 - Forward active** ($U_{BE} \approx 0.7 \text{ V}$; $U_{CE} > 0.2 \text{ V}$): The only plausible solution
- Find the missing quantities
 - $U_{BE} = 0.7 \text{ V}$
 - $I_C = \beta_{DC} I_B = 100 \cdot 13 \mu\text{A} = 1.3 \text{ mA}$
 - $U_{CE} = U_{CC} - I_C R_C = 8.37 \text{ V}$
- Because $I_C > 0$ cut-off can be excluded.
- Because $U_{CE} > 0.2 \text{ V}$ saturation can also be excluded.



Transistor circuit in unknown state (top).
Equivalent circuit assuming forward active mode (bottom).

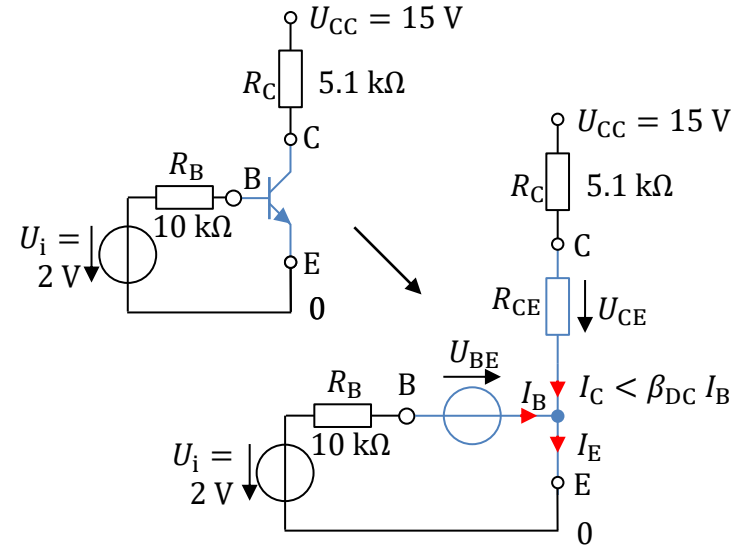
5.1.3 BJT large-signal model

Exercise: Model of assumed states



Solve the transistor circuit by determining unknown voltages U_{BE} and U_{CE} using the large-signal DC BJT model and the method of assumed states. Assume $\beta_{DC} = 100$.

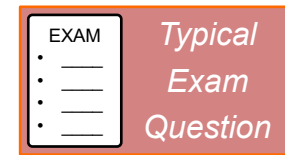
—	First assume a state
⌈	Cut-off: Can't be due to the BE-voltage ($0 < U_{BE} < 2 \text{ V}$)
⌈	Saturation ($U_{BE} \approx 0.7 \text{ V}$; $U_{CE} \approx 0.2 \text{ V}$):
⌈	Base current: $I_B = \frac{U_i - U_{BE}}{R_B} = \frac{2 \text{ V} - 0.7 \text{ V}}{10 \text{ k}\Omega} = 130 \mu\text{A}$
⌈	Collector current in saturation: $I_{Cs} = \frac{U_{CC} - 0.2 \text{ V}}{R_C} = 2.9 \text{ mA}$
⌈	Check if condition $I_{Cs} < \beta_{DC} I_B$ holds:
⌈	$I_{Cs} = 2.9 \text{ mA} < \beta_{DC} I_B = 13 \text{ mA} \rightarrow$ transistor in saturation
—	Find the missing quantities in this order
⌈	$U_{BE} \approx 0.7 \text{ V}$
⌈	$I_C = \beta_{DC} I_B = 100 \cdot 130 \mu\text{A} = 13 \text{ mA}$
⌈	$U_{CE} = U_{CC} - I_C R_C = 210 \text{ mV}$
—	Because $I_C > 0$ cut-off can be excluded.
—	Because $U_{CE} \approx 0.2 \text{ V}$ forward active can also be excluded.



Transistor circuit in unknown state (top).
Equivalent circuit assuming forward active mode (bottom).

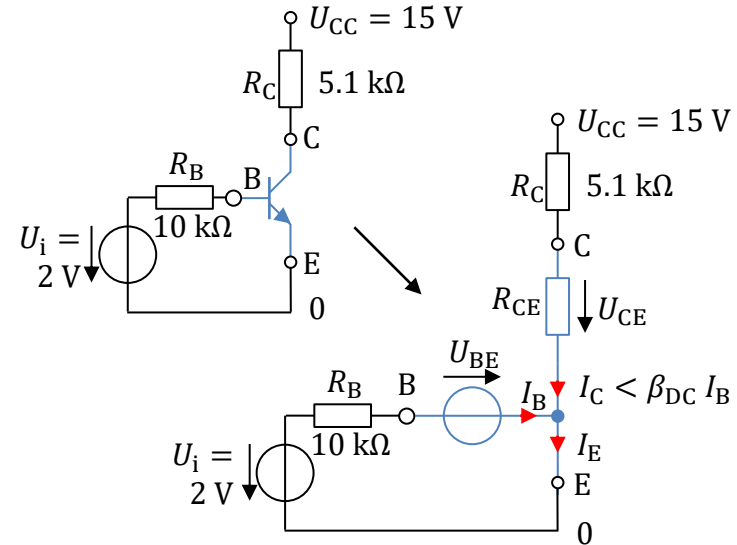
5.1.3 BJT large-signal model

Homework: Model of assumed states



Solve the transistor circuit by determining unknown voltages U_{BE} and U_{CE} using the large-signal DC BJT model and the method of assumed states. Assume $\beta_{DC} = 100$. Start by assuming forward active mode!

- Assuming forward active ($U_{BE} \approx 0.7 \text{ V}$; $U_{CE} > 0.2 \text{ V}$):
 - Base current: $I_B = \frac{U_i - U_{BE}}{R_B} = \frac{2 \text{ V} - 0.7 \text{ V}}{10 \text{ k}\Omega} = 130 \text{ }\mu\text{A}$
 - $I_C = \beta_{DC} I_B = 100 \cdot 130 \text{ }\mu\text{A} = 13 \text{ mA}$
 - Collector-emitter voltage: $U_{CE} = U_{CC} - I_C R_C = -51.3 \text{ V}$
 - Forward active mode is not a plausible option.
 - Saturation is the only plausible option.
- Find the missing quantities in this order
 - $U_{BE} = 0.7 \text{ V}$
 - $U_{CE} = U_{CC} - I_C R_C \approx 210 \text{ mV}$
- Because $I_C > 0$ cut-off can be excluded.
- Because $U_{CE} \approx 0.2 \text{ V}$ forward active can also be excluded



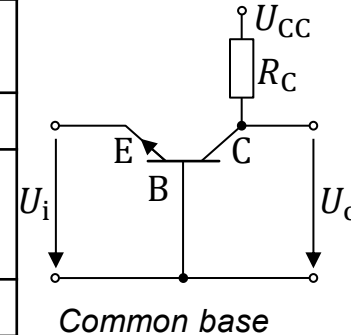
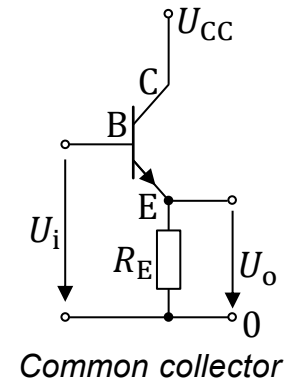
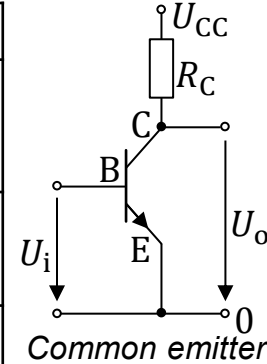
Transistor circuit in unknown state (top).
Equivalent circuit assuming forward active mode (bottom).

5.1.4 BJT basic circuits

5.1.4 BJT basic circuits

Overview

Circuit	Emitter	Collector	Base
Current gain A_i	$\beta = \frac{i_C}{i_B} \gg 1$	$\gamma = \frac{i_E}{i_B} = \beta + 1 \approx \beta$	$\alpha = \frac{i_C}{i_E} = \frac{\beta}{\beta + 1} \approx 1$
Voltage gain A_{uZ}	High, $-\frac{R_C}{\frac{r_{BE}}{\beta} + R_E}$	≈ 1	High, $+\beta \frac{R_C}{r_{BE}}$
Input impedance r_i	$R_B \parallel (r_{BE} + (\beta + 1) R_E)$	$R_B \parallel (r_{BE} + (\beta + 1) R_E \parallel R_L)$	$\frac{r_{BE}}{\beta + 1}$
Output impedance r_o	$R_C \parallel r_o$	$\approx \frac{r_{BE}}{\beta + 1}$	$R_C \parallel r_o$
Output Polarity	Inverted	Non-inverted	Non-inverted
Main property	High voltage gain, output inverted	Buffer, High input, low output impedance	High bandwidth, low input imp
Main Purpose	General purpose	Impedance matching	High-freq. amp.
Prevalence	Most Common	Common	Specialized



To identify the circuit type, find the transistor terminal shared by the input and output and remains at nearly constant voltage.

Basic NPN transistor configurations.

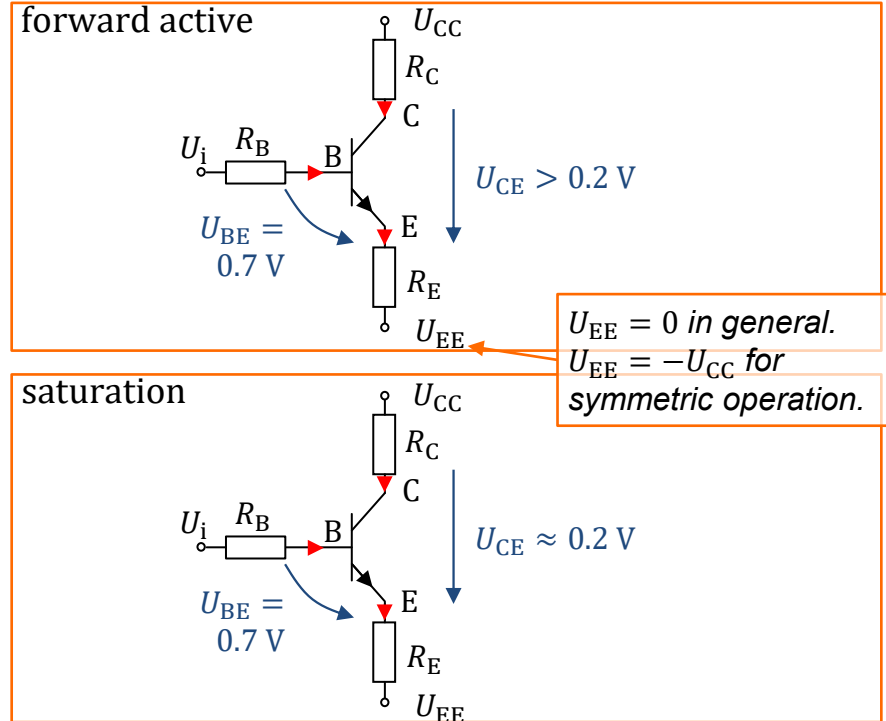
5.1.4 BJT basic circuits

DC biasing: purpose and target conditions

- DC biasing sets the transistor into the desired operating region **before any AC signal** is applied.
- For **forward-active** operation of a silicon BJT:

$U_{BE} \approx 0.7 \text{ V}$ and $U_{CE} > 0.2 \text{ V}$
- For switching operation in **saturation**, use

$U_{BE} \approx 0.7 \text{ V}$ and $U_{CE} \approx 0.2 \text{ V}$
- Choose resistor values such that the operating point Q is stable and lies in the intended region.
- Then apply **Ohm's law**, **KVL**, and **voltage-divider** relations to solve the circuit.



Voltage and current relations for a npn- (top) and a pnp-transistor (bottom) in active state.

5.1.4 BJT basic circuits

DC biasing: General DC relations for npn and pnp

- In **forward-active** mode, the collector current is approximately

$$I_C = \beta_{DC} I_B = \frac{U_{CC} - U_C}{R_C}$$

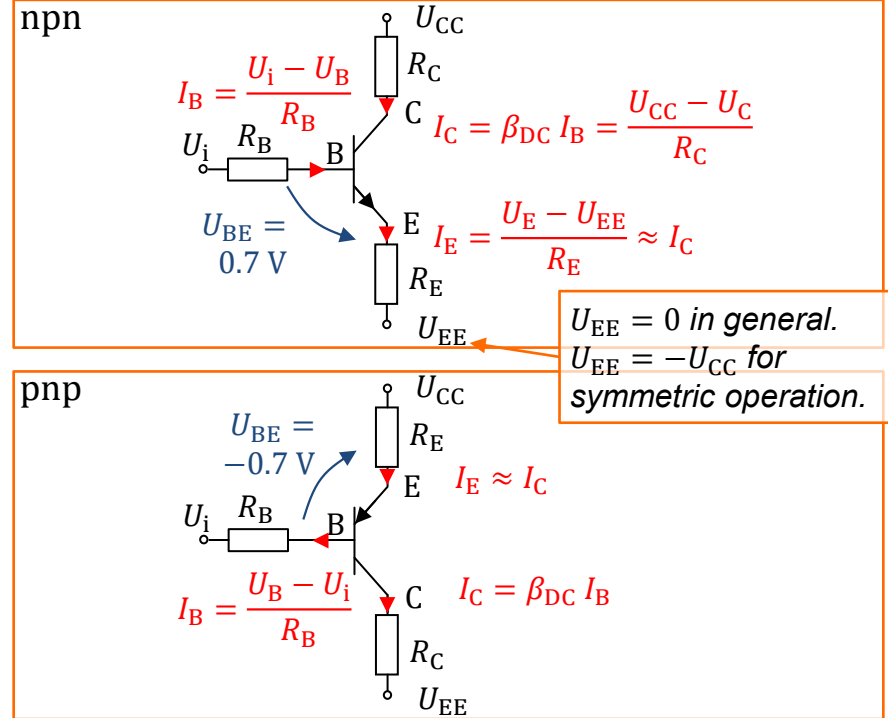
- In **saturation**

$$I_C \leq \beta_{DC} I_B$$

- In both modes, emitter current follows from

$$I_E = I_B + I_C \approx I_C$$

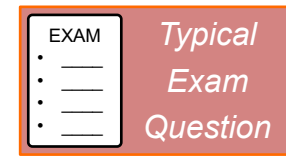
- For the npn transistor, the base is about 0.7 V above the emitter.
- For the pnp transistor, the base about 0.7 V below the emitter.
- Hint: For **symmetric voltage swing** in amplifier operation, choose the **collector-emitter voltage** in the middle of the supply range.*



Current and voltage relationships of an npn (**top**) and pnp-transistor (**bottom**) in **forward active mode**. For saturation $I_C = \beta_{DC} I_B$ does not apply!

5.1.4 BJT basic circuits

Exercise: DC biasing



Design a circuit where a pnp-transistor is in forward active mode and driving a collector current. The following parameters are given: $U_{CC} = 6\text{ V}$, $U_{CE} = -4.5\text{ V}$, $\beta_{DC} = 200$, $I_C = 10\text{ mA}$. *Hint: Use $I_E = I_C$ and $U_{BE} = -0.7\text{ V}$.*

- The emitter is pinned to the supply voltage (common-emitter topology):

$$U_E = U_{CC}$$

- With a collector-emitter voltage $U_{CE} = -4.5\text{ V}$ the collector voltage is:

$$U_C = U_{CC} + U_{CE} = 6\text{ V} - 4.5\text{ V} = 1.5\text{ V}$$

- The collector resistance must be:

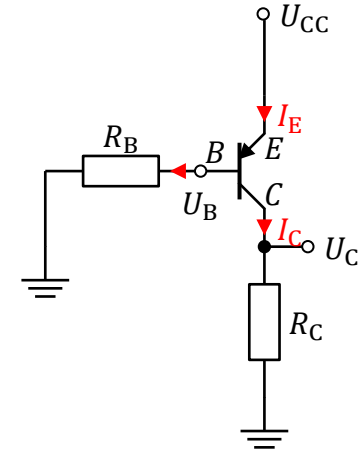
$$R_C = \frac{U_C}{I_C} = \frac{1.5\text{ V}}{10\text{ mA}} = 150\ \Omega$$

- The transistor is active ($U_{BE} = -0.7\text{ V}$), the base must be at:

$$U_B = U_E + U_{BE} = 6\text{ V} - 0.7\text{ V} = 5.3\text{ V}$$

- The base current calculates via β_{DC} and I_C and Ohm's law:

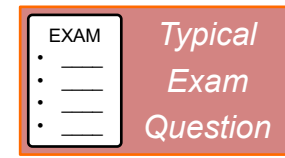
$$I_B = \frac{I_C}{\beta_{DC}} = \frac{U_B}{R_B}$$
$$\therefore R_B = \frac{\beta_{DC} U_B}{I_C} = \frac{200 \cdot 5.3\text{ V}}{10\text{ mA}} = 100\text{ k}\Omega$$



A pnp common-emitter circuit.

5.1.4 BJT basic circuits

Homework: DC biasing



What is the maximum base resistance R_B so that the transistor calculated before goes into saturation. Assume $U_{CE} = -0.1 \text{ V}$. Do not change the collector resistance.

- The collector resistance was calculated to $R_C = 150 \Omega$.
- The base voltage remains unchanged at $U_B = 5.3 \text{ V}$.
- With a collector-emitter voltage $U_{CE} = -0.1 \text{ V}$ the collector voltage is now:

$$U_C = U_{CC} + U_{CE} = 6\text{V} - 0.1 \text{ V} = 5.9 \text{ V}$$

- The collector current is therefore:

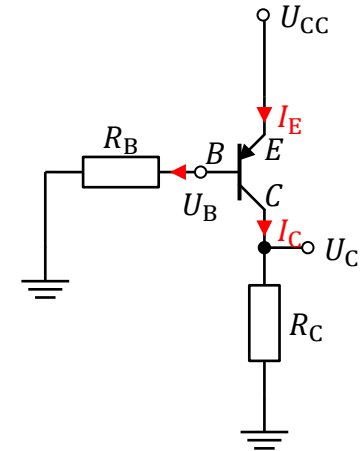
$$I_C = \frac{U_C}{R_C} = \frac{5.9 \text{ V}}{150 \Omega} = 39 \text{ mA}$$

- The inequality for a transistor in saturation must be satisfied:

$$I_C < \beta_{DC} I_B = \beta_{DC} \frac{U_B}{R_B}$$

$$\therefore R_B < \beta_{DC} \frac{U_B}{I_C} = 200 \cdot \frac{5.3 \text{ V}}{39 \text{ mA}} = 27 \text{ k}\Omega$$

- *Note: For robust operation in saturation, factor in an overshoot factor of 2, i. e. double the base current by halving the resistance $R_B^* = 14 \text{ k}\Omega$.*

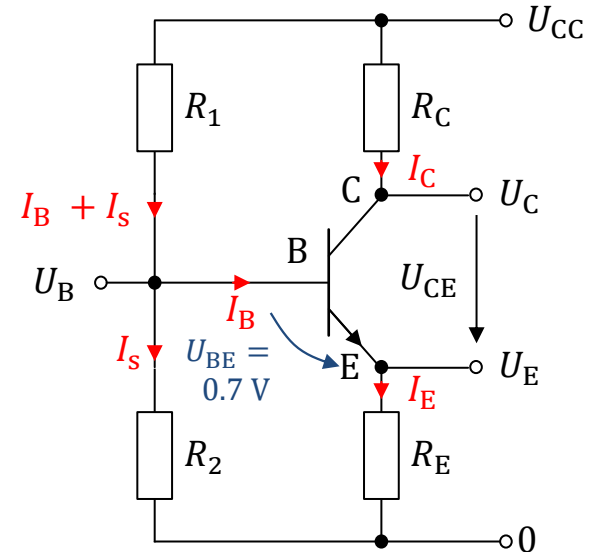


A pnp common-emitter circuit.

5.1.4 BJT basic circuits

Four-resistor bias circuit

- For many applications, a **four-resistors design with single supply voltage** U_{CC} is used to bias a transistor circuit.
- After correct biasing, small-signal s can be coupled in and out via **coupling capacitors** at terminals U_B , U_C and U_E .
- Resistors R_C or R_E can be removed **to pin the voltage** to U_{CC} or ground, respectively.
- In this case, the **circuit is named after the terminal** that is **common to the two others** and where no signal is applied.
 - ⌈ **Common emitter**: Input via U_B output via U_C
 - ⌈ **Common collector**: Input via U_B output via U_E
 - ⌈ **Common base**: Input U_E output via U_C



Generic four-resistor bias circuit.

5.1.4 BJT basic circuits

Four-resistor bias circuit: DC biasing

— CE path

- Define the mode of operation (off, active, saturation)
- Follow the procedure *DC biasing: General DC relations*

— BE path

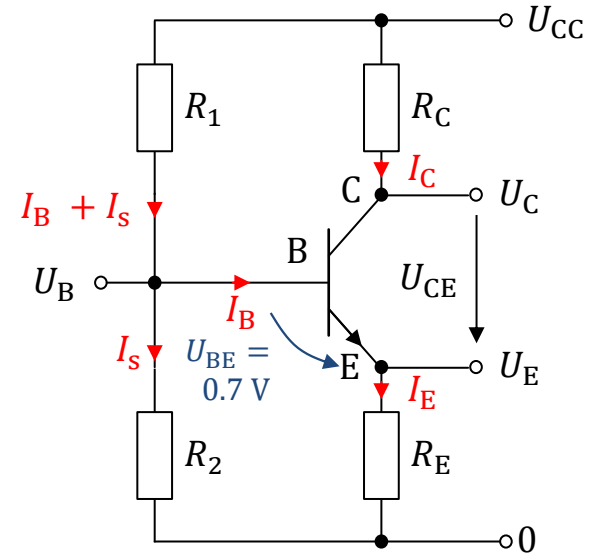
- The resistances R_1 and R_2 are calculated as

$$R_1 = \frac{U_{CC} - U_B}{I_B + I_S} \quad \text{and} \quad R_2 = \frac{U_B}{I_S}$$

With the current I_S through R_2 and $I_S + I_B$ through R_1 .

- If $I_S \gg I_B$ (typically $I_S > 10 I_B$), then R_1 and R_2 effectively form a voltage divider.
- R_1 is then approximately a function of R_2 .

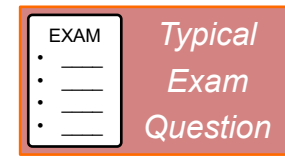
$$U_B = \frac{R_2}{R_1 + R_2} U_{CC} \quad \therefore R_1 = \left(\frac{U_{CC}}{U_B} - 1 \right) R_2$$



Generic four-resistor bias circuit.

5.1.4 BJT basic circuits

Example: Four-resistor circuit design and operating point 1/2



Determine the operating point $Q(U_{CE}, I_C)$ of a four-resistor basic circuit defined by the following parameters: $U_{CC} = 9\text{ V}$, $U_C = 6\text{ V}$, $U_E = 3\text{ V}$, $I_C = 1\text{ mA}$, $\beta_{DC} = 100$, $I_S = 10 I_B$. First calculate the CE-path of the circuit.

1. CE-part: Plug in all known values and calculate R_C , I_E , R_E and U_{CE}

— $R_C = \frac{U_{CC} - U_C}{I_C} = \frac{9\text{ V} - 6\text{ V}}{1\text{ mA}} = 3\text{ k}\Omega$

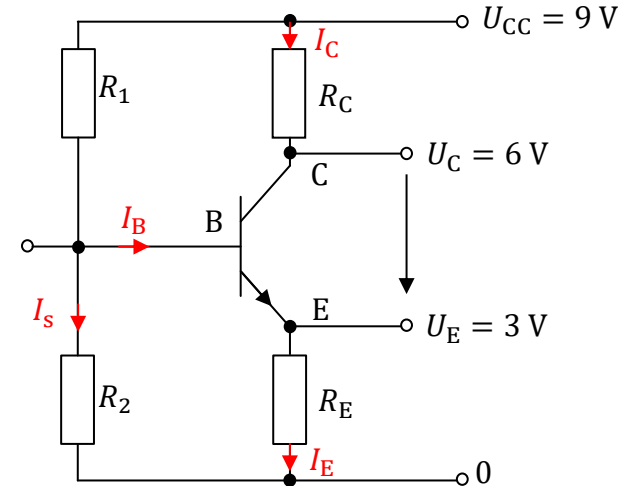
— $I_E = I_C + I_B = I_C + \frac{I_C}{\beta_{DC}} = \left(1 + \frac{1}{100}\right) I_C = 1.01\text{ mA} \approx 1\text{ mA}$

— $R_E = \frac{U_E}{I_E} = \frac{3\text{ V}}{1\text{ mA}} = 3\text{ k}\Omega$

— $U_{CE} = U_C - U_E = 6\text{ V} - 3\text{ V} = 3\text{ V}$

— Operating point: $Q(3\text{ V}; 1\text{ mA})$

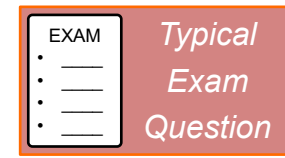
→ the transistor is in forward active state.



Fully-biased four-resistor npn-BJT transistor circuit.

5.1.4 BJT basic circuits

Example: Four-resistor circuit design and operating point 2/2



Determine the operating point $Q(U_{CE}, I_C)$ of a four-resistor basic circuit defined by the following parameters: $U_{CC} = 9\text{ V}$, $U_C = 6\text{ V}$, $U_E = 3\text{ V}$, $I_C = 1\text{ mA}$, $\beta_{DC} = 100$, $I_S = 10 I_B$. Use the exact and the simplified calculation to identify R_1 and R_2 . Check your calculations using Ltspice *BJT_basic_circuit.asc*

2. BE-part

1. Base current: $I_B = \frac{I_C}{\beta_{DC}} = \frac{1\text{ mA}}{100} = 10\text{ }\mu\text{A}$

2. Side current: $I_S = 10 \cdot I_B = 100\text{ }\mu\text{A}$

3. R_2 : Transistor is in active state: $U_{BE} = 0.7\text{ V} \quad \therefore \quad U_B = 3.7\text{ V}$

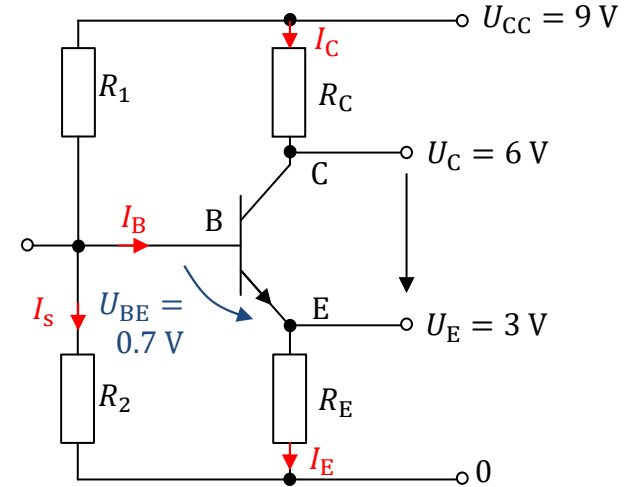
$$\therefore R_2 = \frac{U_B}{I_S} = 37\text{ k}\Omega$$

4. Exact: $R_1 = \frac{U_{CC} - U_B}{I_B + I_S} = \frac{9\text{ V} - 3.7\text{ V}}{10\text{ }\mu\text{A} + 100\text{ }\mu\text{A}} = 48\text{ k}\Omega$

5. Simplified

$$U_B = \frac{R_2}{R_1 + R_2} U_{CC} \quad \therefore \quad R_1 = \left(\frac{U_{CC}}{U_B} - 1 \right) R_2 = 56\text{ k}\Omega$$

3. The simplified version underestimates the base and collector current.



Fully-biased four-resistor npn-BJT transistor circuit.

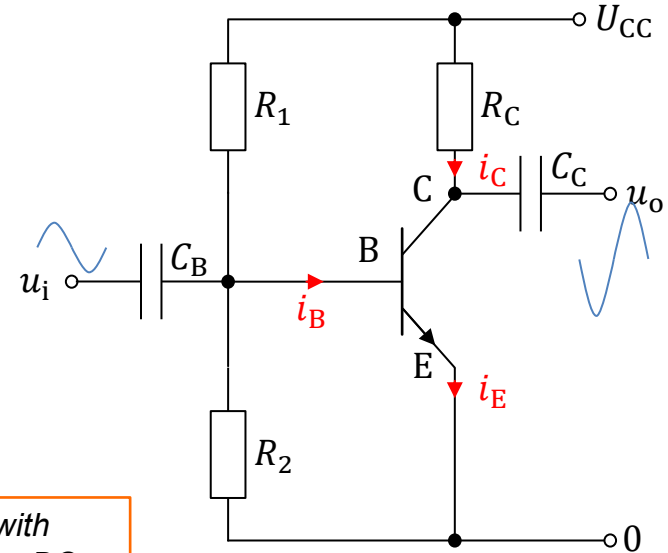
5.1.4 BJT basic circuits

Common emitter circuit

- A transistor topology where both the base and the collector refer to a **common emitter**.
- A **common emitter circuit** is biased via the **collector resistors** R_C and R_1 and R_2 forming a **voltage divider**.
- **Small input signals** u_i are coupled into the **base** via a **coupling capacitor** C_B .
- **Output signals** u_o are coupled out at the **collector**.
- The output voltage is inverted relative to the input; for sinusoidal small-signal s this means a phase shift of 180° .
- The current gain $\beta = \frac{i_B}{i_C}$ is around $-50 \dots -100$.

As the gain might vary with frequency, commonly the DC gain is provided in data sheets.

- The voltage gain depends on the resistance R_C and is high.
- BJT are very **sensitive to temperature** → under operation they get hot, the i_C **increases** and the **Q-point shifts** → non-linear distortion.



Common emitter circuit without stabilization.

5.1.4 BJT basic circuits

Common emitter circuit with Q-point stabilization

- To stabilize the Q-point, an **emitter resistor** $R_E \approx r_{BE}$ is added.
- Now, the **emitter follows the base** ($U_{BE} \approx 0.7 \text{ V}$).
- If the collector current increases $\rightarrow$ emitter voltage is pulled up $\rightarrow$ base-emitter voltage is reduced $\rightarrow$ base current is reduced $\rightarrow$ collector current remains stable $\rightarrow$ **negative current feedback**
- To avoid negative feedback and maximise the gain for AC signals, the **emitter is pulled to ground via a capacitor in parallel**.
- The **voltage gain** is defined by the resistors R_C and R_E , only

$$A_u = \frac{u_o}{u_i} = -\frac{R_C}{R_E}$$

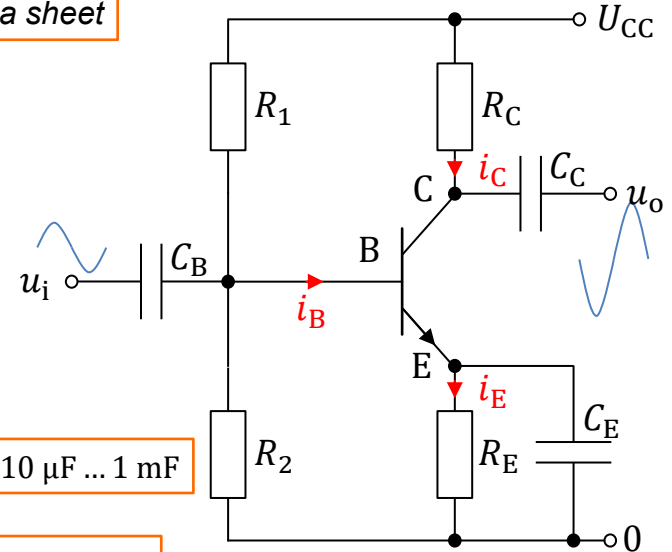
Signal is *inverted*

Quick and dirty intuition:
 R_C up $\therefore I_C$ const. $\therefore u_o$ down
 R_E down $\therefore I_C$ up $\therefore u_o$ down

- **Note: a second output can be generated at the emitter.**

$10 \Omega \dots 1 \text{ k}\Omega$
Data sheet

$C_E = 10 \mu\text{F} \dots 1 \text{ mF}$



Common emitter circuit with negative current feedback.

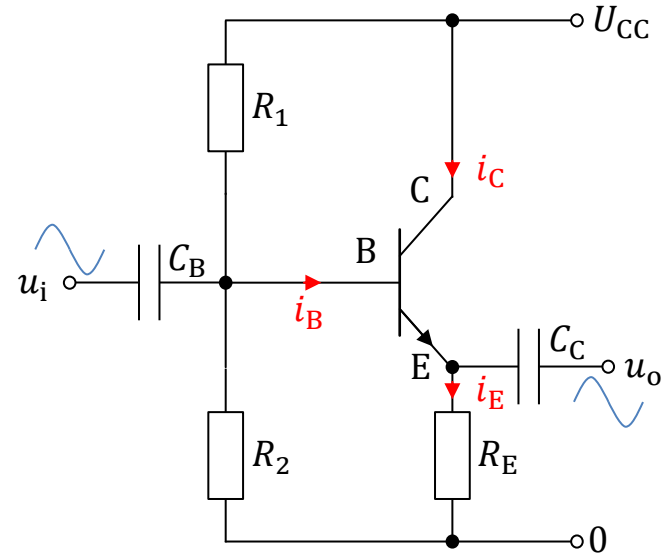
5.1.4 BJT basic circuits

Common collector circuit

- A circuit where the **collector is common** both to the base and the emitter.
- Biased via the emitter resistor R_E and R_1/R_2 forming a voltage divider.
- The emitter resistor enables **negative feedback** (analogue to Q-point stabilization in the common emitter circuit) both for DC large and AC small-signal s and can not be suppressed.
- Other properties
 - ⌈ Voltage gain $A_u = \frac{u_o}{u_i} \approx 1$
 - ⌈ Current gain $\gamma = \frac{i_o}{i_i} = \frac{i_E}{i_B} = 1 + \beta \approx \beta$
 - ⌈ Large input impedance, small output impedance → impedance converter
- To allow for symmetric operation U_E should be set to about $U_E \approx 0.5 U_{CC}$ and $R_E \approx R_L$ to maximise the power (maximum power theorem).

The emitter voltage follows the base

$$i_E = i_B + i_C \\ = (1 + \beta_{DC}) i_B$$



5.1.4 BJT basic circuits

Converting a four-resistor to a three-resistor design

- Instead of using a **four-resistor-design** with **one voltage source**, a **three-resistor-design** with **two voltage sources** can be used.
- Apply **Thévenin theorem** to convert one to another.

- Thévenin resistance R_B at the base

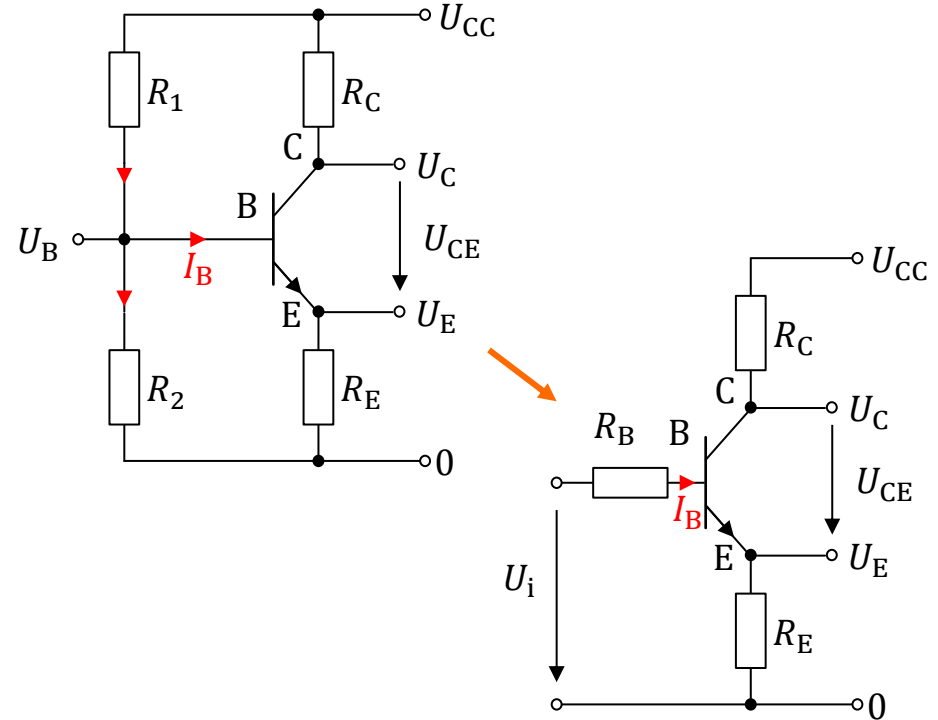
$$R_B = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}$$

- Thévenin voltage U_i under open-load condition

$$U_i = U_B = \frac{R_2}{R_1 + R_2} U_{CC}$$

- Alternative: calculation via KVL

$$U_i = U_{RB} + U_{BE} + U_E = R_B I_B + U_{BE} + R_E I_E$$



5.1.4 BJT basic circuits

Example: Four-resistor circuit design and operating point

Convert the four-resistor circuit to a three-resistor equivalent. The parameters from before were calculated as: $R_1 = 48 \text{ k}\Omega$, $R_2 = 37 \text{ k}\Omega$, $U_{CC} = 9 \text{ V}$, $I_B = 10 \text{ }\mu\text{A}$, $U_B = 3 \text{ V}$, $U_E = 3 \text{ V}$. Use LTSpice to compare the result.

- Calculate the equivalent base resistance

$$R_B = \frac{R_1 R_2}{R_1 + R_2} = \frac{48 \text{ k}\Omega \cdot 37 \text{ k}\Omega}{48 \text{ k}\Omega + 37 \text{ k}\Omega} = 20.894 \text{ k}\Omega \approx 21 \text{ k}\Omega$$

- Determine the input voltage using Thévenin theorem:

$$U_i = \frac{R_2}{R_1 + R_2} U_{CC} = \frac{37 \text{ k}\Omega}{37 \text{ k}\Omega + 48 \text{ k}\Omega} 9 \text{ V} = 3.92 \text{ V}$$

- Check: Calculate the equivalent input voltage using KVL

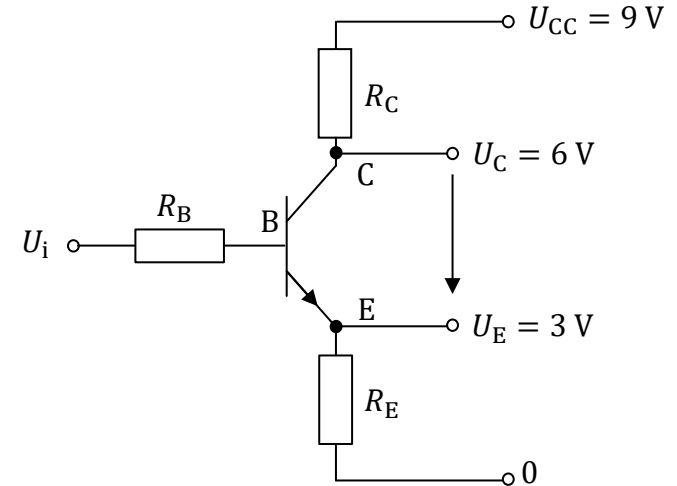
$$\begin{aligned} U_i &= R_B I_B + U_{BE} + U_E \\ &= 21 \text{ k}\Omega \cdot 10 \text{ }\mu\text{A} + 0.7 \text{ V} + 3 \text{ V} = 3.91 \text{ V} \end{aligned}$$

- Check the result: The base voltage must remain at $U_B = 3.7 \text{ V}$

$$I_B = \frac{U_i - U_B}{R_B}$$

$$\therefore U_B = U_i - R_B I_B = 3.91 \text{ V} - 21 \text{ k}\Omega \cdot 10 \text{ }\mu\text{A} = 3.7 \text{ V}$$

- Except for the missing side current, the three-resistor equivalent is equal to the four-resistor circuit.



Fully-biased four-resistor npn-BJT transistor circuit.

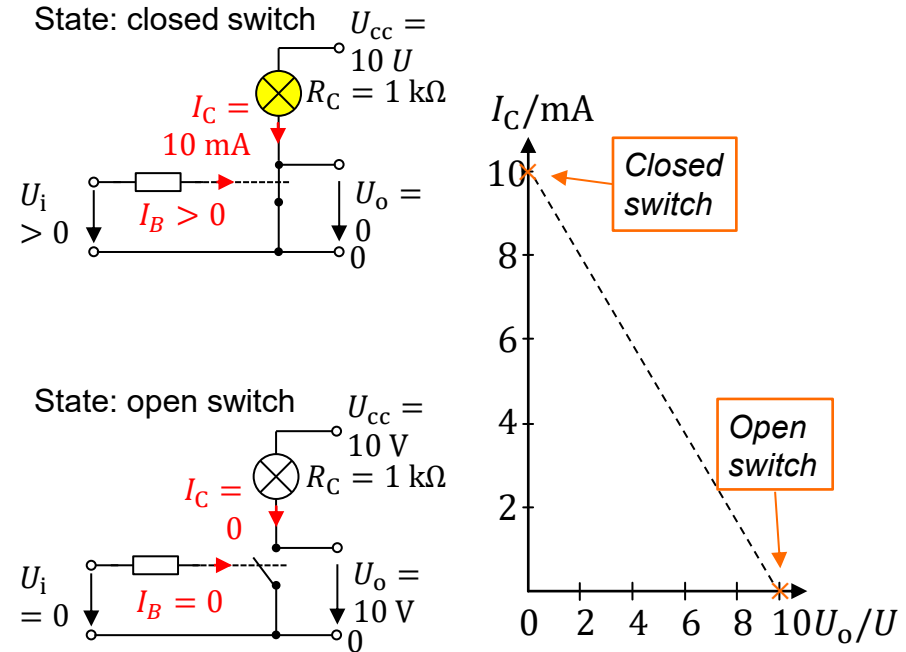


5.1.5 Transistor as switch

5.1.5 Transistor as switch

About switches

- Definition of a switch: exactly two states
 - ⌈ Open
 - ⌈ Closed
- Corresponds to exactly two discrete states on the load line
- The trigger can be
 - ⌈ Physical force (“light switch”)
 - ⌈ Current (BJT)
 - ⌈ Voltage (FET)



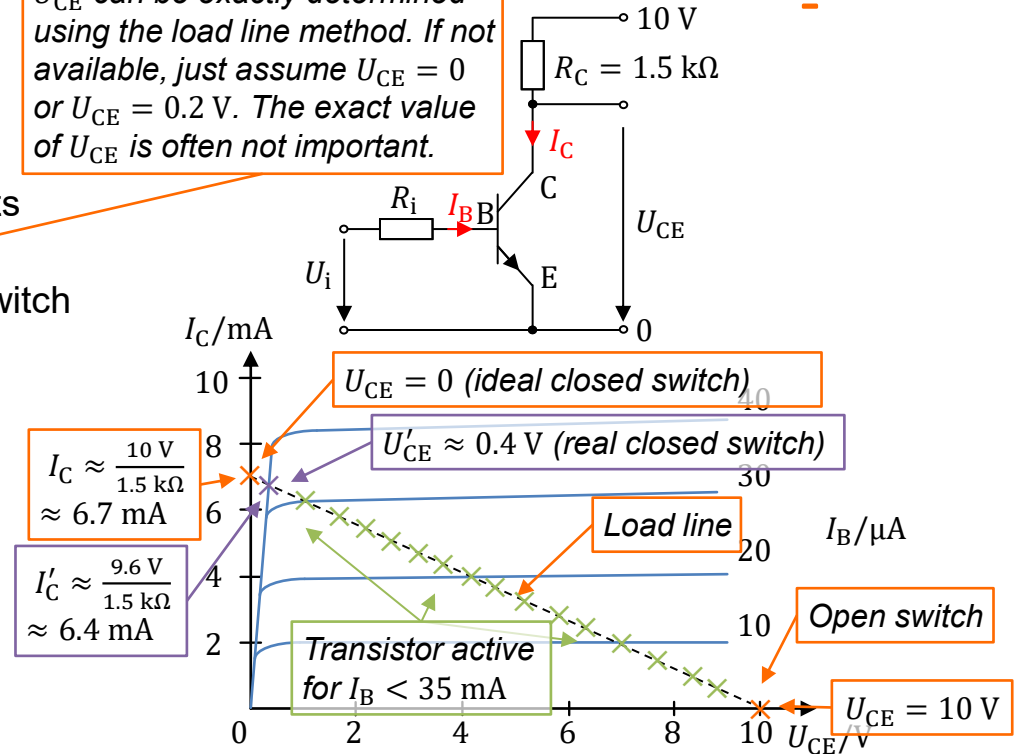
Left: Equivalent circuits of a voltage-controlled switch in on- and off state. **Right:** Load line with open and closed state.

5.1.5 Transistor as switch

Inverter circuit with input resistor

- A transistor switching between **cut-off** and **saturation** acts as an (almost) ideal switch.
- In common emitter topology, the circuit inverts the input signal.
 - 7 $U_i = \text{high} \rightarrow I_B = \text{high} \rightarrow$ **closed** switch
 $\rightarrow U_{CE} \approx 0.2 \text{ V}$
 - 7 $U_i = \text{low} \rightarrow I_B = 0 \rightarrow$ **open** switch
 $\rightarrow U_{CE} = U_{CC}$
- Note: If I_B is too small, the transistor state is forward active, not in saturation.

U_{CE} can be exactly determined using the load line method. If not available, just assume $U_{CE} = 0$ or $U_{CE} = 0.2 \text{ V}$. The exact value of U_{CE} is often not important.



Left: common-emitter circuit utilized as switch.

Top: characteristic curve and load line including operating points P0 and P1.

5.1.5 Transistor as switch

Exercise: Real vs. Ideal transistor in saturation.

Compare the collector current I_C for the transistor in saturation by applying the graphical load line method and by assuming $U_{CE} = 0$ V.

1. Load line: The CE-path is replaced by a short and an open:

Short: $I_{Cs} = \frac{U_{CC}}{R_C} = \frac{12 \text{ V}}{1.5 \text{ k}\Omega} = 8 \text{ mA}$

Open: $U_{CEo} = U_{CC} = 12 \text{ V}$

Draw the load line by connecting both points

The intersect with the characteristic curve for $I_B = 40 \mu\text{A}$ and larger is the exact operating point of the transistor in saturation

Measure the current and voltage.

U: 10 V correspond to 6.78 cm

I: 10 mA corresponds to 4.38 cm

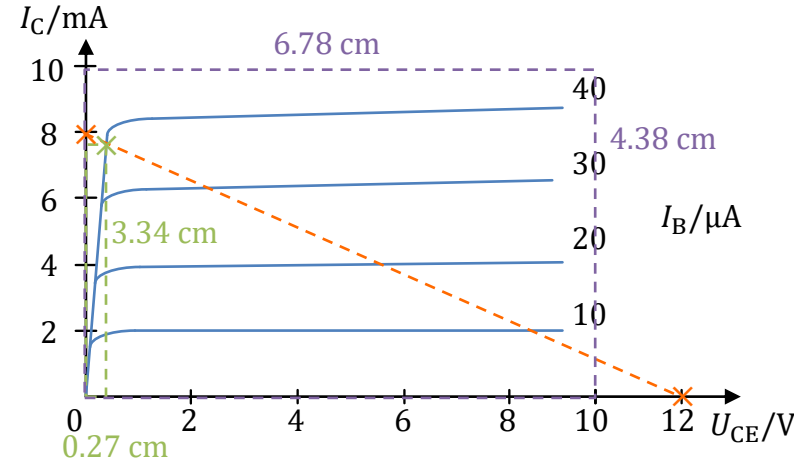
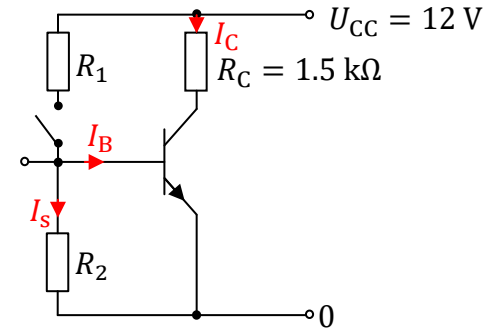
The operating point at $x = 0.27 \text{ cm}$, $y = 3.34 \text{ cm}$ corresponds

$$U_{CE} = \frac{0.27 \text{ cm}}{6.78 \text{ cm}} \cdot 10 \text{ V} = 0.398 \text{ V} \approx 0.4 \text{ V}$$

$$I_C = \frac{3.34 \text{ cm}}{4.38 \text{ cm}} \cdot 10 \text{ mA} = 7.626 \text{ mA} \approx 7.6 \text{ mA}$$

2. Assuming $U_{CE} = 0$ V, the collector current is equal to the short circuit current, or $I_C = 8 \text{ mA}$

3. Comparison: $Q_{\text{real}}(0.4 \text{ V}, 7.6 \text{ mA})$ versus $Q_{\text{ideal}}(0; 8 \text{ mA})$



5.1.5 Transistor as switch

Electrical design of an inverter circuit

1. Choose the collector current in saturation:

$$I_C = \frac{U_{CC} - U_{CE,sat}}{R_C} \approx \frac{U_{CC}}{R_C}$$

Because of the overshoot factor, $U_{CE,sat}$ can be often ignored.

1. Compute the minimum base current:

$$I_B = \frac{I_C}{\beta_{DC}}$$

3. Add an overshoot factor, typically 2, to ensure robust saturation, this is called **overdrive**.

$$I_B^* = b I_B = 2 I_B$$

3. Use KVL to calculate R_i

$$U_i - I_B^* R_i - U_{BE} = 0$$

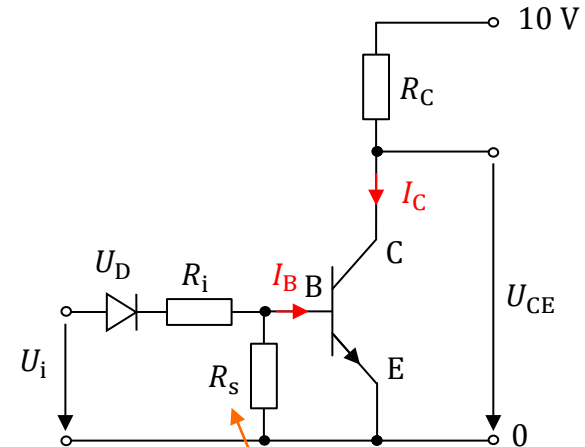
$-U_D$ if a diode in series is added.

5. Verify the saturation condition

$$I_C \leq \beta_{DC} I_B$$

— Optional:

- 7 series diode U_D to raise turn-on threshold
- 7 large side resistor R_s to bypass leakage in cut-off



Cut-off: $R_s \ll r_{BE}$ to bypass residual current via R_s
Saturation: $R_s \gg r_{BE}$ to reduce leakage via R_s

Inverter basic circuit with input resistance R_i , side resistance R_s and series diode.

5.1.5 Transistor as switch

Exercise: Recap BJT in saturation biasing

Determine the minimum base I_B and overdrive current I_B^* to bring the transistor in saturation. Determine R_1 and R_2 . *Note: Assume the $U_{CE} = 0$, $I_S = 10 I_B^*$ and determine β from the graph.*

- Minimum Base current

$$I_B = \frac{I_C}{\beta} = \frac{U_{CC} - U_{CE}}{\beta R_C} = \frac{12 \text{ V} - 0 \text{ V}}{200 \cdot 1.5 \text{ k}\Omega} = 40 \mu\text{A}$$

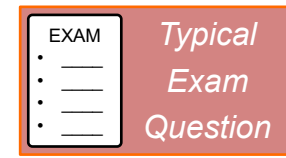
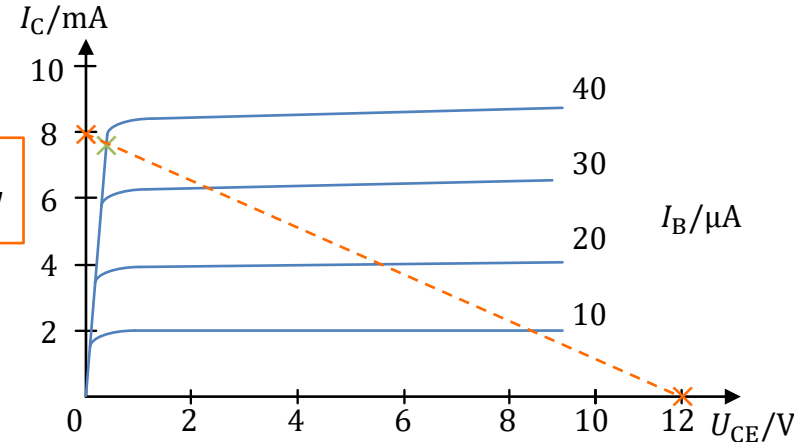
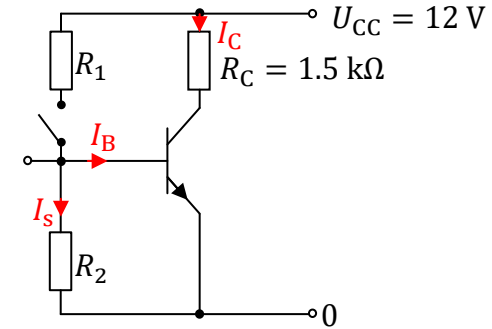
- Overdrive current: $I_B^* = 2 \cdot I_B = 2 \cdot 40 \mu\text{A} = 80 \mu\text{A}$
- Common emitter configuration with base voltage $U_B = 0.7 \text{ V}$.
- Determine R_2 for $I_S = 10 I_B = 10 \cdot 80 \mu\text{A} = 800 \mu\text{A}$

$$R_2 = \frac{U_B}{I_S} = \frac{0.7 \text{ V}}{800 \mu\text{A}} = 875 \Omega \approx 0.91 \text{ k}\Omega$$

- Determine R_1

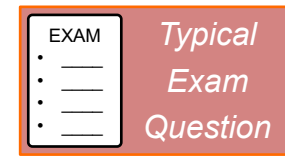
$$R_1 = \frac{U_{CC} - U_B}{I_B + I_S} = \frac{12 \text{ V} - 0.7 \text{ V}}{80 \mu\text{A} + 800 \mu\text{A}} = 12.8 \text{ k}\Omega \approx 13 \text{ k}\Omega$$

Next in the E24 series!



5.1.5 Transistor as switch

Homework: Recap BJT in saturation biasing



Determine the minimum base I_B and overdrive current I_B^* to bring the transistor in saturation. Determine R_1 and R_2 . This time assume $U_{CE} = 0.4 \text{ V}$ from the load line and round to the next value in the E24 series.

- Determine β : The collector current in forward active is approx. 200 times larger than the base current: $\beta \approx 200$

- Minimum Base current

$$I_B = \frac{I_c}{\beta} = \frac{U_{CC} - U_{CE}}{\beta R_C} = \frac{12 \text{ V} - 0.4 \text{ V}}{200 \cdot 1.5 \text{ k}\Omega} = 38.667 \mu\text{A} \approx 39 \mu\text{A}$$

- Overdrive current: $I_B^* = 2 \cdot I_B = 2 \cdot 39 \mu\text{A} = 78 \mu\text{A}$
- Common emitter configuration with base voltage $U_B = 0.7 \text{ V}$.
- Determine R_2 for $I_S = 10 I_B = 10 \cdot 78 \mu\text{A} = 780 \mu\text{A}$

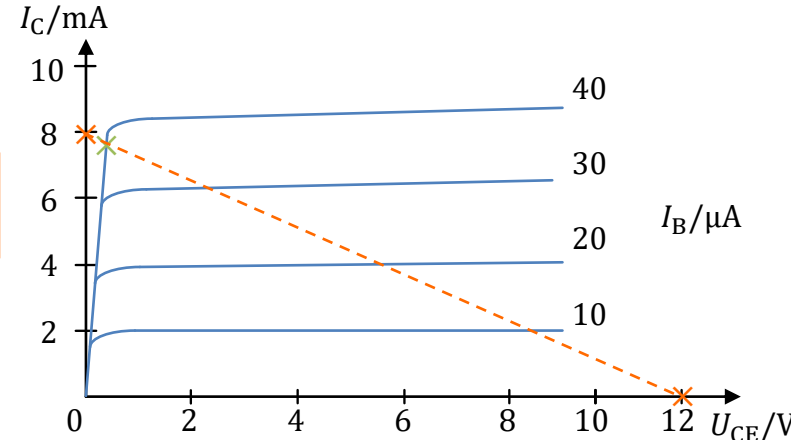
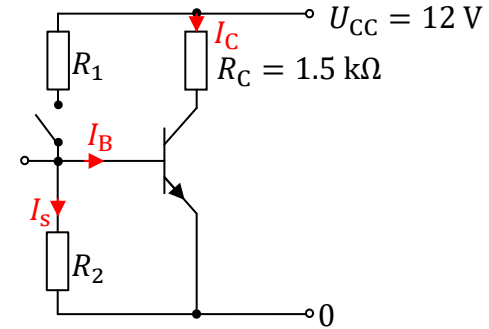
$$R_2 = \frac{U_B}{I_S} = \frac{0.7 \text{ V}}{780 \mu\text{A}} = 897 \Omega \approx 0.91 \text{ k}\Omega$$

- Determine R_1

$$R_1 = \frac{U_{CC} - U_B}{I_B + I_S} = \frac{12 \text{ V} - 0.7 \text{ V}}{78 \mu\text{A} + 780 \mu\text{A}} = 13.170 \text{ k}\Omega \approx 13 \text{ k}\Omega$$

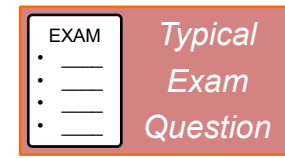
- In practice both results are equal. The real voltage drop of the transistor can be ignored in most cases.

Next in the E24 series!



5.1.5 Transistor as switch

Homework: Electrical design of an inverter circuit 1/2



You are given the following inverter circuit. Please parametrize I_B and R_i and output voltage U_o for $U_{iH} = 3.5 \text{ V}$ and $U_{iL} = 0.4 \text{ V}$ for the following parameters: $U_{CC} = 10 \text{ V}$, $R_C = 1 \text{ k}\Omega$, $R_S = 470 \text{ k}\Omega$, $\beta_{DC} = 100$, $b = 2$.

— H-level input: $U_{iH} = 3.5 \text{ V}$

— Calculate: I_B^* (base current in overdrive)

$$I_B = \frac{I_C}{\beta_{DC}} = \frac{10 \text{ mA}}{100} = 0.1 \text{ mA}$$

$$I_B^* = b I_B = 2 \cdot 0.1 \text{ mA} = 0.2 \text{ mA}$$

I_{RS} can be ignored because r_{BE} is much smaller than R_S .

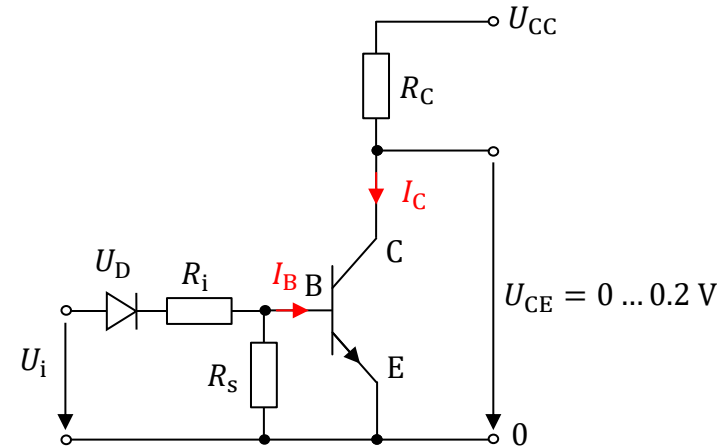
$$R_i = \frac{U_i - U_D - U_{BE}}{I_B^* + I_{RS}} = \frac{3.5 \text{ V} - 0.7 \text{ V} - 0.7 \text{ V}}{0.2 \text{ mA}} = 10 \text{ k}\Omega$$

— Confirm that the condition for saturation $I_C < \beta_{DC} I_B^*$ is satisfied:

$$I_C = \frac{U_{CC} - U_{CE \text{ sat}}}{R_C} \approx \frac{U_{CC}}{R_C} = \frac{10 \text{ V}}{1 \text{ k}\Omega} = 10 \text{ mA} < \beta_{DC} I_B^* = 20 \text{ mA}$$

— With the transistor in saturation, the output voltage is about:

$$U_o = U_{CE \text{ sat}} \approx 0.2 \text{ V}$$



5.1.5 Transistor as switch

Exercise: Electrical design of an inverter circuit 2/2

EXAM

Typical
Exam
Question

You are given the following inverter circuit. Please find the high and low-level output U_o for the following parameters: $U_{CC} = 10\text{ V}$, $R_C = 1\text{ k}\Omega$, $R_S = 470\text{ k}\Omega$, $U_{iH} = 3.5\text{ V}$, $U_{iL} = 0.4\text{ V}$, $\beta_{DC} = 100$, $b = 2$.

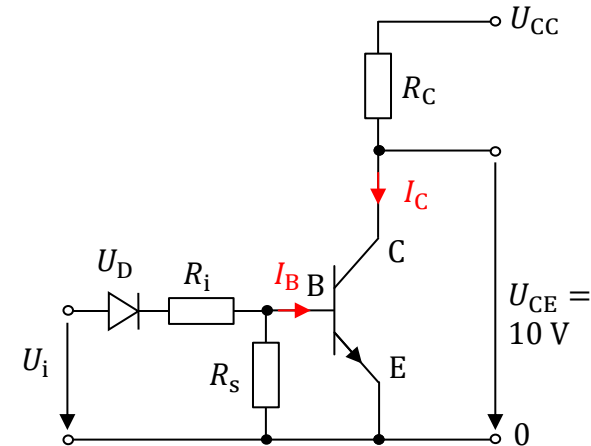
- L-level input: $U_{iL} = 0.4\text{ V}$
- Because the diode and the BE-path are below the threshold voltage and the resistances are in the Mega-Ohm range (open switches). → The current I_B is negligible.
- The side resistance R_S is small compared to the base-emitter differential resistance r_{BE} of the BE-diode in cut-off and bypasses any residual base current through the series diode.
- We can conclude that the transistor is off.

$$\therefore I_B = 0$$

$$\therefore I_C = 0$$

$$\therefore U_o = U_{CE} = U_{CC} = 10\text{ V}$$

- The output voltage is the inverse of the input voltage.

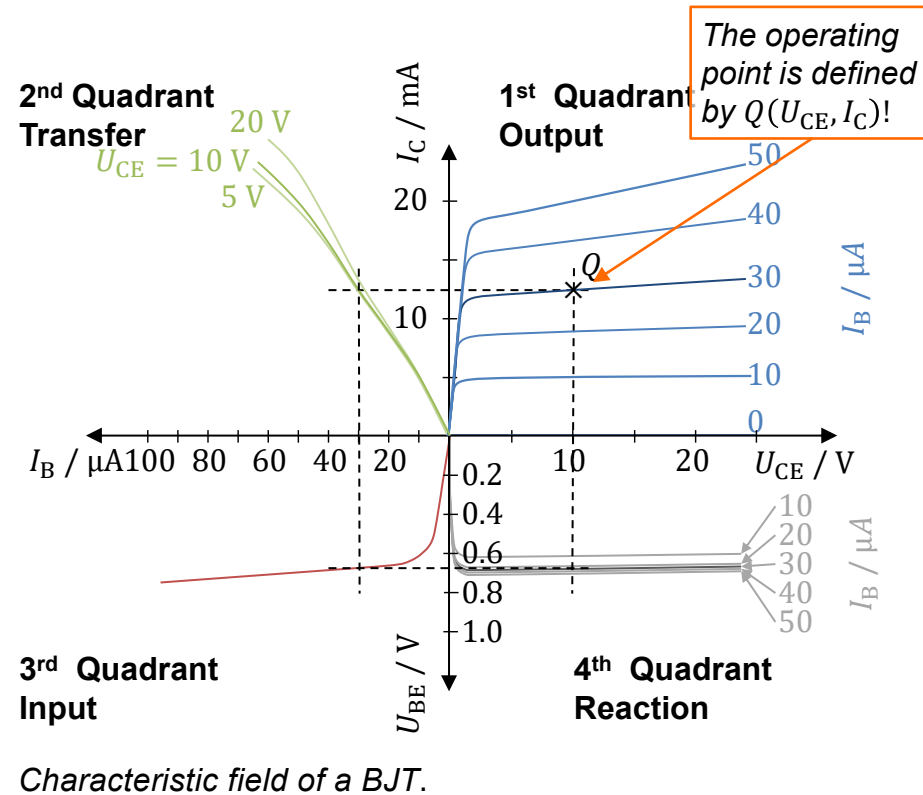


5.1.6 Transistor characteristic field

5.1.6 Transistor characteristic field

About four-quadrant characteristic fields

- The four main factors describing a transistor are
 - 1 **Collector current** I_C
 - 1 **Collector-emitter voltage** U_{CE}
 - 1 **Base current** I_B
 - 1 **Base-emitter voltage** U_{BE}
- The relation between these parameters can be visualized by a four-quadrant characteristic field
 - 1 **Q1 output:** Relation I_C to U_{CE}
 - 1 **Q2 transfer:** Relation of I_C to I_B , current gain β
 - 1 **Q3 input:** Relation U_{BE} to I_B
 - 1 **Q4 reaction:** Feedback U_{CE} to U_{BE}
- The four-quadrant field provides a unified picture of output, input, transfer and feedback behaviour
- It is the basis for amplifier design and small-signal models.



5.1.6 Transistor characteristic field

Output characteristics

- For each **base current** I_B exists a characteristic curve where the **collector current** I_C is a function of **collector-emitter voltage** U_{CE} :

$$I_C = f(U_{CE})$$

- Each **operating point** on the characteristic curve has a distinct **differential output resistance**:

$$r_{CE} = \frac{\Delta U_{CE}}{\Delta I_C}$$

- In forward active: almost flat / constant $\rightarrow r_{CE} \rightarrow \infty$
- In saturation: linear slope from the origin $\rightarrow$ Ohmic
- Example: Calculate the differential output resistance r_{CE} at operating point $Q_f(10 \text{ V}, 12.5 \text{ mA})$ and $Q_s(1 \text{ V}, 10 \text{ mA})$.*

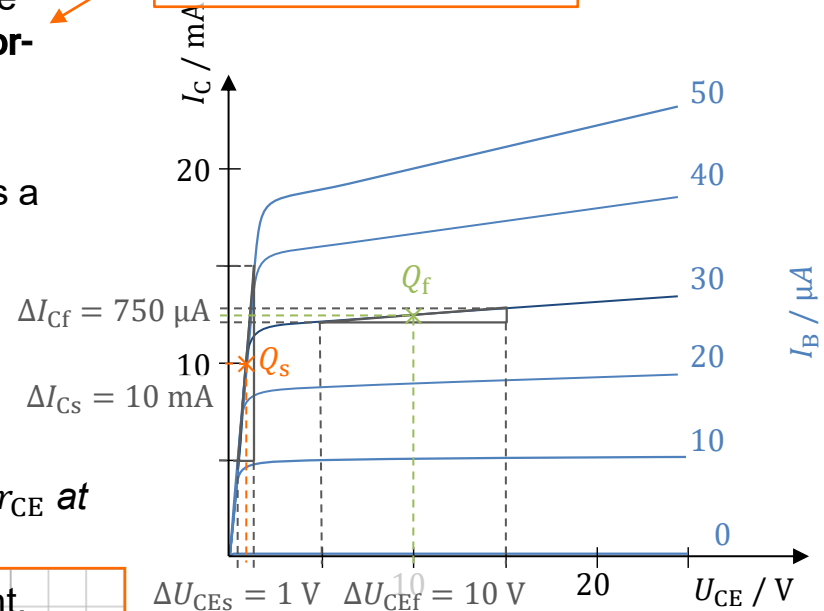
- Symmetrically extrapolate around the operating point.

$$r_{CEf} = \frac{10 \text{ V}}{750 \mu\text{A}} = 13 \text{ k}\Omega \quad r_{CEs} = \frac{1 \text{ V}}{10 \text{ mA}} = 100 \Omega$$

In forward mode r_{CE} is in the Kiloohm range.

In saturation r_{CE} is in the Ohm range and Ohmic: $r_{CEs} = R_{CEs}$

Only in the mathematical sense. Ideally I_C is constant!



Output characteristics for different base currents I_B .

5.1.6 Transistor characteristic field

Input characteristics

- The BE-path has the same input characteristic as a diode, where the **base current** I_B is a function of the **collector-emitter voltage** U_{BE} :

$$I_B = f(U_{BE})$$

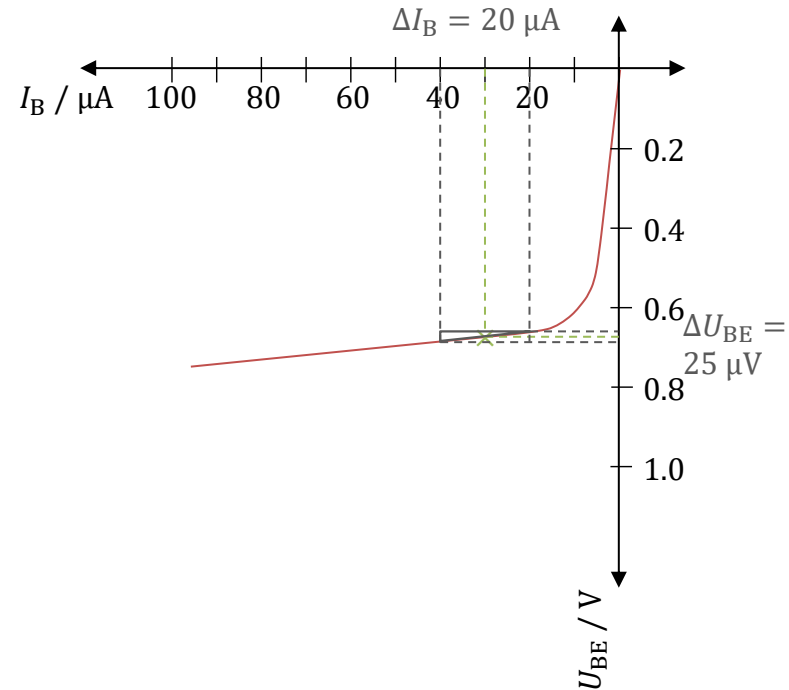
- *Note: we'll later see in the feedback characteristics, that U_{CE} also plays a marginal, mostly negligible, role.*
- Each operation point on the characteristic curve has a distinct differential **input resistance**

$$r_{BE} = \frac{\Delta U_{BE}}{\Delta I_B}$$

- *Note: ideally the differential input resistance is small $\rightarrow$ impact of I_B on U_{BE} is small.*
- *Example: Calculate the differential input resistance r_{BE} at $Q(10\text{ V}, 12.5\text{ mA})$ corresponding to $I_B = 30\text{ }\mu\text{A}$.*

$$r_{BE} = \frac{\Delta U_{BE}}{\Delta I_B} = \frac{25\text{ }\mu\text{V}}{20\text{ }\mu\text{A}} = 1.3\text{ }\Omega$$

In forward mode, r_{BE} is typically a few ohms, only.



Input characteristics of a transistor operated at $U_{CE} = 5\text{ V}$.

5.1.6 Transistor characteristic field

Transfer characteristics

- For each **collector-emitter voltage** U_{CE} exists a characteristic curve where the **collector current** I_C is a function of the **base current** I_B :

$$I_C = f(I_B)$$

- Determined experimentally from each $I_C - I_B$ -pair
- The **large-signal DC current gain** is the slope of the line through the origin.

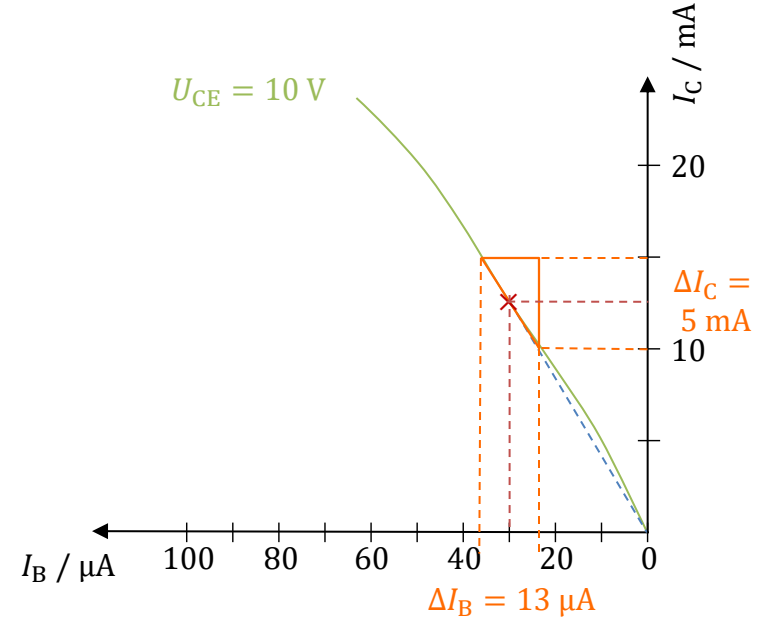
$$\beta_{DC} = I_C / I_B$$

- For **small AC signals** the **differential current gain** is the slope around the operation point:

$$\beta_{AC} = \Delta I_C / \Delta I_B$$

- Note: The values of β_{AC} and β_{DC} are mostly very similar.*
- Example: Calculate the current gain β_{DC} and the differential current gain β_{AC} at $Q(10\text{ V}, 12.5\text{ mA})$.*

$\beta_{DC} = \frac{I_C}{I_B} = \frac{12.5\text{ mA}}{30\text{ }\mu\text{A}} = 420$	$\beta_{AC} = \frac{\Delta I_C}{\Delta I_B} = \frac{5\text{ mA}}{13\text{ }\mu\text{A}} = 380$
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Transfer characteristics of a transistor operated at $U_{CE} = 5\text{ V}$.

5.1.6 Transistor characteristic field

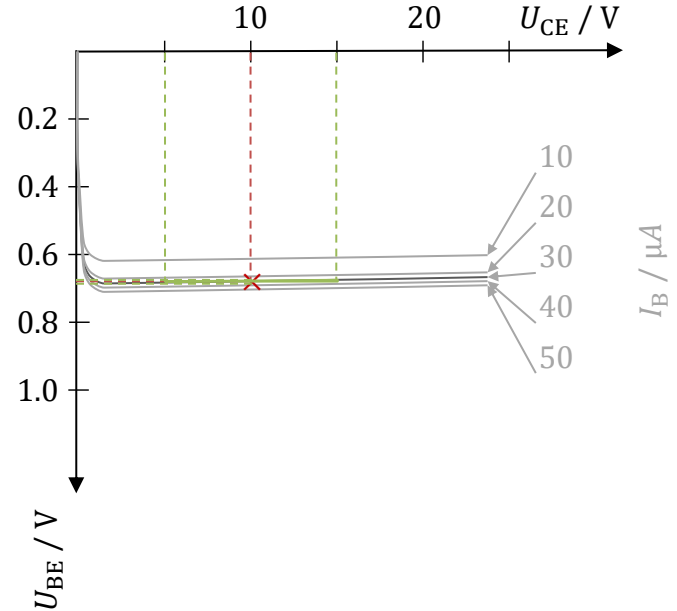
Feedback characteristics

- Feedback is the process where a fraction of the **output signal** U_{CE} has **influence on input** U_{BE} , resulting in a **reduction** of the base-emitter voltage.
- The differential feedback gain of U_{CE} to U_{BE} for a given base current I_B is:

$$D = \frac{\Delta U_{BE}}{\Delta U_{CE}}$$

- In forward mode, this ratio is very small, and the influence of the collector-voltage to the base-voltage mostly negligible.
- *Example: Calculate the differential feedback gain at Q_f (10 V, 12.5 mA).*

- The slope at the point (10 V, 0.65 V) is almost horizontal.
- The differential feedback gain is in the range of $D \approx -10^{-3}$.
- This means, an increase $\Delta U_{CE} = 10$ V reduces U_{BE} by just 10 mV.

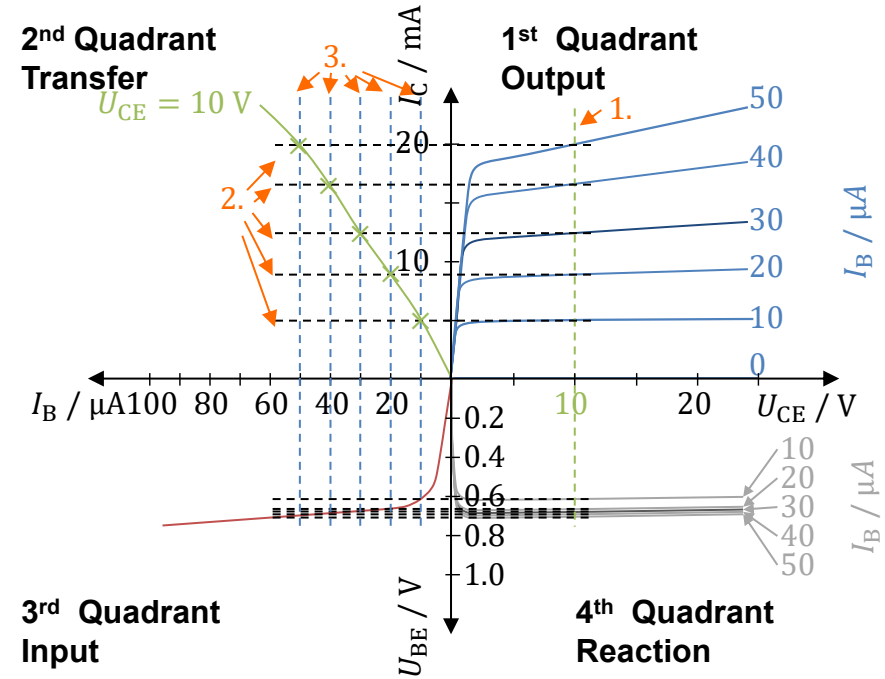


Feedback characteristics for different base currents I_B .

5.1.6 Transistor characteristic field

Constructing the static transfer characteristics

- The **transfer characteristic** curve for a fixed **collector-emitter voltage** U_{CE} can be constructed from **output curves** at **given base currents** I_B .
 1. Vertical line into the 1. quadrant at the **selected collector-emitter-voltage** U_{CE} .
 2. Horizontal lines through intersection of the **output curves** and the vertical line into the 2. quadrant. → collector current I_C
 3. Vertical line at the values of the **base current** representing the **output curves**
 4. Intersection of 2. and 3. gives the **transfer characteristics** at the **selected collector-emitter-voltage** U_{CE} .
- *Note: Analogue procedure using the input curve for the feedback characteristics.*

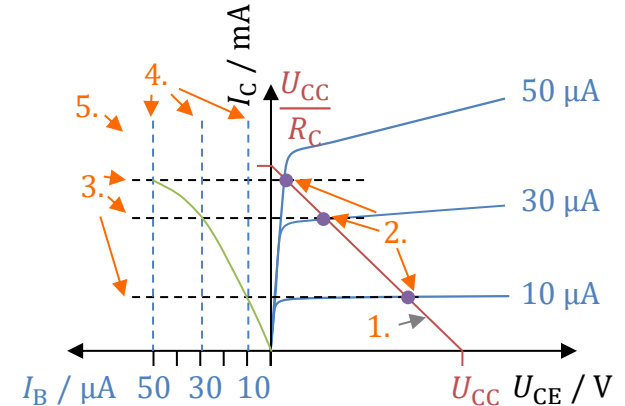
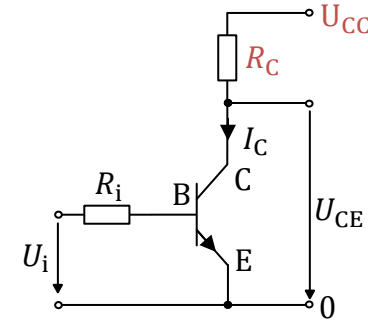


Construction of the transfer characteristics of a transistor.

5.1.6 Transistor characteristic field

Constructing the **dynamic** transfer characteristics

- A transistor in a circuit usually does not operate at a fixed collector-emitter voltage U_{CE} .
- The operating points of each output curve define the **dynamic characteristic**.
 1. Construct the **load line** in the output quadrant for a given circuit.
 2. The intersection with the **output characteristics** define the **operating points** for each base current I_B .
 3. Horizontal line through each **operating point**.
 4. Vertical line at the values of the **base current** representing the **output curves**.
 5. Intersection of 2. and 3. gives the **dynamic transfer characteristics** for your circuit.
- *Note: Did you notice the non-linear gain for high currents? Here we drive the transistor in saturation with a large base current.*



Construction of the dynamic characteristics of a transistor connected in a circuit.

5.1.6 Transistor characteristic field

Constructing the field of operation – power and current limit

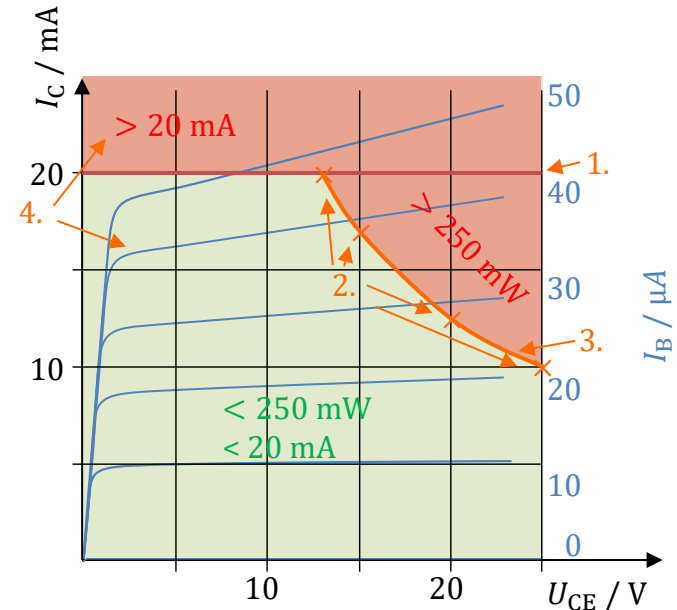
- Each transistor can only **dissipate a maximum power** and **carry a maximum current** as specified in the **data sheet**.
→ **limits of operation**

- Construction of a safe operating area in the output characteristics.

1. Mark the **maximum collector current** $I_{\max}$.
2. Find the maximum power rating of your transistor and for various collector-emitter voltages U_{CE} calculate the permissible collector current and put these points in the graph:

$$I_{C\max} = \frac{P_{\max}}{U_{CE}}$$

3. Construct the **maximum power line** by connecting the calculated current-voltage relations in the graph.
4. The maximum powerline and collector current divides the characteristics field into an **unsafe** and **safe** area.



Maximum power line for a transistor rated at $P_{\max} = 250 \text{ mW}$ and a maximum collector current rating $I_{C\max} = 20 \text{ mA}$.

5.1.7 Transistor as amplifier

Basics of electronics and components
© Prof. Dr. Henning Lormann, 2026

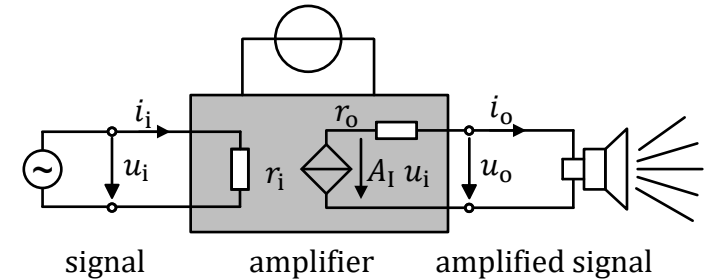
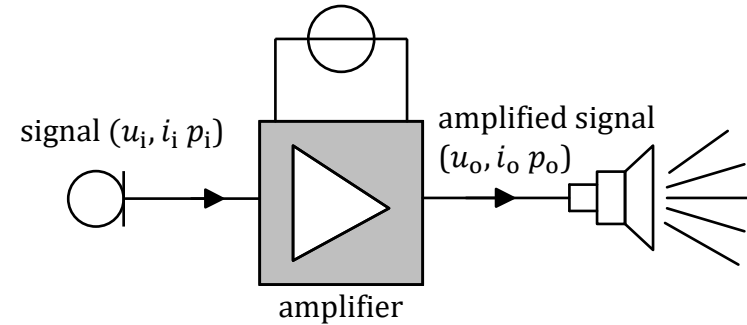


Car Audio Extreme Bass

5.1.7 Transistor as amplifier

Concept of amplification

- An amplifier uses a DC supply to produce a **larger output signal** from a smaller input signal.
- An ideal voltage amplifier is modelled as a voltage-controlled voltage source.
 - ⌈ In a BJT, the input voltage produces a base current.
 - ⌈ In a FET, the input voltage acts at the gate; the gate current is ideally zero.
- The gain A is the ratio of output quantity to input quantity.
 - ⌈ Voltage gain: $A_u = \frac{u_o}{u_i}$
 - ⌈ Current gain: $A_i = \frac{i_o}{i_i}$
 - ⌈ Power gain: $A_p = \frac{p_o}{p_i}$
- The input impedance is defined as: $r_i = \frac{u_i}{i_i}$
- The output impedance is defined as: $r_o = \frac{u_o}{i_o}$

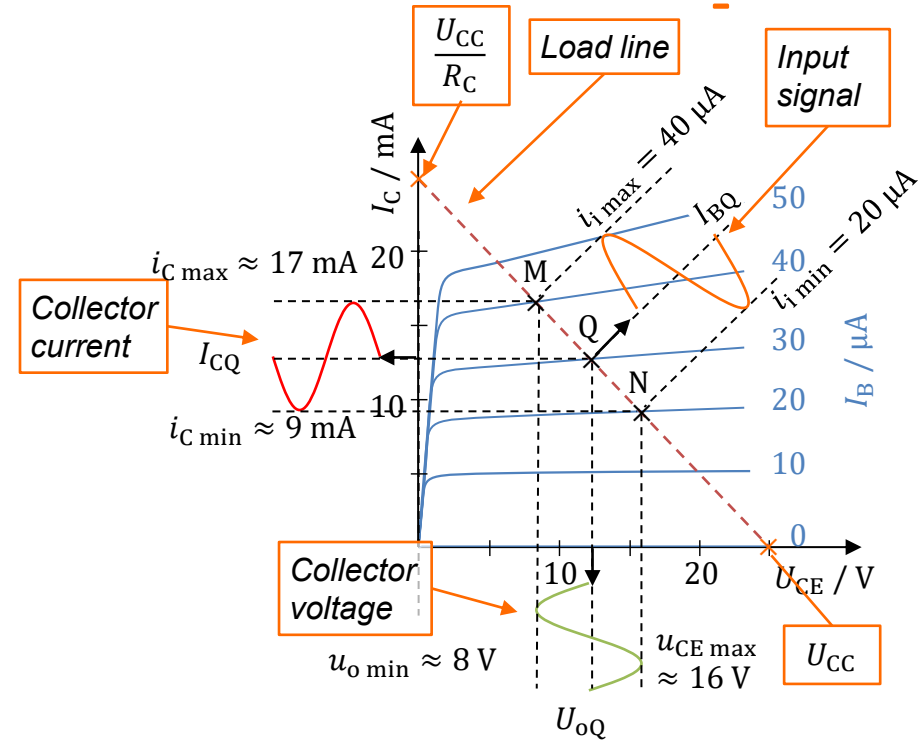
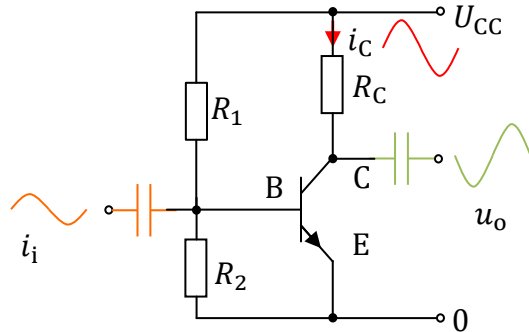


Simplified schematic of a voltage-controlled amplifier (top) and equivalent amplification circuit (bottom).

5.1.7 Transistor as amplifier

Application as amplifier

- Amplifies small variations of the input voltage in a biased circuit → **small-signal model**
- Characteristics are manifold and depend on the **circuit topology**.
- A **transistor alone is no amplifier**. An external circuit is required.
- A **common emitter circuit** is chosen for voltage and current amplification.
- The field of operation is determined using the load line method.



Left: Common emitter circuit for the amplification of small input signals. **Top:** Output curve including load line and signal transfer.

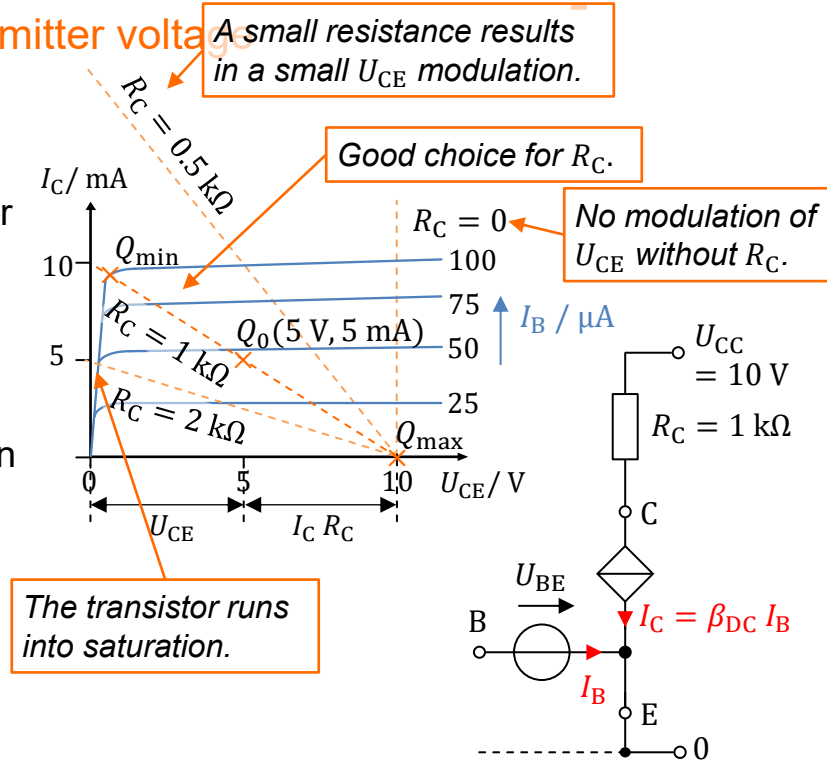
5.1.7 Transistor as amplifier

Step-by-step procedure: Circuit design – 1. Collector emitter voltage

- Choose a **suitable topology** (here: **common emitter**)
- Define the desired operating point Q_0 .
- The range for the collector current I_C and therefore collector resistance R_C between the operating points is determined via **KVL** or the **load line method**

$$I_C = \frac{U_{CC} - U_{CE}}{R_C}$$

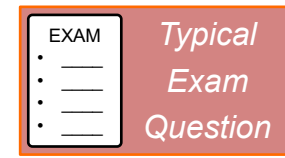
- The collector resistor R_C should enable symmetric operation between $Q_{\min}$ and $Q_{\max}$ when modulated by an AC signal
 - ⌋ To small: low/no modulation of U_{CE}
 - ⌋ To large: transistor runs into saturation.
- Remember: In the forward active region, I_C is a function of $\beta_{DC} I_B$. Changing R_C will change the transistor's operating point so that I_C is maintained.



*Top: Output characteristics including load line.
Bottom: Transistor equivalent circuit.*

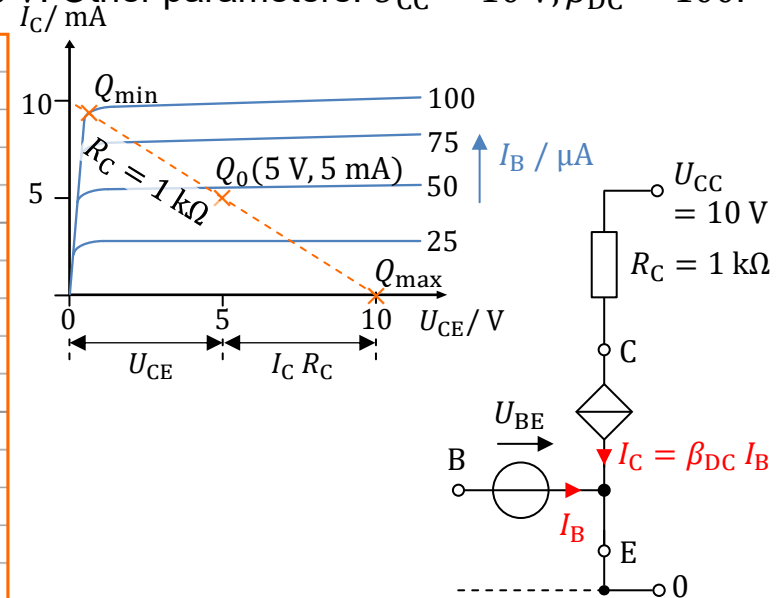
5.1.7 Transistor as amplifier

Exercise: Circuit design – 1. Collector emitter voltage



You are given the circuit as shown. Please find the collector current I_C , the base current I_B for the operating points $Q_0, Q_{\min}, Q_{\max}$ for $U_{CE0} = 5 \text{ V}, U_{CE\min} = 0.5 \text{ V}, U_{CE\max} = 10 \text{ V}$. Other parameters: $U_{CC} = 10 \text{ V}, \beta_{DC} = 100$.

—	Determine and draw load line between (0,10mA) and (10 V, 0)
Q_0 :	
—	Graphically: $U_{CE0} = 5 \text{ V}, I_C = 5 \text{ mA}, I_B \approx 45 \mu\text{A}$
—	Algebraically: $I_C = \frac{10 \text{ V} - 5 \text{ V}}{1 \text{ k}\Omega} = 5 \text{ mA} \therefore I_B = \frac{5 \text{ mA}}{100} = 50 \mu\text{A}$
$Q_{\min}$:	
—	Graphically: $U_{CE\min} = 0.5 \text{ V}, I_C = 9.5 \text{ mA}, I_B = 100 \mu\text{A}$
—	Algebraically: $I_C = \frac{10 \text{ V} - 0.5 \text{ V}}{1 \text{ k}\Omega} = 9.5 \text{ mA} \therefore I_B = \frac{9.5 \text{ mA}}{100} = 95 \mu\text{A}$
$Q_{\max}$:	
—	Graphically: $U_{CE\max} = 10 \text{ V}, I_C = 0, I_B = 0$
—	Algebraically: $I_C = \frac{U_{CC} - U_{CE}}{R_C} = \frac{10 \text{ V} - 10 \text{ V}}{1 \text{ k}\Omega} = 0 \therefore I_B = \frac{I_C}{\beta_{DC}} = \frac{0}{100} = 0$
→	Because the value of β_{DC} does not exactly match, the calculated value for I_B slightly differs from the experimental curve.



Top: Output characteristics including load line.
Bottom: Transistor equivalent circuit.

5.1.7 Transistor as amplifier

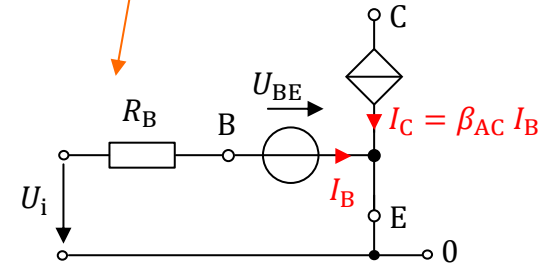
Step-by-step procedure: Circuit design – 2. Base current

- With knowledge of the DC collector current, the corresponding base current I_B can be calculated
 - With β_{DC} from the data sheet.
 - Graphically from the characteristic output curve and the load line.
- The transistor is then biased via U_i and R_B (for the base resistor setup) or U_{CC} and R_1/R_2 (for the base voltage divider setup) according to 5.1.4 BJT basic circuits.
- For the large-signal the following applies in particular:

$$U_i = I_B R_B + U_{BE} + U_E$$

- Note: For fine-tuning of R_B often a fixed resistor is used in series with a potentiometer.*

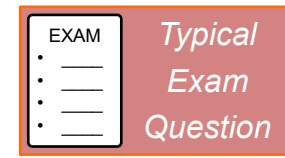
Either bias the base via a separate equivalent input voltage U_i and a base resistor R_B or via the supply voltage U_{CC} and a voltage divider R_1 and R_2 .



Large-signal model for accurate biasing an amplifier circuit.

5.1.7 Transistor as amplifier

Exercise: Circuit design – 2. Base current



Please calculate the required series resistance R_B for a npn transistor with the following boundary conditions: $U_i = 10\text{ V}$, $U_{BE} = 0.7\text{ V}$, $\beta_{DC} = 100$ if a collector current $I_C = 5\text{ mA}$ is required. Reconstruct the four-resistor base circuit from the Thévenin equivalent using $U_{CC} = 12\text{ V}$.

- Calculate the base current:

$$I_B = \frac{I_C}{\beta_{DC}} = \frac{5\text{ mA}}{100} = 50\text{ }\mu\text{A}$$

- Calculate the equivalent series resistance R_B

$$R_B = \frac{U_i - U_{BE}}{I_B} = 19\text{ k}\Omega$$

Next in the E12 series would be 18 k Ω .

- Calculate the side current to be about five times the base current. This defines R_2

$$R_2 = \frac{U_B}{5 I_B} = \frac{0.7\text{ V}}{5 \cdot 50\text{ }\mu\text{A}} = 3\text{ k}\Omega$$

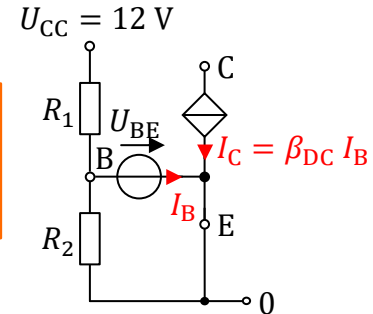
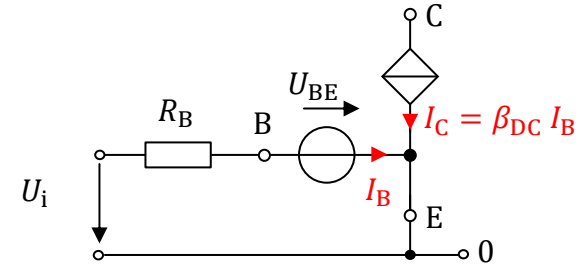
- Calculate R_1 via KCL and Ohm.

$$I_{R1} = I_{R2} + I_B = 6 I_B = 300\text{ }\mu\text{A}$$
$$R_1 = \frac{U_{CC} - U_B}{I_{R1}} = \frac{(12\text{ V} - 0.7\text{ V})}{300\text{ }\mu\text{A}} = 38\text{ k}\Omega \text{ (E12: } 39\text{ k}\Omega)$$

Via voltage divider (ignoring I_B)

$$R_1 = R_2 \left(\frac{U_{CC}}{U_{BE}} - 1 \right) = 48\text{ k}\Omega$$

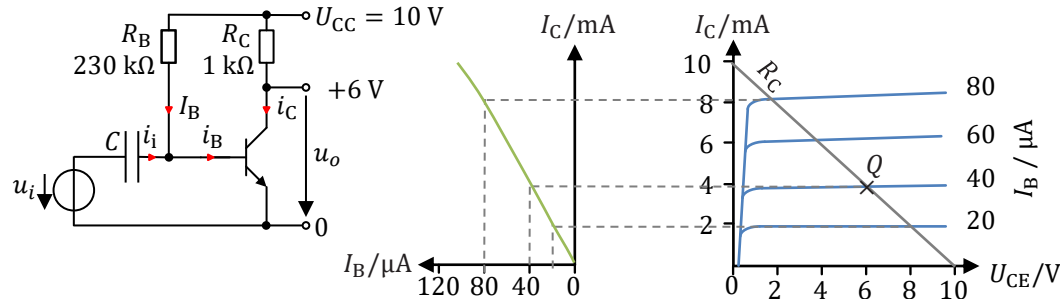
(E12: 47 k Ω)



Appendix to 5.1.7 Transistor as amplifier: Graphical AC signal analysis

Appendix to 5.1.7 Transistor as amplifier:

Step-by-step procedure: AC signal analysis 1/3 (circuit biasing)



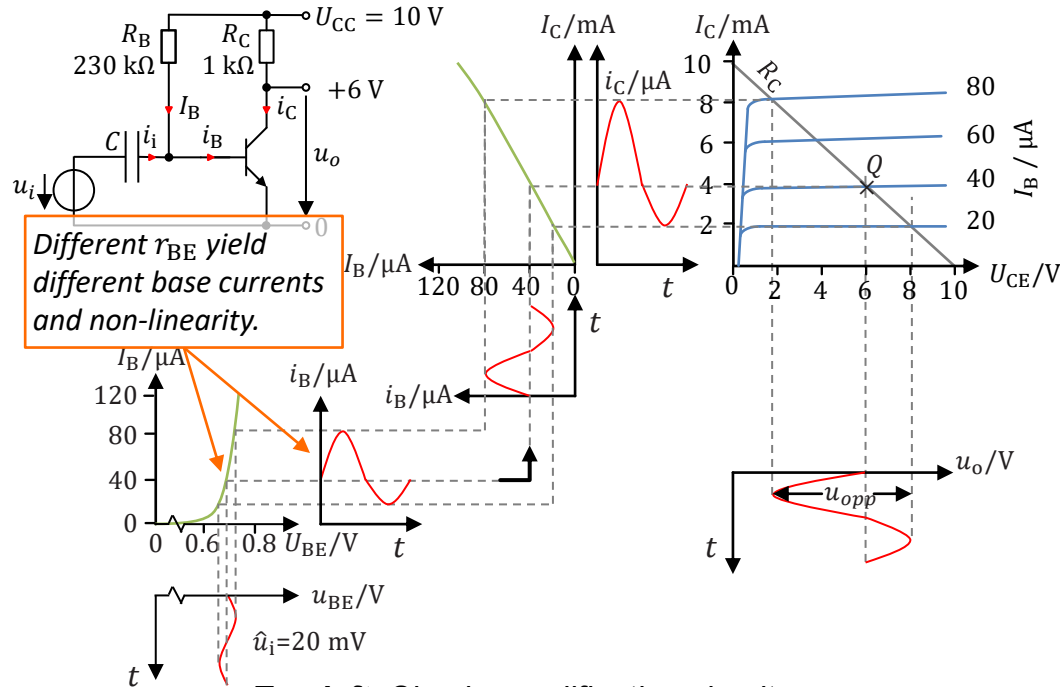
1. Bias the circuit via R_B and R_C
 - For $Q(6 \text{ V}, 4 \text{ mA}), \beta = 100$:
 - $I_B = 40 \mu\text{A} \therefore R_B = 230 \text{ k}\Omega$
 - $R_C I_C = U_{CC} - U_C = 4 \text{ V} \therefore R_C = 1 \text{ k}\Omega$
2. Construct the load line for R_C .
3. Construct the transfer curve via the characteristic output field.

Top left: Simple amplification circuit.

Bottom left to bottom right: AC signal analysis.

Appendix to 5.1.7 Transistor as amplifier:

Step-by-step procedure: AC signal analysis 2/3 (signal tracing)



1. Enter signal u_i via a coupling capacitor:

$$u_{BE} = U_{BE} + \hat{u}_i \sin \omega t$$
2. Mirror u_{BE} at the characteristic input curve to yield i_B :

$$i_B = I_B + i_i$$

*I_B : DC base current
 i_i : AC modulation
 i_B : DC+AC current*
3. Mirror i_B at the transfer characteristic curve to yield i_C :

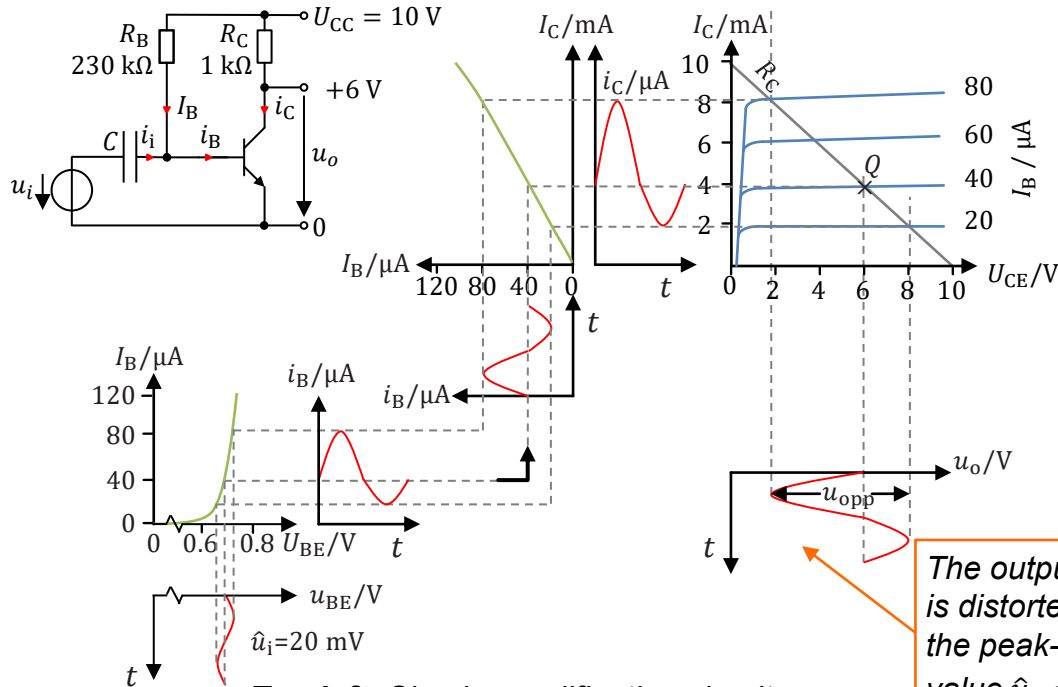
$$i_C = I_C + i_c$$

Non-sinusoidal alternating current
4. Mirror i_C at R_C load line to yield u_{CE} :

$$u_{CE} = U_{CE} + u_{ce}$$

Appendix to 5.1.7 Transistor as amplifier:

Step-by-step procedure: AC signal analysis 3/3 (analysis)



Top left: Simple amplification circuit.

Bottom left to bottom right: AC signal analysis.

1. The output voltage is inversed.
2. Voltage amplification
$$|A_u| = \frac{u_{opp}}{2 \hat{u}_i} = \frac{6.2\text{ V}}{40\text{ mV}} = 155 = 44\text{ dB}$$
3. Output signal u_o is distorted by higher harmonics:
 - non-linear input curve
 - slightly non-linear transfer curve.
4. Note: improve simplified circuit e. g. via operating point stabilization (see: 5.1.4 BJT basic circuits – Q-point stabilization)



The output signal is distorted. Use the peak-to peak value $\hat{u}_o = \frac{u_{opp}}{2}$

Graphical method is insightful but approximate, whereas the small-signal model introduced next provides a compact analytical approximation around the operating point.

5.1.8 BJT small-signal model

5.1.8 BJT small-signal model

Superposition theorem on a biased BJT circuit

- A transistor at operating point can be described by the superposition of two models

Large-signal model in forward mode:

- ⌈ Describes the properties of the biased transistor (U_{BE} , U_{CE} , I_B etc.).
- ⌈ Modelled by an ideal voltage source and a current controlled current source.

Small-signal model around operating point.

- ⌈ Input signal u_i
- ⌈ Output signal u_o
- ⌈ Input resistance r_{BE}
- ⌈ Output resistance $r_{CE} \parallel R_C$



r_{BE} and r_{CE} behave Ohmic for small AC signals around Q-point!

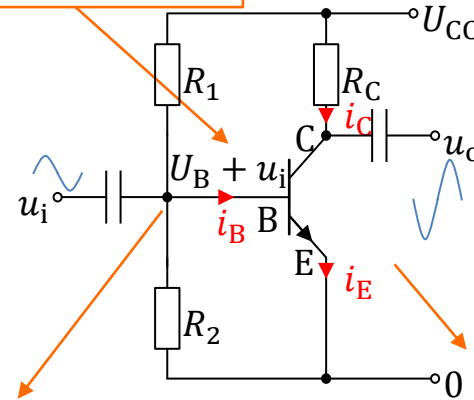
- Both models can be handled separately (superposition theorem):

Remove all DC components

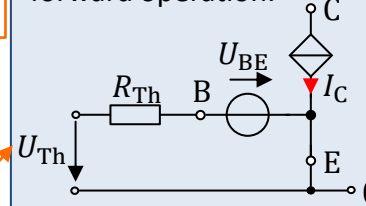
Thévenin equivalent voltage source.

Analyse the small- AC input vs. output signal

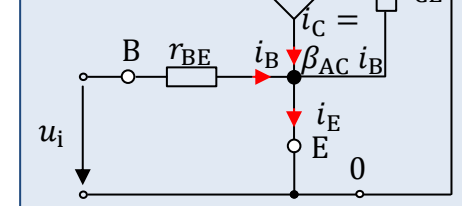
The input signal is the superposition of the DC bias voltage U_B and the AC signal u_i .



Large-signal model in forward operation.



Small-signal model.



Common-emitter circuit (**top**) and representation of the DC-part (**left**) as well as the AC-part (**right**).

5.1.8 BJT small-signal model

Procedure for an ideal input voltage source

- Remove the coupling capacitors ($Z_C \rightarrow 0$).
- Ignore the DC-part by shorting the DC voltage source (BEE “superposition principle”).
- Rearrange the circuit.
- Apply the transistor AC small-signal -model.
- BE: Input
 - Base-emitter resistance $r_{BE} = \frac{U_T}{I_B}$
 - Input resistance $r_i = r_{BE}$
 - Input voltage $u_i = r_{BE} i_B$
- CE: Output
 - Output resistance is: $r_o = R_C || r_{CE} \approx R_C$
 - Current-controlled current source: $i_C = \beta_{AC} i_B$
 - Output voltage $u_o = r_o i_C = R_C || r_{CE} i_C$
- The voltage gain is the ratio of the peak voltages

$$A_U = \frac{\hat{u}_o}{\hat{u}_i} = \frac{r_o \hat{i}_C}{r_{BE} \hat{i}_B} = \beta_{AC} \frac{r_o \hat{i}_B}{r_{BE} \hat{i}_B} = \beta_{AC} \frac{R_C || r_{CE}}{r_{BE}}$$

Part of the small-signal i_{RB} enters the base resistor.

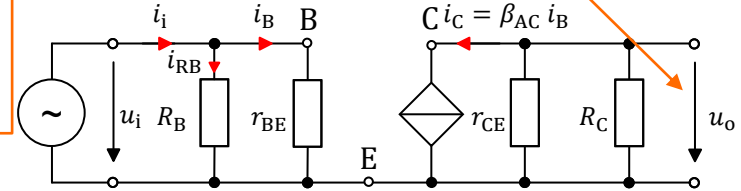
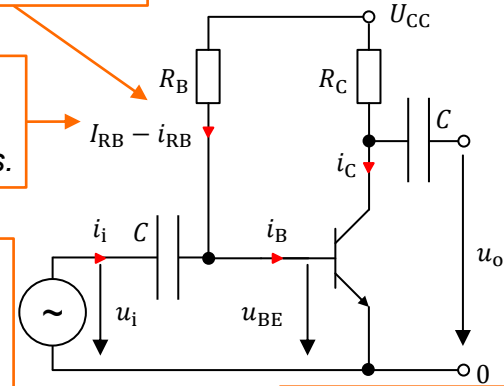
The DC part I_{RB} can be ignored for the small-signal analysis.

Differential resistance at operating point with the thermal voltage $U_T \approx 25.7 \text{ mV}$ at room temperature. (Chapter 4.2.2 Diode modelling.)

R_B is parallel to u_i and obsolete

$R_C || r_{CE} || R_L$ with load. In forward active: $r_{CE} \gg R_C$

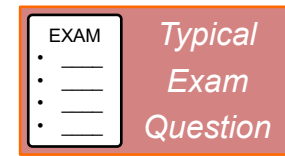
Voltage opposite to current direction $\rightarrow u_o < 0!$



Top: Transistor amplifier circuit. **Bottom:** Small AC signal equivalent circuit.

5.1.8 BJT small-signal model

Exercise



Please draw an equivalent circuit and then calculate the parameters A_u , r_i , r_o of the amplifier at operating point $U_{CE} = 6 \text{ V}$. Transistor characteristics: $\beta_{AC} = 100$, $r_{CE} = 50 \text{ k}\Omega$. *Reminder:* $U_T \approx 25.7 \text{ mV}$ at room temperature.

- Collector current at operating point

$$I_C = \frac{U_S - U_{CE}}{R_C} = \frac{(10 \text{ V} - 6 \text{ V})}{1 \text{ k}\Omega} = 4 \text{ mA}$$

- Base-emitter resistance at operating point

$$r_{BE} = \frac{U_T}{I_B} = \beta_{AC} \frac{U_T}{I_C} = 100 \cdot \frac{25.7 \text{ mV}}{4 \text{ mA}} = 642.5 \Omega$$

- Voltage gain (*Remember: emitter-circuit $\rightarrow$ gain is negative*)

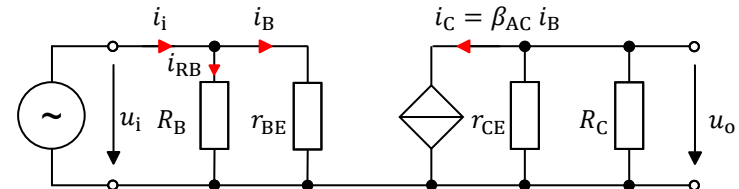
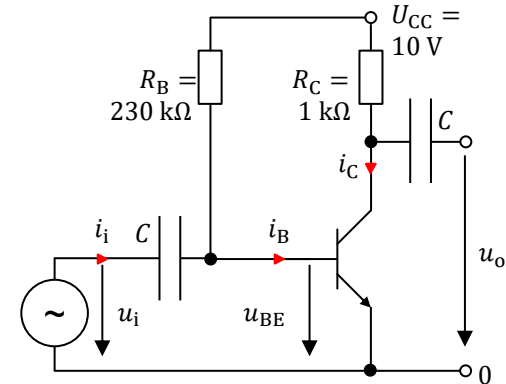
$$A_u = \frac{\hat{u}_o}{\hat{u}_i} = -\frac{r_o i_C}{r_{BE} i_B} = -\beta_{AC} \frac{r_o i_B}{r_{BE} i_B} = -\beta_{AC} \frac{R_C || r_{CE}}{r_{BE}} = -100 \cdot \frac{1 \text{ k}\Omega || 50 \text{ k}\Omega}{642.5 \Omega} = -152.6$$

- Input resistance

$$r_i = R_B || r_{BE} = 230 \text{ k}\Omega || 642.5 \Omega \approx 642.5 \Omega$$

- Output resistance

$$r_o = R_C || r_{CE} = 1 \text{ k}\Omega || 50 \text{ k}\Omega \approx 1 \text{ k}\Omega$$



5.1.8 BJT small-signal model

Procedure for real voltage sources

- For real voltage sources (with R_i) the base voltage u_{BE} is not equal to the input voltage u_i .
- Replace the source by its Thévenin equivalent.
- Calculate the Thévenin resistance

$$R_{Th} = R_i \parallel R_B = \frac{R_i R_B}{R_i + R_B}$$

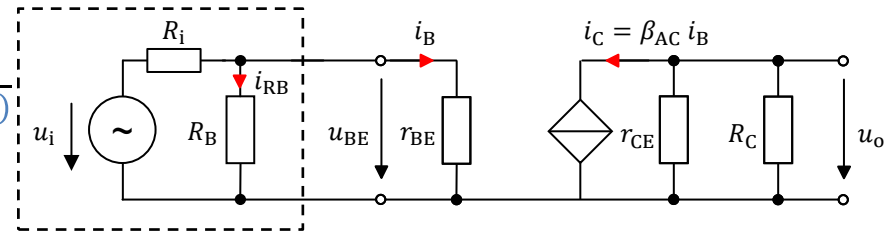
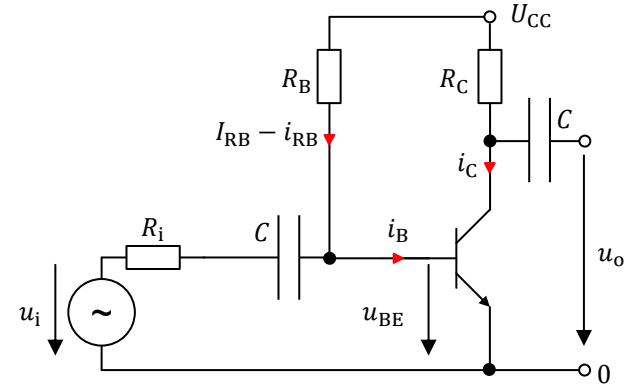
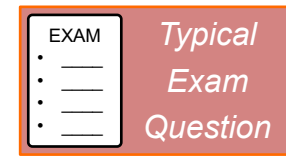
- Calculate the Thévenin voltage and solve for u_i

$$u_{Th} = \frac{R_B}{R_i + R_B} u_i \quad \therefore u_i = \frac{R_i + R_B}{R_B} u_{Th} = \frac{R_i}{R_{Th}} u_{Th}$$

- Proceed analogue to the previous example.

$$A_U = \frac{\hat{u}_o}{\hat{u}_i} = \frac{R_{Th} r_o \hat{i}_C}{R_i u_{Th}} = \frac{R_{Th} R_C \parallel r_{CE} \beta_{AC} i_B}{R_i i_B (R_{Th} + r_{BE})} = \beta_{AC} \frac{R_{Th} R_C \parallel r_{CE}}{R_i (R_{Th} + r_{BE})}$$

$$\therefore A_U \approx \beta_{AC} \frac{R_{Th} R_C}{R_i (R_{Th} + r_{BE})}$$

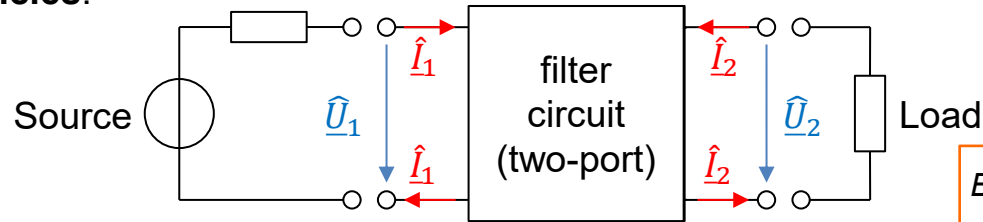


5.1.9 Frequency response of transistor amplifiers

5.1.9 Frequency response of transistor amplifiers

Recap: IRO1 BEE 5.3.1 Frequency response and filters (analogue BRO1)

Frequency selective circuits are circuits that **pass** only those input **signals** to the output that reside in a desired **range of frequencies**.



The **frequency response** $\underline{H}(j\omega)$ is the ratio of the filter output to its input.

$$\underline{H}(j\omega) = \frac{\hat{X}_2(j\omega)}{\hat{X}_1(j\omega)} = |\underline{H}(j\omega)| e^{j\phi(\omega)}$$

Because $\frac{\hat{X}_2(j\omega)}{\hat{X}_1(j\omega)} = \frac{X_2(j\omega)}{X_1(j\omega)}$ the frequency response is valid for both the amplitude $\hat{X}$ and the effective value X !

- Where $\hat{X}$ can be **either a voltage $\hat{U}$ or a current $\hat{I}$** amplitude or its effective value $\underline{U}$ and $\underline{I}$!
- The function is **frequency dependent**.
- The absolute value of the frequency response $|\underline{H}(j\omega)|$ is called **amplitude response** or **gain**.
- The phase angle $\phi(\omega)$ of the frequency response is called **phase response**.

Frequency selective circuits are also called **filters**.

- **Attenuate input signals** with frequencies outside a particular frequency.
- **Filters are passive** if they depend only on **passive elements** (i.e. R, L, C).

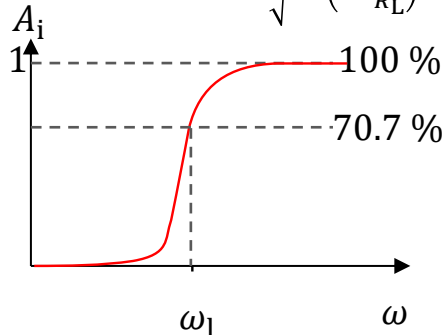
5.1.9 Frequency response of transistor amplifiers

Frequency response – lower limit

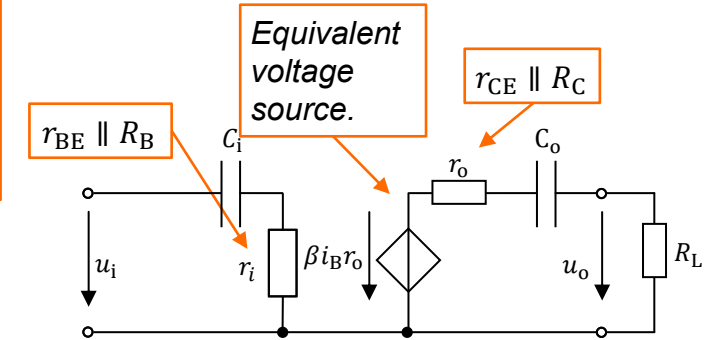
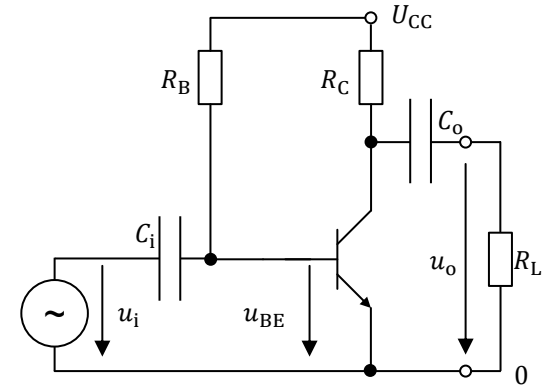
- The combination of resistors and coupling capacitors in amplifier circuits generate a **high-pass filter**.
- The frequency limit is defined by a drop of 3 dB (for more details check IRO1, Basics of electrical engineering).
- Current lower frequency limit

Input: $f_{li} = \frac{1}{2 \pi C_i r_i}$ ← Ordinary series RC high-pass.

Output: $f_{lo} = \frac{1}{2 \pi C_o R_L \sqrt{2 - \left(1 + \frac{r_o}{R_L}\right)^2}}$ ← Series RC high-pass with additional series resistance. If the load is too small compared to the output resistance, the drop in gain is always larger than 3 dB!



Simplified frequency response.



Transistor circuit (**top**) in common-emitter topology and equivalent circuit for the calculation of the frequency limit (**bottom**).

5.1.9 Frequency response of transistor amplifiers

Exercise: Lower frequency limit

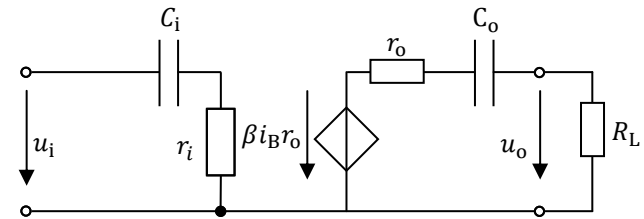
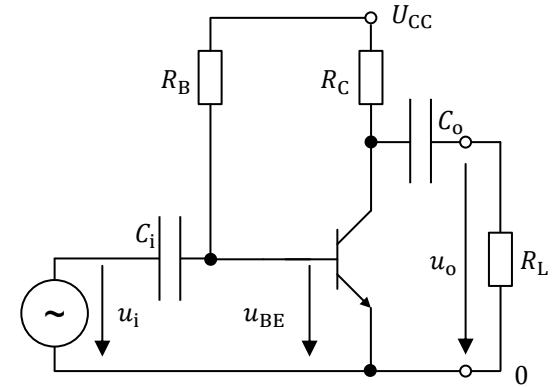
For the given circuit please calculate the lower frequency limit. Input: $C_i = 10 \text{ nF}$, $r_i = 750 \Omega$; Output: $C_o = 10 \text{ nF}$, $r_o = 1 \text{ k}\Omega$, $R_L = 5 \text{ k}\Omega$

— Input

$$f_{li} = \frac{1}{2 \pi C_i r_i} = \frac{1}{2 \pi \cdot 10 \text{ nF} \cdot 750 \Omega} = 21 \text{ kHz}$$

— Output

$$f_{lo} = \frac{1}{2 \pi C_o R_L \sqrt{2 - \left(1 + \frac{r_o}{R_L}\right)}} \\ = \frac{1}{2 \pi \cdot 10 \text{ nF} \cdot 5 \text{ k}\Omega \sqrt{2 - \left(1 + \frac{1 \text{ k}\Omega}{5 \text{ k}\Omega}\right)^2}} = 4.3 \text{ kHz}$$



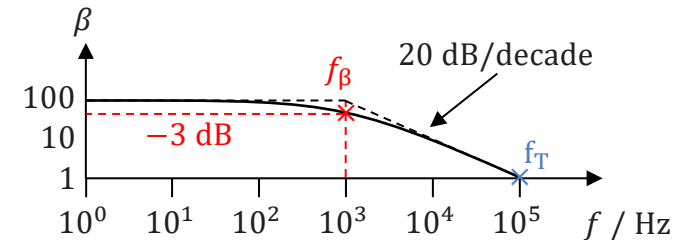
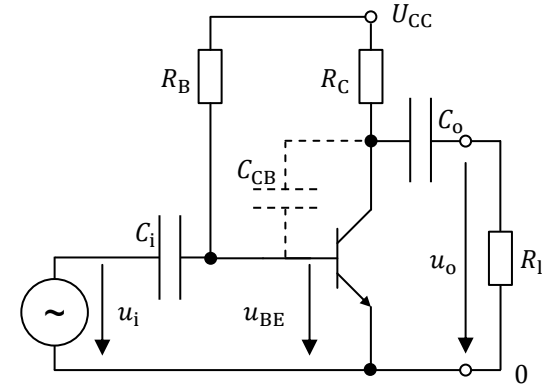
5.1.9 Frequency response of transistor amplifiers

Upper frequency limit

- The collector-base diode of a transistor forms a capacitance (Miller-effect)
- For high frequencies, it bypasses the BC-diode and reduces the output voltage by pulling the collector to the base.
- This results in a decline in amplification β for high frequencies $\rightarrow$ the transistor acts as **low-pass filter**.
- The frequency for which the gain is one is called **transit frequency** f_T .
- The upper **frequency limit** f_β where the current gain drops by 3 dB is defined by the current gain β and the

$$f_\beta = \frac{\text{transit frequency}}{\text{current gain}} = \frac{f_T}{\beta}$$

- Below f_β , the gain is considered constant.
- Above f_β , the gain is reduced by 20 dB/decade.
- *Note: an additional collector-base capacitor can be used to tune the upper frequency limit*



Top: Schematic of the effect of C_{CB} . Bottom: Frequency dependency of the current gain β .

5.1.9 Frequency response of transistor amplifiers

Recap: IRO1 BEE, chapter 5.3 *Frequency response and filters (Analogously BRO1)*

- **Level differences** (i. e. the ratio of input to output signals) are often specified in **decibels**.
- The decibel is a **relative measure** and very **useful** when the ratio spans **several orders of magnitude**.
- For power $|S|$, the **power level difference** is defined as

$$L_P = 10 \log \left(\frac{|S_2|}{|S_1|} \right) \text{ dB}$$

- 10 dB corresponds to a change in power by a factor of 10.
- Since $|S| \propto U^2 \propto I^2$, the **voltage and current level difference** is expressed as:

$$L_U = 20 \log \left(\frac{U_2}{U_1} \right) \text{ dB} \quad \text{and} \quad L_I = 20 \log \left(\frac{I_2}{I_1} \right) \text{ dB}$$

- A change in amplitude by a factor of 10 corresponds to a level change of 20 dB.
- The decibel is a **dimensionless suffix** to a number, **not a physical unit**.

dB	Power ratio	Amplitude ratio
20	100	10
10	10	≈ 3
6	≈ 4	≈ 2
3	≈ 2	$\approx \sqrt{2}$
1	1.3	1.1
0	1	1
-1	0.8	0.9
-3	$\approx 1/2$	$\approx \sqrt{1/2}$
-6	$\approx 1/4$	$\approx 1/2$
-10	0.1	$\approx 1/3$
-20	0.01	0.1

Calculation path: $L_U = 10 \log \left(\frac{U_2^2}{U_1^2} \right) \text{ dB} =$
 $10 \log \left(\frac{U_2}{U_1} \right)^2 \text{ dB} = 20 \log \left(\frac{U_2}{U_1} \right) \text{ dB}$

5.1.9 Frequency response of transistor amplifiers

Exercise: Upper frequency limit

The manufacturers data sheet provides the following info: $f_T = 70 \text{ MHz}$, $\beta = 100$ (@1 kHz). Please calculate the frequency where the collector current is dropped to 70.7 % ($\beta = -3 \text{ dB}$) at a constant base current.

$$f_\beta = \frac{\text{transit frequency}}{\text{current gain}} = \frac{f_T}{\beta} = \frac{70 \text{ MHz}}{100} = 700 \text{ kHz}$$

Chapter outline

Fundamental properties

Negative feedback: linear amplifier

Positive feedback: switch

5.2 Operational amplifier

Learning objectives

Students can describe the **operating principle** of an OP amp in **open loop** and **closed loop configuration**.

Students can **calculate** the **transfer characteristics**.

Students can **draw** the **line diagram** and **transfer characteristics** of OP amps in open and closed loop configuration.

Students can design the **upper and lower switching thresholds** of Schmitt triggers.

Students **never forget**:

- The operation of an OP amp **requires a positive and negative supply voltage!**
- The **golden rules** of OP amps under closed circuit operation.
- The difference between OP amps in **open loop and closed loop** mode.

5.2 Operational amplifier

Chapter outline

Symbols and pin configuration

Basic properties in open loop configuration

OP amp design

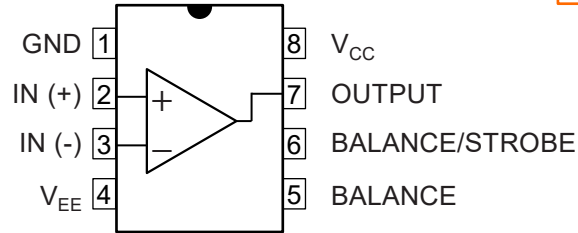
5.2.1 Fundamental properties

5.2.1 Fundamental properties

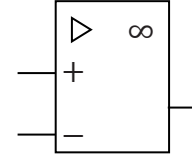
Symbols and pin configuration

- OP amp circuits are complicated → own symbol
- Naming conventions
 - Inverting** input IN (-), U_i^-
 - Non-inverting** input IN (+), U_i^+
 - Output** / OUT, U_o, V_o
 - Positive **supply voltage** $U^+, V^+, U_{CC}, V_{CC}, U_s$
 - Negative supply voltage $U^-, V^-, U_{EE}, V_{EE}, -U_{CC}, -U_s$
 - Ground** GND

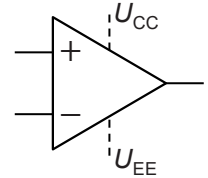
Don't forget to power up your device!



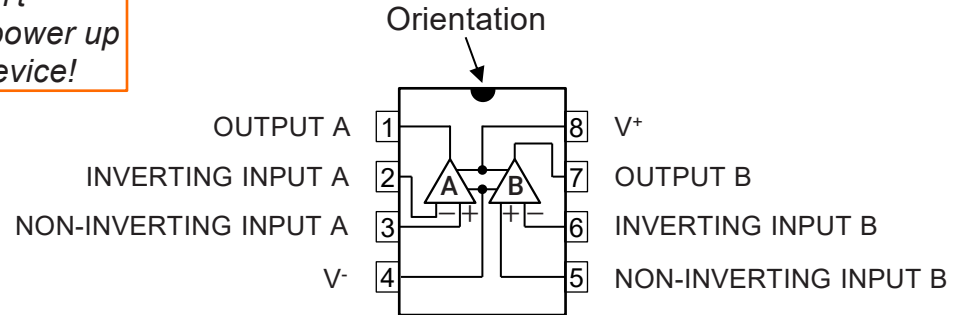
LM311 Single comparator pin configuration.



Symbol according EN-60617.



Practical symbol. (Operation voltage in dashed lines).



TL082 Dual JFET Input OA pin configuration.

5.2.1 Fundamental properties

Basic properties in open loop configuration

- An OP amp **amplifies the voltage difference** at the input

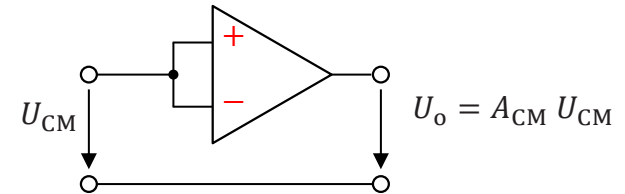
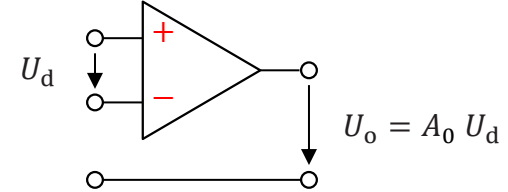
$$U_o = A_0 U_d = A_0 (U_i^+ - U_i^-)$$

- Very high gain ($A_0 > 100\,000$)
 - **very small voltage differences** U_d to bring OP amp in saturation (typically $U_o \approx 0.8 U_S$)
 - Without feedback loop an OP amp operates as a switch
 - Negative feedback required when using as amplifier
- **High input impedance**
 - Negligible input current

- When the two inputs are on the **same node**, the output **voltage is ideally zero** (common mode rejection ratio):

$$\text{CMRR(dB)} = 20 \log \frac{A_0}{A_{\text{CM}}} = 20 \log \frac{A_0 U_{\text{CM}}}{U_o}$$

***THE** essential equation for the understanding an OP amp!*



*OP am in open circuit configuration (**top**) and common mode operation (**bottom**).*

5.2.1 Fundamental properties

Basic properties in open loop configuration

The transient characteristic of an OP amp depends on which input the voltage U_i is applied to.

- **Non-inverting input → Non-inverted output**
- **Inverting input → Inverted output**

Linear range

- Output is a linear function of input $U_o = A_0 U_d$
- Due to the high gain, **no sensible operation** in open loop mode w/o feedback (on/off behaviour)

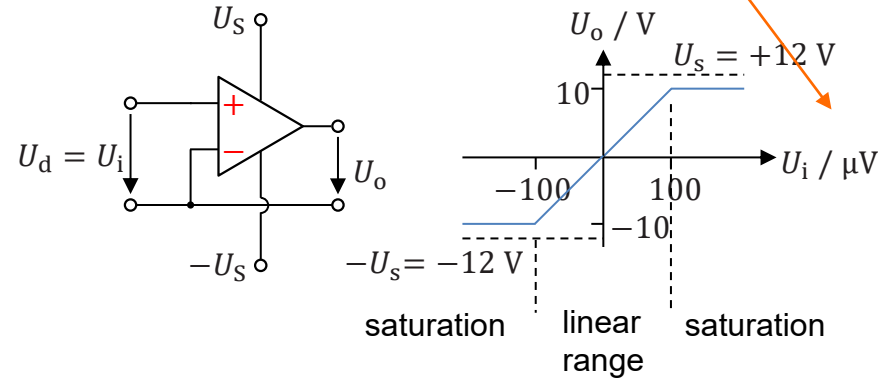
Beyond linear range → **saturation**

- Output voltage is **limited by supply voltage** U_s
- Typically, $U_o \approx 0.8 U_s$ due to internal losses.

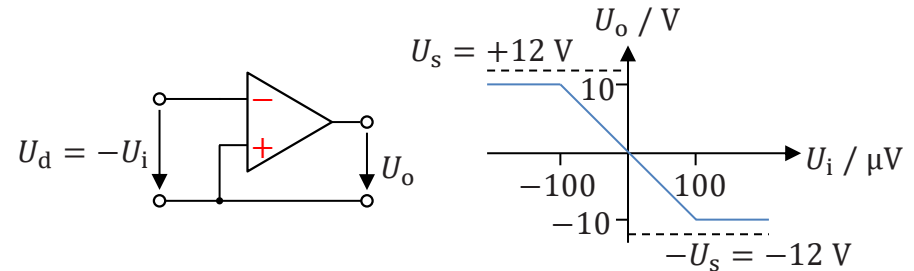
Example: Calculate the voltage U_d to bring an OP amp into saturation for $U_s = \pm 12 \text{ V}$, $A_0 = 100 \text{ dB}$.

$$\therefore U_i = \frac{U_o}{A_0} \approx \frac{0.8 U_s}{A_0} = \frac{9.6 \text{ V}}{100\,000} = 96 \mu\text{V}$$

Look at the scale.
Basically, we see binary operation, here.



OPA transfer characteristics. Non-inverting input.



OPA transfer characteristics. Inverting input.

5.2.1 Fundamental properties

Design

Input stage

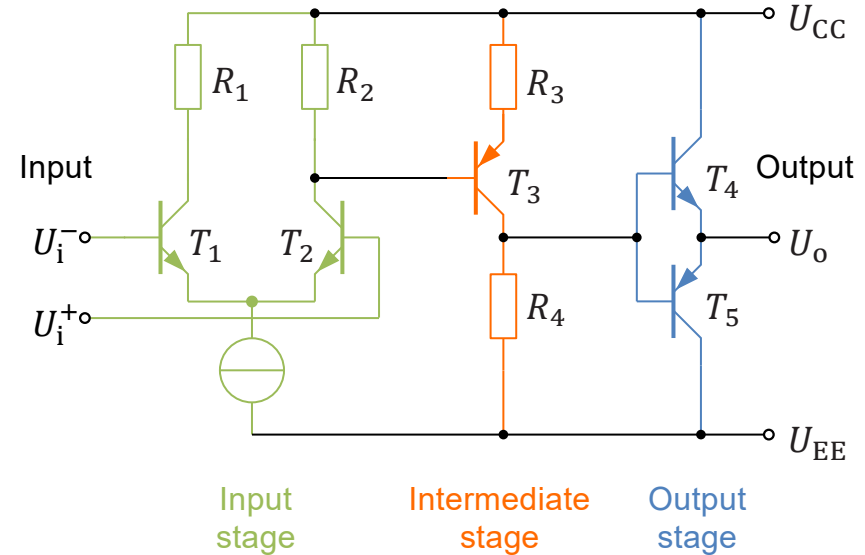
- Differential amplifier with inverting (U_i^-) and non-inverting (U_i^+) input
- Amplifies signal differences and suppresses common-mode signals (CMMR)
- High input impedance

Intermediate stage

- Amplification via pnp common emitter topology
→ stable, voltage gain depends on ratio of R_3 and R_4 .
- Darlington configuration for high gain.

Output stage (Push-pull amplifier)

- Positive input signal: T_4 open, T_5 closed → U_o pushed to U_{CC}
- Negative input signal: T_4 closed, T_5 open → U_o pulled to U_{EE}



Schematic circuit of an Operational amplifier.

5.2.1 Fundamental properties

Homework: Operating principle

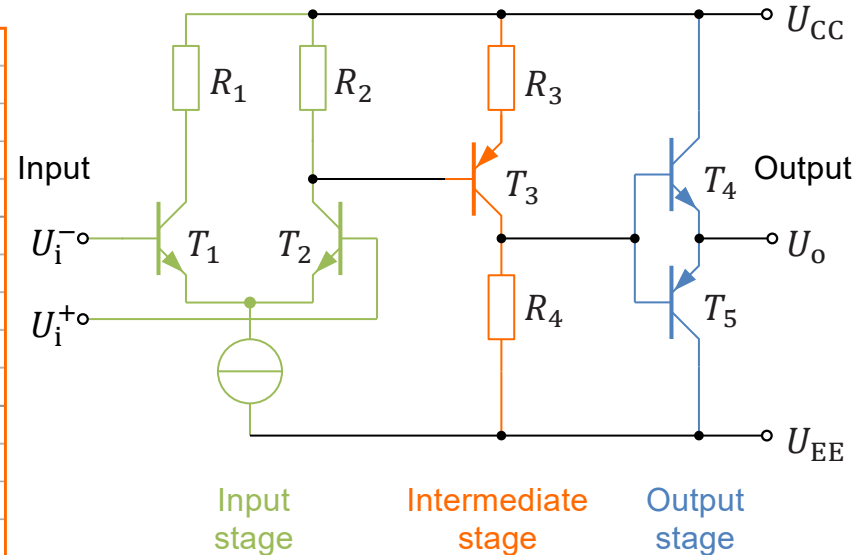
Trace a high input signal at the inverting and non-inverting input to the output of the following circuit.

U_i^- high:

- T_1 is conducting and pulls up the common emitter
- T_2 is non-conducting
- The base voltage of T_3 pulled up to U_{CC}
 - T_3 (npn type) is non-conducting
 - The common base of T_4 and T_5 is pulled down to U_{EE}
 - T_4 is non conducting
 - T_5 (pnp) is conducting
 - U_o is pulled down to U_{EE}

U_i^+ high:

- opposite behaviour, except that T_3 is pushed to U_{CC} –
$$\frac{R_3}{(R_3 + R_4)} (U_{CC} - U_{EE}).$$



5.2.1 Fundamental properties

Example: Differential amplifier

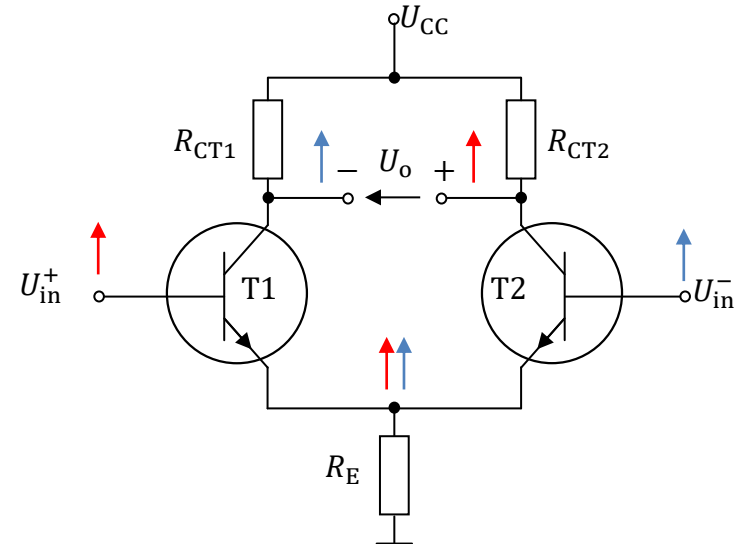
Please analyse the following circuit. Can you identify the two basic circuits inside? What happens at the output if $U_{in}^+ > U_{in}^-$? What happens if $U_{in}^+ < U_{in}^-$? Please describe the application of this circuit.

Case $U_{in}^+ > U_{in}^-$

- Transistor T1 is in emitter-follower configuration. With increasing U_{in}^+ the emitter is also pulled up.
- $U_{BE\ T2}$ in transistor T2 decreases and limits the current $I_{C\ T2}$.
- With decreasing $I_{C\ T2}$ the voltage U_o is pulled up to U_{CC} . → Non-inverting input.

Case $U_{in}^+ < U_{in}^-$:

- Opposite behaviour: A positive input signal gives a negative output signal → inverting input
- Summary: The output voltage is equal to the voltage difference times a (yet unknown) gain A .
- $U_o = A (U_i^+ - U_i^-)$
- Potential application as input stage of an Operational amplifier.



Differential amplifier.

5.2.1 Fundamental properties

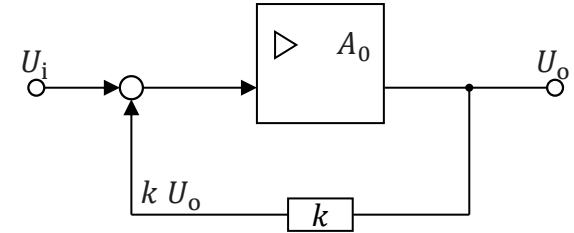
Overview

	Ideal OPA	Real OPA
Open loop gain A_0	∞	100 000 – 1 000 000
Input resistance R_i	∞	1 M Ω – 1 000 M Ω
Lower frequency limit $f_{\min}$	0 Hz	0 Hz
Common mode amplification A_{CM}	0	≈ 0.2
Common mode rejection rate CMRR	∞	94 dB
Noise	0	$\approx 3 \mu\text{V}$

5.2.1 Fundamental properties

Closed loop operation

- A fraction of the output voltage $k U_o$ is fed back to the input, where $0 \leq k \leq 1$.
- **Negative feedback**
 - ⌈ The feedback signal is subtracted from the input
 - ⌈ Opposes the input signal variation
 - ⌈ Stabilizes gain and reduces distortion
 - ⌈ Application: linear amplifier (chapter 2)
- **Positive feedback**
 - ⌈ The feedback is added to the input signal
 - ⌈ Reinforces the input signal and increases overall gain
 - ⌈ Application: switching (chapter 3)



Summary

- An Operational amplifier multiplies the voltage difference at the input by the open loop voltage gain:
- $U_o = A_0 U_d = A_0 (U_i^+ - U_i^-)$
- Ideal property of an OP amp
 - ┐ Infinite open loop voltage gain: $A_0 \rightarrow \infty$
 - ┐ Infinite input impedance: $R_i \rightarrow \infty$
- An OP amp can only run if powered by negative and positive supply power!
- An OP amp in open circuit (comparator) switches between positive and negative supply voltage.
- Positive feedback: adds signal, used for switching
- Negative feedback: subtracts signal, stabilizes gain

5.2.1 Fundamental properties

Chapter outline

Golden rules

Inverting amplifier

Non-inverting amplifier

Impedance converter

5.2.2 Linear amplifier

5.2.2 Linear amplifier

Golden rules

When placing an OP amp in a feedback circuit, i. e. use it as linear amplifier, the following idealized properties apply:

1. The output voltage U_o adjusts so that the voltage difference at the inputs U_d approaches zero.

$$U_d \rightarrow 0 \quad \rightarrow \text{simplifies KVL}$$

2. The input impedance is infinity.

$$R_i \rightarrow \infty$$

3. The input currents are negligible.

$$I_i^{+/-} \rightarrow 0 \quad \rightarrow \text{simplifies KCL}$$

3. The open loop voltage gain is infinity.

$$A_o \rightarrow \infty$$



5.2.2 Linear amplifier

Negative feedback

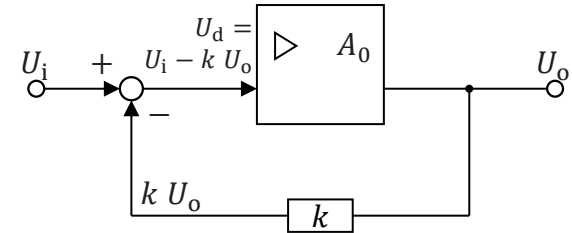
- **Closed loop operation:** a fraction of the output voltage $k U_o$ ($0 \leq k \leq 1$) is feed back into the input.
- **Negative feedback:** The feedback is subtracted from the input

$$U_d = U_i - k U_o = U_i - k A_0 U_d$$
$$\therefore U_i = (1 + k A_0) U_d$$

- For very large open loop amplification $A_0 \rightarrow \infty$ the, closed loop amplification A is therefore

$$A = \frac{U_o}{U_i} = \frac{A_0 U_d}{(1 + k A_0) U_d} = \frac{A_0}{1 + k A_0} \approx \frac{1}{k}, \quad \text{for } k A_0 \gg 1$$

The **closed loop amplification** only depends on the **coupling coefficient** k !



The coupling coefficient is defined by the ratio of two resistors.

Closed loop feedback in a signal flow diagram

5.2.2 Linear amplifier

Inverting amplifier

— Loop I

$$R_1 I_1 - U_d - U_i = 0 \quad \text{with } U_d = 0 \quad \boxed{\therefore I_1 = \frac{U_i}{R_1}} \quad (I)$$

— Loop II

$$R_2 I_2 - U_d - U_o = 0 \quad \text{with } U_d = 0 \quad \boxed{\therefore I_2 = \frac{U_o}{R_2}} \quad (II)$$

— Node P (virtual ground)

$$I_1 + I_2 - I_i^- = 0 \quad \text{with } I_i^- = 0 \quad \boxed{\therefore I_1 = -I_2} \quad (III)$$

— Plug (I), (II) in (III) and solve for U_o

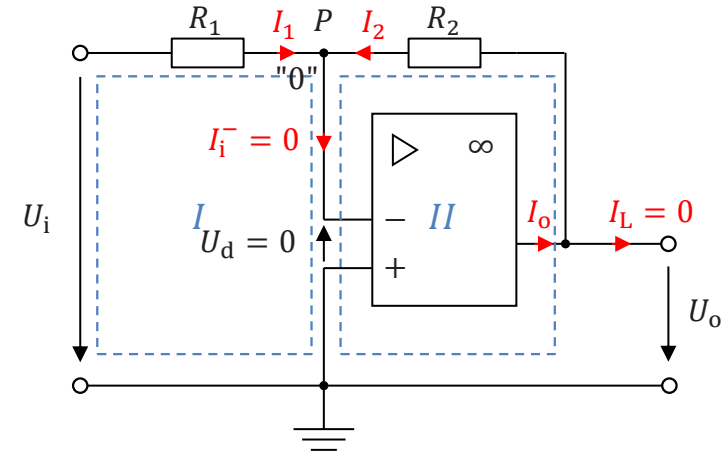
$$U_o = -\frac{R_2}{R_1} U_i$$

— Or in terms of the voltage gain A and coupling coefficient k

$$\boxed{A = \frac{U_o}{U_i} = \frac{1}{k} = -\frac{R_2}{R_1}}$$

→ **Amplification** depends only on R_1 and R_2 .

→ Signal is **inverted**!



Inverting amplifier. The non-inverting input is on ground. The inverting input is on virtual ground "0".

5.2.2 Linear amplifier

Non-inverting amplifier

Node P

$$I_2 - I_1^- - I_1 = 0 \quad \text{with } I_1^- = 0 \quad \boxed{\therefore I_1 = I_2 = I} \quad (I)$$

Loop I

$$-U_i + U_d + R_1 I = 0 \quad \text{with } U_d = 0 \quad \boxed{\therefore I = \frac{U_i}{R_1}} \quad (II)$$

Loop II

$$-U_o + R_1 I + R_2 I = 0 \quad \boxed{\therefore I = \frac{U_o}{R_1 + R_2}} \quad (III)$$

Plug (II) in (III) and solve for U_o :

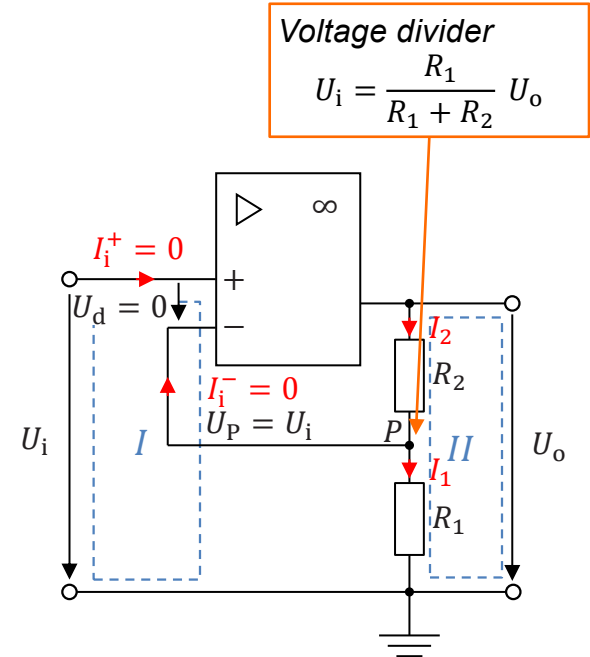
$$U_o = \frac{R_1 + R_2}{R_1} U_i$$

Or in terms of the voltage gain A and coupling coefficient k :

$$\boxed{A = \frac{U_o}{U_i} = \frac{1}{k} = \frac{R_2}{R_1} + 1}$$

→ Amplification depends **only on R_1 and R_2** .

→ Signal is **not inverted**



Non-inverting amplifier. The voltage at node P is pulled to the input voltage U_i .

5.2.2 Linear amplifier

Homework: Impedance converter

An impedance converter is a variant of a non-inverting amplifier, where $R_1 \rightarrow \infty$ is replaced by an open circuit and $R_2 = 0$ is replaced by a short circuit. Discuss the properties of this circuit. In particular, discuss the voltage gain A , the internal resistance R_i of the OP amp and limits of operation. *Hint: work with the loop $U_i/U_{OPA}/U_o$.*

- R_1 removed $\rightarrow R_1 \rightarrow \infty$
- R_2 relaced by short $\rightarrow R_2 = 0$
- The gain of the circuit is

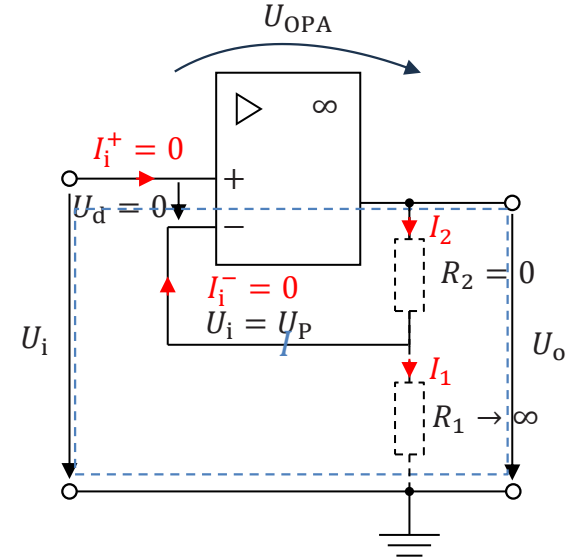
$$A = \frac{U_o}{U_i} = \frac{1}{k} = 1$$

- The output voltage is equal the input voltage $\rightarrow$ **Voltage follower.**
- Applying KVL to the loop

$$U_i - U_{OPA} - U_o = 0$$
$$\therefore U_{OPA} = 0 \quad \therefore R_i = 0$$

- Ideal voltage source.
- Limitation: Requires high currents for small loads to maintain condition:

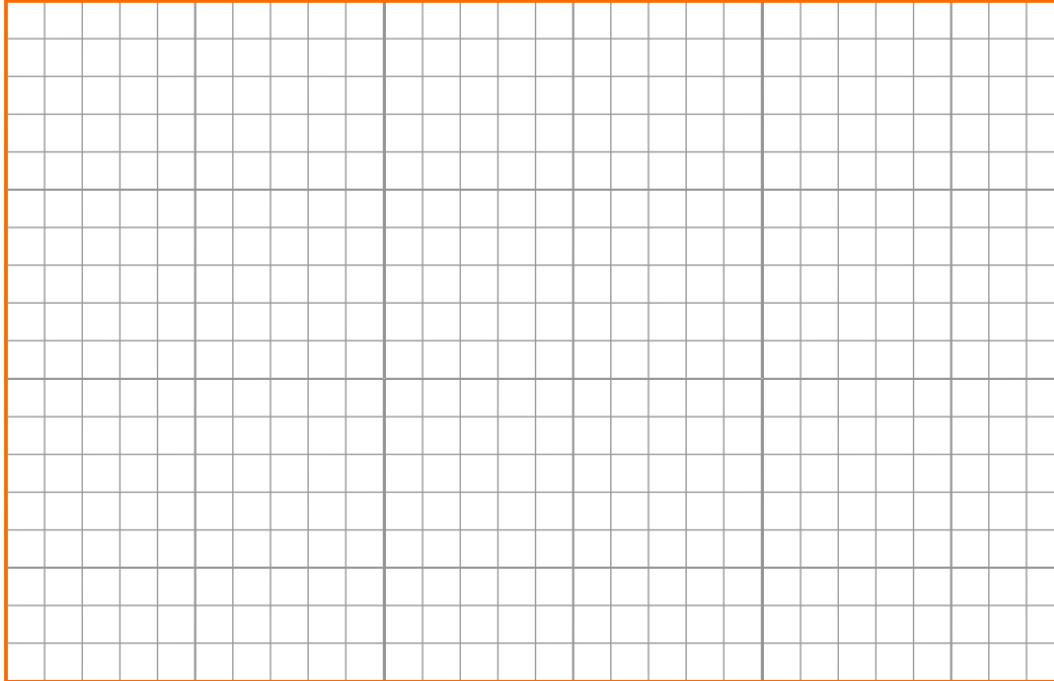
$$U_o = U_i = R_i \cdot I_o$$



5.2.2 Linear amplifier

Homework 2: Impedance converter

For an input voltage $U_i = 10 \text{ V}$ and an LM471 op amp, find the minimal load resistance R_L to maintain an output voltage $U_o \geq 0.9 U_i$. Also find the general equation for the internal resistance R_{OPA} of the op amp.



Summary

- Linear amplification requires negative feedback: Some part of the output signal is plugged back into the inverting input.
- With negative feedback
 - ┐ the OP amp adjusts so that $U_d = U_i^+ - U_i^- \rightarrow 0$
 - ┐ The open loop current gain A_o is not relevant.
 - ┐ The system amplification A only depends on the ratio of the input- and the feedback resistors R_1 and R_2 .
- An impedance converter can be considered an ideal voltage source with not internal resistance $R_i = 0$.
- When the maximum output current is reached, the impedance converter can not maintain the condition $U_o = U_i \rightarrow$ the internal resistance increases.

5.2.2 Linear amplifier

Chapter outline

Comparator

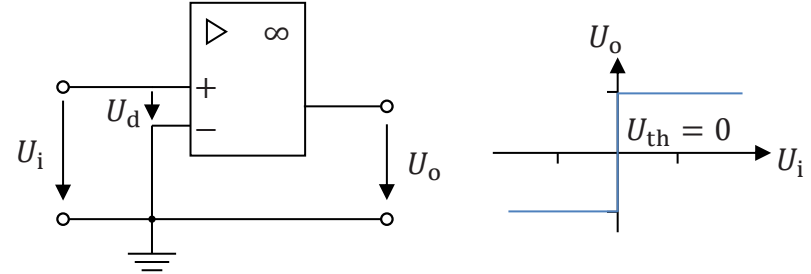
Schmitt-trigger

5.2.3 Switch

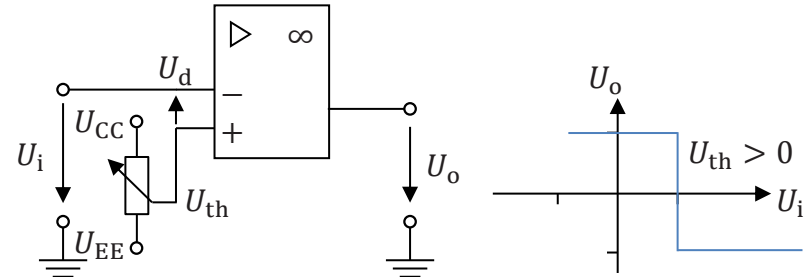
5.2.3 Switch

Comparator

- Operated in **open-loop** → output is always in saturation (high or low), negligible linear region.
- Non-inverting** comparator
 - U_i at **non-inverting** input **above** threshold
 $U_i > U_{th} \rightarrow U_o = A_0 U_d \approx +0.8 U_s = \text{high}$
 - U_i at **non-inverting** input U_i **below** threshold
 $U_i < U_{th} \rightarrow U_o = A_0 U_d \approx -0.8 U_s = \text{low}$
- Inverting** comparator: opposite behavior
 - U_i at **inverting** input **above** threshold $U_{th} \rightarrow$
 $U_o = A_0 U_d \approx -0.8 U_s = \text{low}$
 - U_i at **inverting** input U_i **below** threshold $U_{th} \rightarrow$
 $U_o = A_0 U_d \approx +0.8 U_s = \text{high}$
- Problematic: Input voltage U_i around threshold voltage → oscillation of output voltage U_o
- Comparator with hysteresis → Schmitt-trigger



Non-inverting comparator at unbiased threshold voltage $U_{th} = 0$.



Inverting comparator with threshold voltage $U_{th} > 0$.

5.2.3 Switch

Inverting Schmitt-trigger

- Operated in **positive feedback** → output is always in saturation (high or low), no linear region.
- Difference to comparator: **hysteresis** between two states
- The OP amp switches if

$$U_{i,on} > U_{th,h} \text{ and } U_{i,off} < U_{th,l}$$

- U_{th} is calculated via the voltage divider rule

$$U_{th} = \frac{R_1}{R_1 + R_2} U_o$$

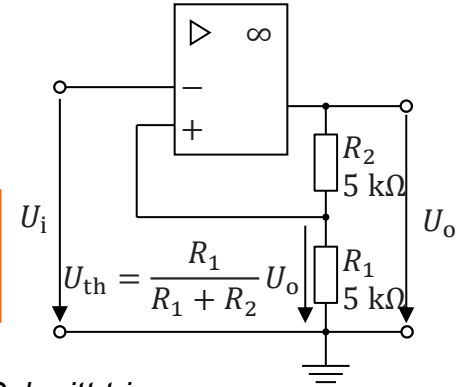
- The upper and lower thresholds are therefore

$$U_{th,h} = \frac{R_1}{R_1 + R_2} U_{o,max} \text{ and } U_{th,l} = \frac{R_1}{R_1 + R_2} U_{o,min}$$

- Sequence for $U_i < U_{th,h}$

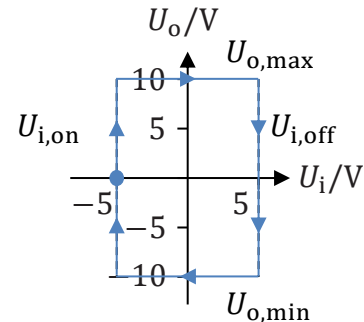
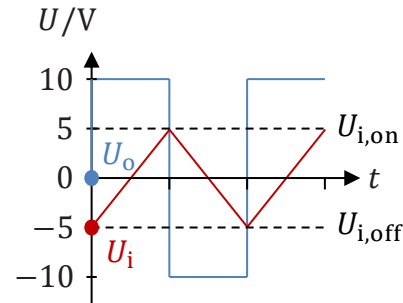
- Input below threshold $U_i < U_{th,h} \rightarrow U_o = \text{high}$
- Output U_o stays high until input is above threshold $U_i > U_{th,h} \rightarrow U_o = \text{low}$
- Repeat

Mnemonic: Feedback loop in the positive input → positive feedback/trigger
Feedback in the negative input → negative feedback/linear amplifier



Do not confuse with non-inverting amplifier! What is the subtle difference?

Inverting Schmitt trigger.

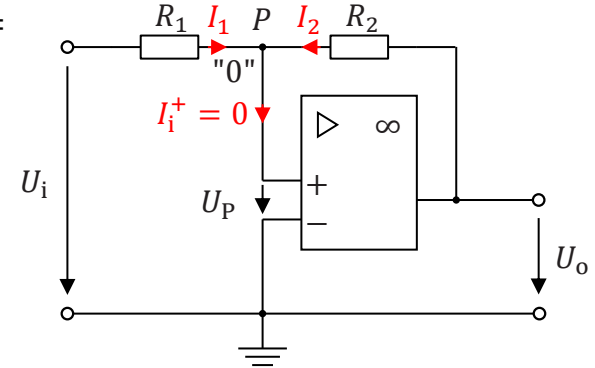


Left: Line chart. Right: Transfer characteristic.

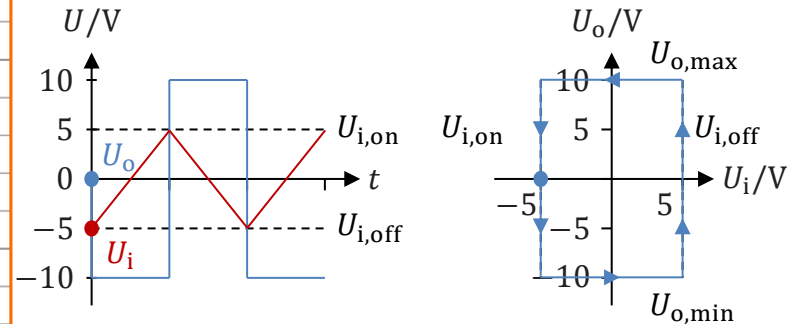
5.2.3 Switch

Homework: Non-inverting Schmitt trigger

Design a non-inverting Schmitt trigger using the following parameters. $U_o = \pm 10 \text{ V}$, $U_{th} = \pm 5 \text{ V}$. Draw the line diagram and transfer characteristic.



Non-inverting Schmitt trigger.



Left: Line chart. Right: Transfer characteristic.

- The output voltage switches at $U_P = 0$
 - $U_P > 0 \therefore$ the output voltage yields $U_{o,max}$
 - $U_P < 0 \therefore$ the output voltage yields $U_{o,min}$
- To find the threshold voltage first analyze node P using KCL

$$I_1 + I_2 - I_i^+ = 0 \quad \text{with } I_i^+ = 0 \quad \therefore I_1 = -I_2$$

$$\frac{U_i - U_P}{R_1} = \frac{U_P - U_o}{R_2} \quad (I)$$

- With P on virtual ground it follows that $U_P = 0$.
- The upper and lower threshold therefor is

$$U_{i,on} = -\frac{R_1}{R_2} U_{o,min} \therefore \frac{R_1}{R_2} = -\frac{U_{i,on}}{U_{o,min}} = -\frac{5 \text{ V}}{-10 \text{ V}} = \frac{1}{2}$$

$$U_{i,off} = -\frac{R_1}{R_2} U_{o,max} \therefore -\frac{U_{i,off}}{U_{o,max}} = -\frac{-5 \text{ V}}{10 \text{ V}} = \frac{1}{2}$$

5.2.3 Switch

Constructing a signal generator circuit

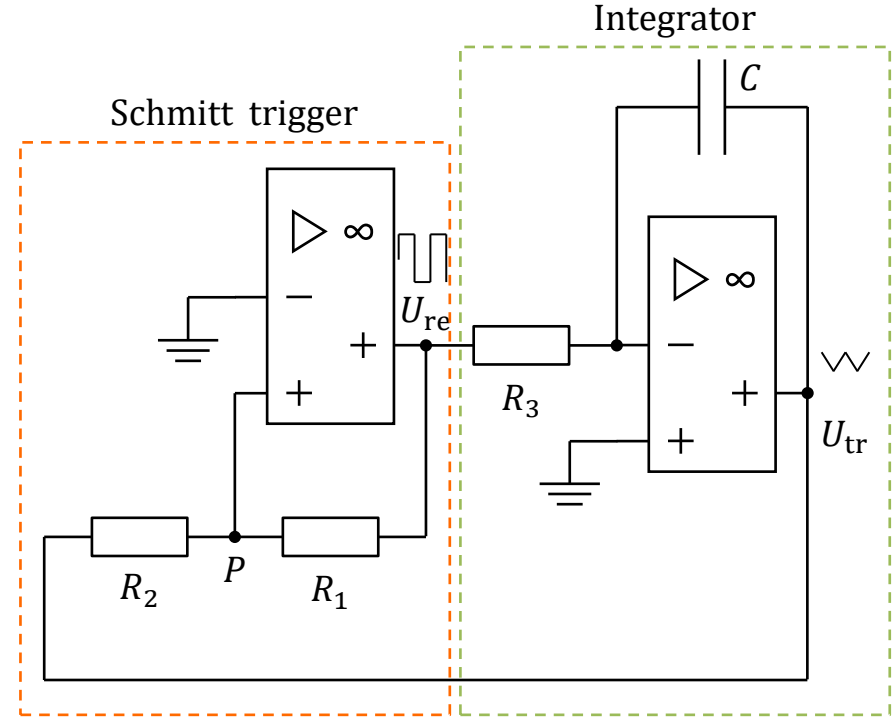
Consists of a non-inverting Schmitt trigger and an integrator (negative feedback via capacitor)

- Initialize: Tiny voltage fluctuation at $P \rightarrow$ output U_{re} is either **high** $U_{re,max}$ or **low** $U_{re,min}$.
- For the **first cycle** assume a **high signal** at P
- Output of the Schmitt trigger is high $U_{re,max}$.
- $U_{re,max}$ enters the **inverting input** of the **integrator** and is **inverted** at the output.
- Capacitor is discharging ($\tau = R_3 C$) and pulling down U_{tr} with a linear voltage ramp.
- U_{tr} is fed back via coupling resistor R_2 and pulls down the voltage at P .
- Once the threshold defined by the ratio $U_{i,off} = -\frac{R_1}{R_2} U_{re,max}$ is reached, the Schmitt trigger immediately switches the output to $U_{re,min}$
- Cycle repeats

Basics of electronics and components

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Notice that except for the DC supply voltage no external voltage is required for the signal generator. In particular **no AC voltage!**

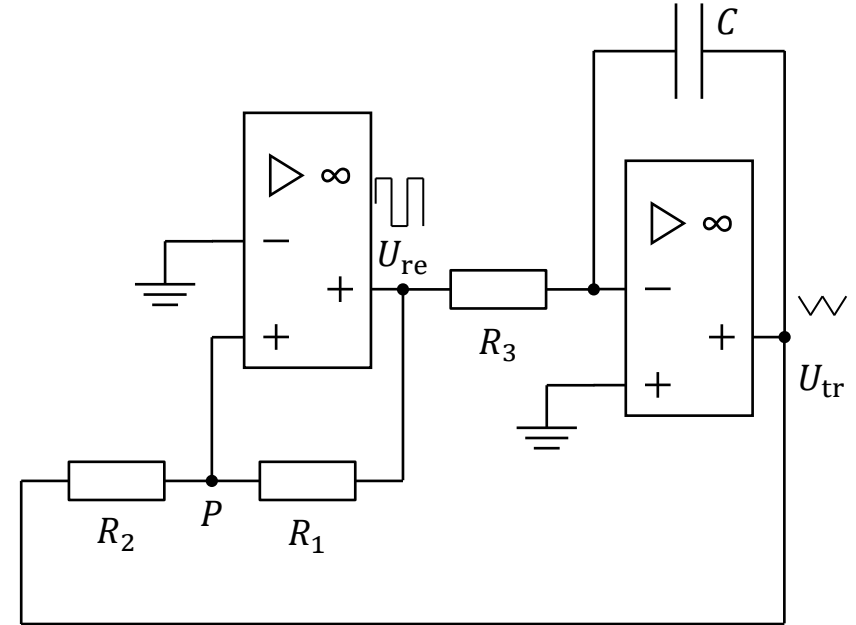
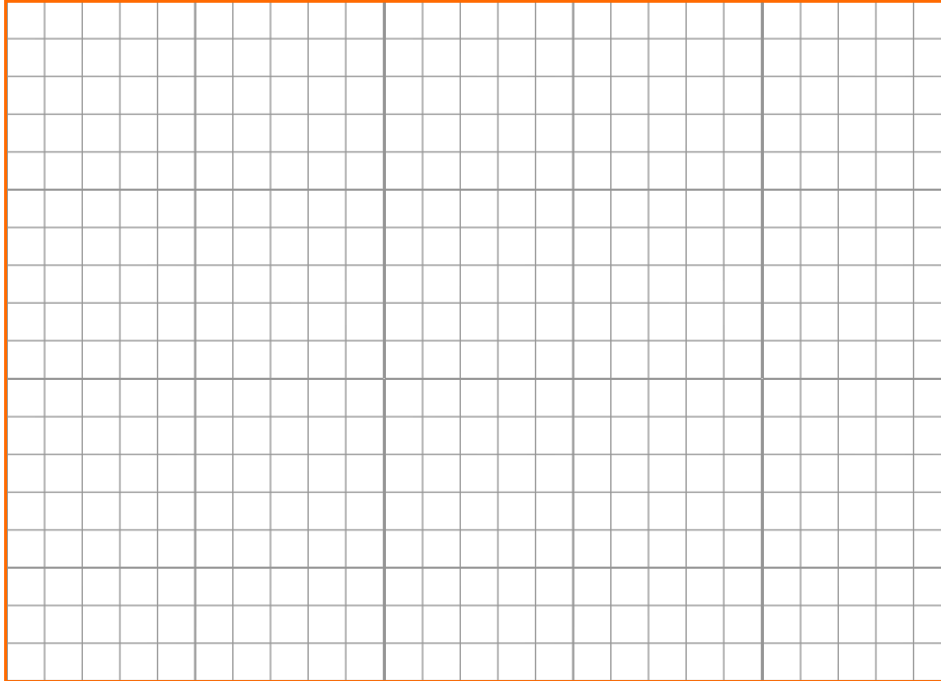


Schematic of a self-triggering multivibrator circuit.

5.2.3 Switch

Homework: Design a multivibrator.

Design a multivibrator with the following properties: $U_S = 15\text{ V}$, $U_{re} = 0.8 \cdot U_S$, $U_{tr} = 5\text{ V}$, $C = 0.22\text{ }\mu\text{F}$, $f = 5\text{ kHz}$



Summary

An OP amp can be used as binary switch in two configurations

1. In open-loop configuration: Comparator, without hysteresis, switches at threshold U_{th} .
2. In positive feedback configuration: Schmitt trigger, with hysteresis, switches at high and low threshold voltage $U_{th,h}$ and $U_{th,l}$.

The design is identical to linear amplifier, except for the feedback into the positive input.

Feedback loop in the positive input → positive feedback/trigger

Feedback in the negative input → negative feedback/linear amplifier

A signal generator (multivibrator) can be constructed using a Schmitt trigger and an integrator.

The frequency and amplitude is defined by the resistors and capacitor. No external clock signal required!

5.2.3 Switch

Chapter outline

JFET

MOSFET

CMOS

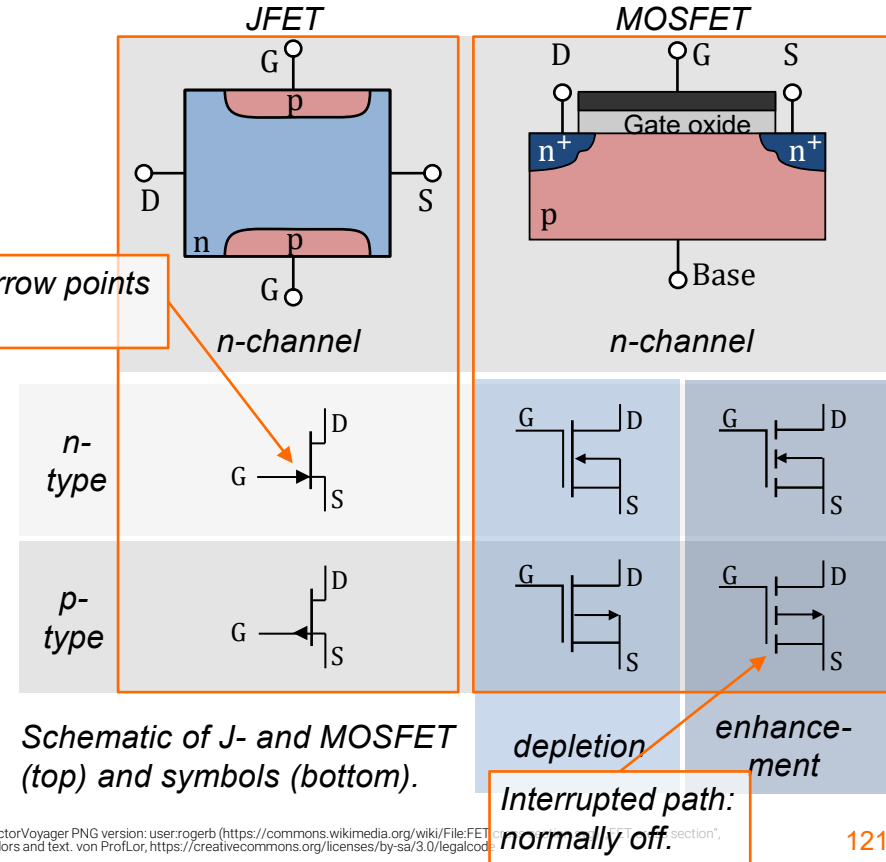
5.3 Field effect transistors

5.3.1 About field effect transistors

5.3 Field effect transistors

About field effect transistors

- Field effect transistors FET use an **electric field** to control a **current** through a semiconductor-channel of the transistor.
- Common FET types**
 - Junction field effect transistors **JFET**
 - Metal-oxide-semiconductor field-effect transistor **MOSFET**
 - (Insulated-gate bipolar transistor **IGBT**)
- Unipolar device: either **electrons (n-type)** or **holes (p-type)** are the only charge carriers
- Terminals
 - Gate G**
 - Drain D**
 - Source S**
- The **drain current** I_D is controlled by the **voltage between gate and source** U_{GS} .

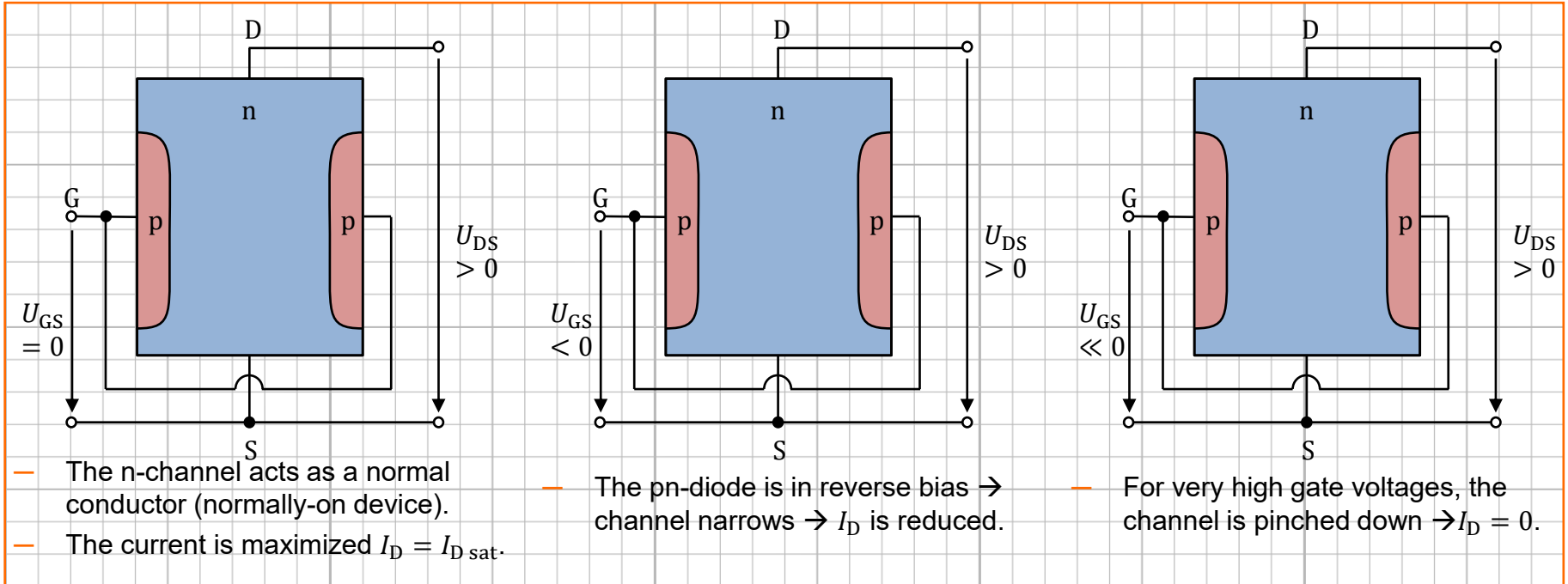


5.3.2 Junction field effect transistors JFET

5.3.2 JFET

Operating principle of a n-type junction field effect transistor JFET

Below is a schematic of a n-type JFET. A positive drain-source voltage U_{DS} is applied. What happens if three different gate voltages $U_{GS} = 0$, $U_{GS} < 0$ and $U_{GS} \ll 0$ are applied?



5.3.2 JFET

n-type junction field effect transistor JFET

Layout

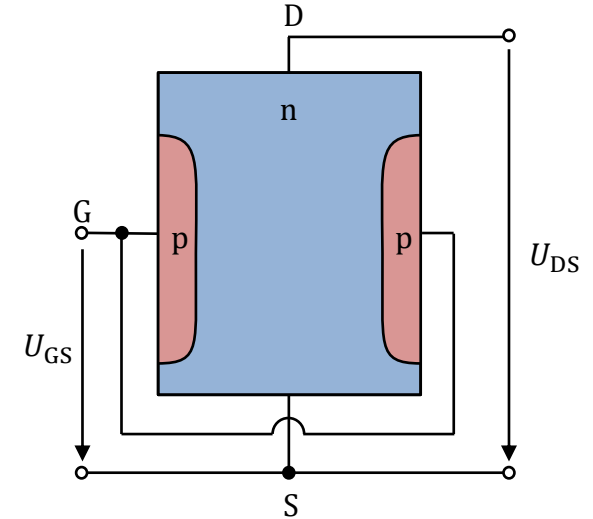
- N-type semiconductor two contacts (**source S** and **drain D**) forming an electron-conducting channel $\rightarrow$ normally-on
- P-region confining the channel: **gate G**
- The width of the **depletion layer** is defined by the voltage between gate and source U_{GS}

Modes of operation

- **Positive drain-source voltage U_{DS} : e^- current**
 - ⌋ Small U_{DS} : Ohmic resistance, I_{DS} proportional to U_{DS}
 - ⌋ Moderate U_{DS} : depletion layer near the drain broadens $\rightarrow$ resistance increases proportional to U_{DS} $\rightarrow$ drain current I_D is constant.

Negative gate-source voltage U_{GS}

- Gate and source form a PN-diode in reverse bias.
 - ⌋ $U_{GS} < 0$: channel width **controlled by U_{GS}** $\rightarrow$ control of I_D
 - ⌋ $U_{GS} \ll 0$: channel is **pinched-off**: $I_D = 0$
- Electrons injected into p-region $\rightarrow$ **small leakage current**



Schematic of a n-type junction field effect transistor JFET.

5.3.2 JFET

Modes of operation

- Active or **saturation region**: The drain current I_D is (almost) constant and independent of the drain-source voltage U_{DS}

$$I_D = I_{Dsat} \left(1 - \frac{U_{GS}}{U_{th}} \right)^2,$$

where I_{Dsat} is the maximum current without any gate-control voltage and U_p is the pinch-off voltage.

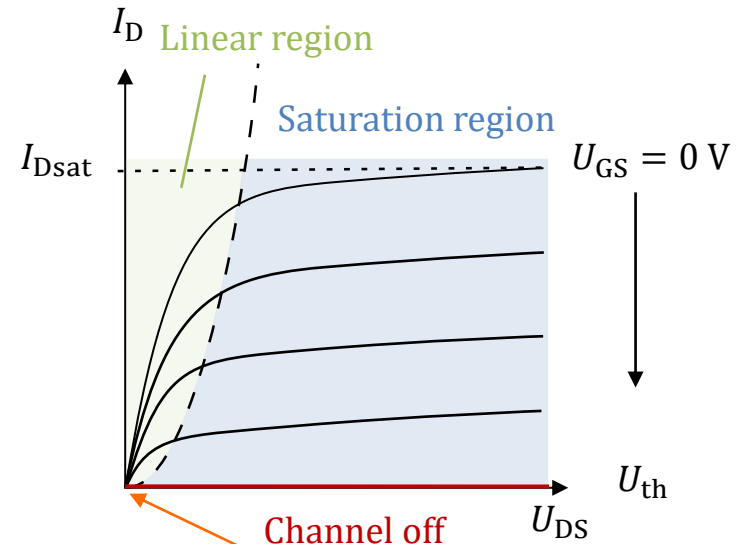
- The small-signal resistance in the saturation region is

$$r_{DS} = \Delta U_{DS} / \Delta I_D$$

- Linear ohmic region**: for a small drain-source voltage U_{DS} current I_D is defined by the ohmic resistance of the channel.

- Channel off**

- Channel is fully pinched-off for a gate-source voltage at the pinch-off voltage $U_{th} \rightarrow$ no current flow
- Typically, between $-0.3 \text{ V} \leq U_p \leq -10 \text{ V}$

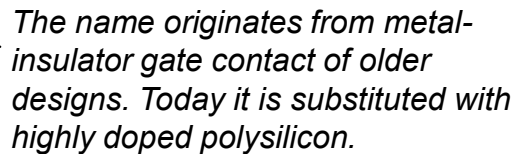


Because the channel width and resistance changes with U_{GS} the characteristic curves already separate at the origin.

I - U characteristics and output plot of an n-channel JFET:

5.3.3 Metal-oxide-semiconductor field-effect transistor MOSFET

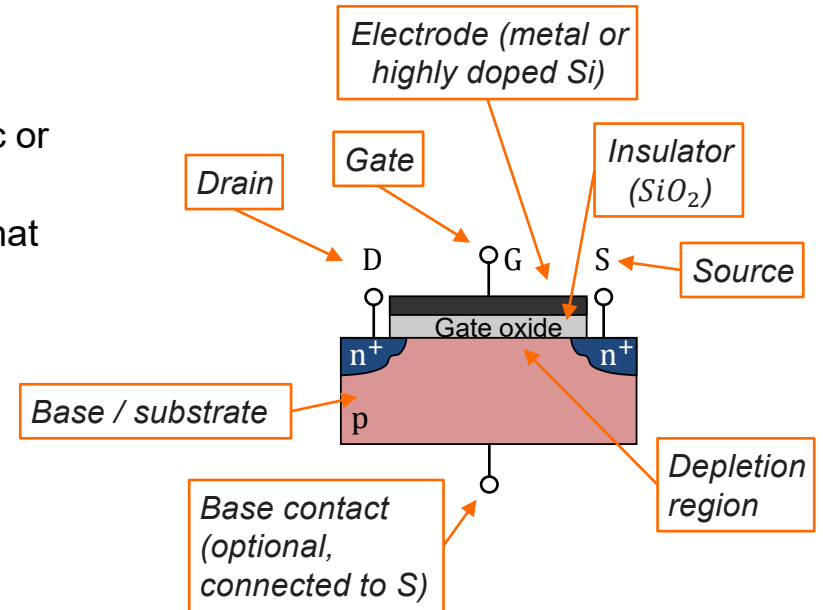
- Field-effect transistor utilizing the voltage U_{GS} of an **insulating gate** to control the drain current I_D .
- Advanced design compared to JFET because the **gate** is **galvanically insulated** from the substrate.
- Application
 - ┐ The most used transistor for **integrated circuits**
 - ┐ Used as power MOSFET in discrete circuits (same principle, advanced thermal management)
- Basic MOSFET types
 - ┐ **Charge carrier**: N- and p-type
 - ┐ **Depletion type**: normally-on
 - ┐ **Enhancement type**: normally-off (more common)



5.3.3 MOSFET

Design of a n-channel enhancement type

- **Lightly p-doped Si** as **substrate**
- Two **heavily n⁺-doped** regions form the **source S** and **drain D** contact.
- A **npn-junction** forms between S, substrate and D
- A **thin SiO₂ layer** between S and D **isolates** the metallic or highly doped **gate G** contact.
- **Gate / insulator / substrate** contact form a **capacitor** that is used to generate or pinch-off a conductive channel between drain and source.



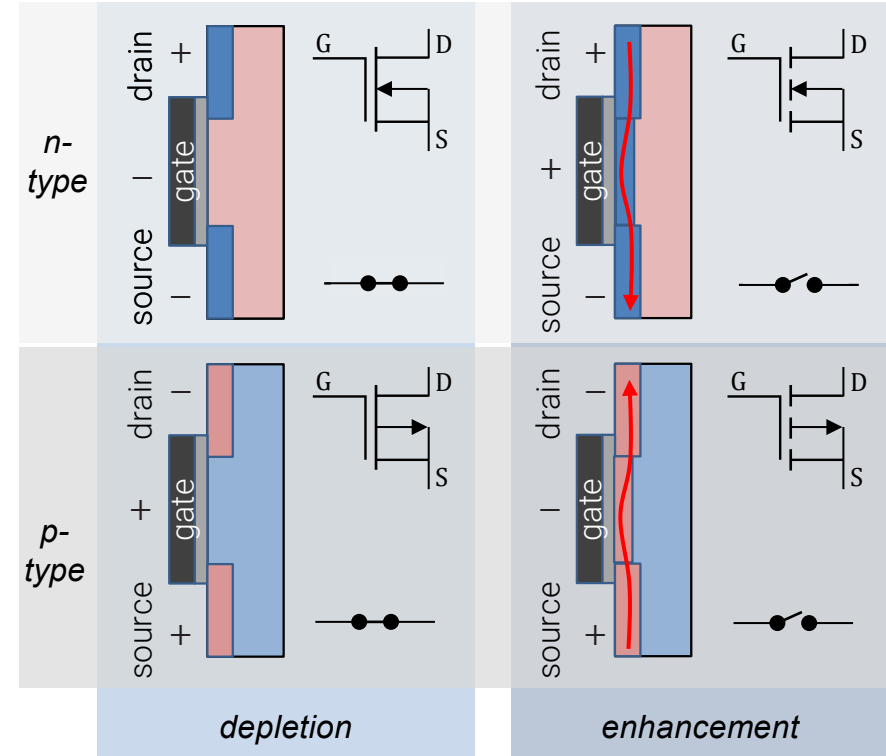
Construction of an n-channel enhancement type MOSFET.

5.3.3 MOSFET

Operating principle of different MOSFETs

- MOSFETs fall into **2x2 categories**
- **Enhancement type / normally-off:**
 - 7 **Conductive channel** between source and drain **builds up** with increased gate voltage.
 - 7 **Gate polarity opposite** to channel type
- **Depletion type / Normally-on:**
 - 7 **Channel** depletes of charge carriers and becomes **non-conductive** off with increasing gate voltage.
 - 7 **Gate polarity equal** to the channel type.
- *Example: What polarity is needed for*
 1. *Normally-off, n-type in active mode and*
 2. *Normally-on, p-type in cut off mode.*

- | | |
|----|-------------------------------------|
| 1. | Active with positive gate voltage. |
| 2. | Cut-off under positive gate voltage |

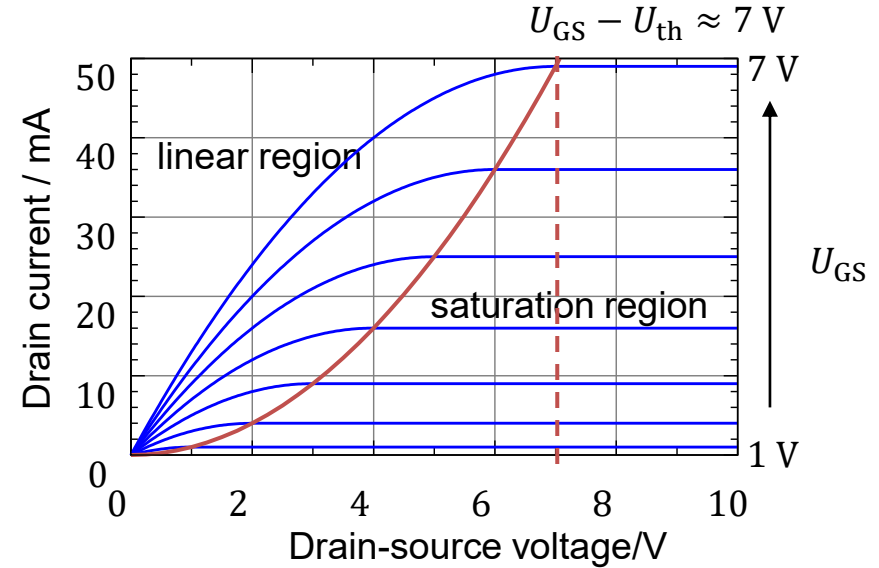


Operating principle of the four principal MOSFET types.

5.3.3 MOSFET

Characteristic curve

- Characteristic curve and modes of operation like JFET
 - ⌈ Cut-off
 - ⌈ Linear region
 - ⌈ Saturation
- Depending on the MOSFET type (depletion or accumulation) the drain current I_D is reduced or increased with the gate-source voltage U_{GS}
- The gate-capacitor is typically in the pF range → the device **can be switched even with very small charging currents** → works with diodes in reverse, capacitors and resistors in series with the gate contact.



MOSFET drain current vs. drain-source voltage for several values of U_{GS} for a normally-off depletion type MOSFET.

5.3.3 MOSFET

Homework nMOS vs BJT

Compare the properties of a NMOS and a BJT by testing the following circuit for the two transistor types and input resistances of 100 k Ω , 1 M Ω and 10 M Ω . Before you start the simulation, please discuss, what results you expect. Bonus: What happens for the NMOS if replace the resistor with a diode in reverse?

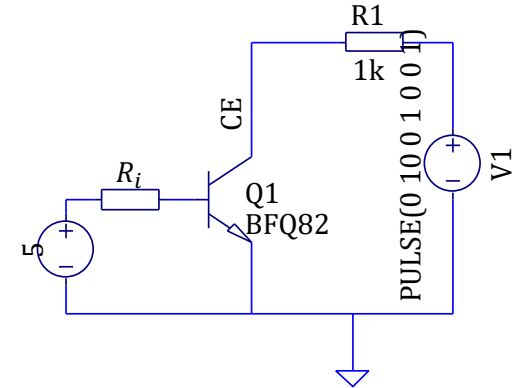
LTspice simulations: Compare_nMOS.asc and Compare_BJT.asc

— BJT:

- base current is reduced inversely w/ increasing R_i
- collector current drops accordingly
- E. g. 10 times resistance $\rightarrow$ 1/10 collector current

— MFET:

- A small current is sufficient to charge the gate capacitor $\rightarrow$ behaviour independent of resistance
- Even a diode works
- Note in the simulation: you can even discharge the capacitor by placing your finger at the gate $\rightarrow$ contact sensor



Introduction into CMOS or complementary MOS circuits

Justify the success of CMOS integrated circuits

Example for simple logic functions

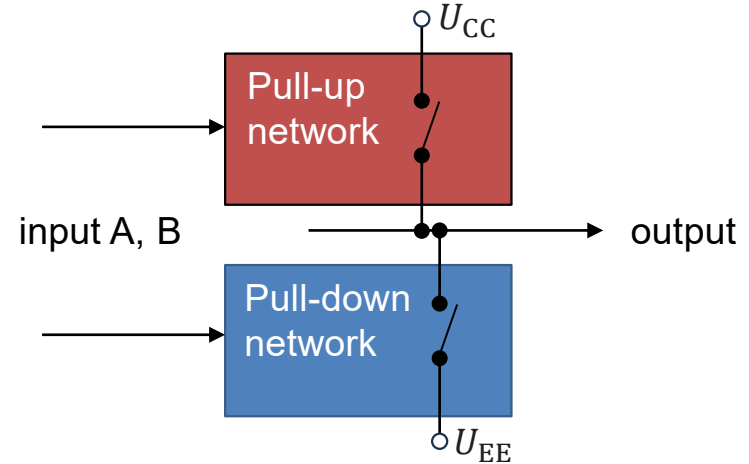
5.3.4 Complementary MOSFET – CMOS

5.3.4 CMOS

Logic circuits

- Complementary MOSFET circuits or CMOS comprise some combination of nMOS and pMOS forming two complementary sides.
 - ┐ Pull-up network → logic high output or 1
 - ┐ Pull-down network → logic low output or 0
- CMOS can be used to construct logic gates.

		AND	OR	NAND	NOR
A	B	$A \wedge B$	$A \vee B$	$A \bar{\wedge} B$	$A \bar{\vee} B$
0	0	0	0	1	1
0	1	0	1	1	0
1	0	0	1	1	0
1	1	1	1	0	0



Concept of a static logic circuit

5.3.4 CMOS

NOR gate

— Pull-up network

- └ Q_{P1} and Q_{P2} are pMOS: if A AND B is **0** then Y is **1**

$$Y = \bar{A} \wedge \bar{B} \rightarrow \bar{Y} = A \vee B \rightarrow Y = A \bar{B}$$

— Pull-down network

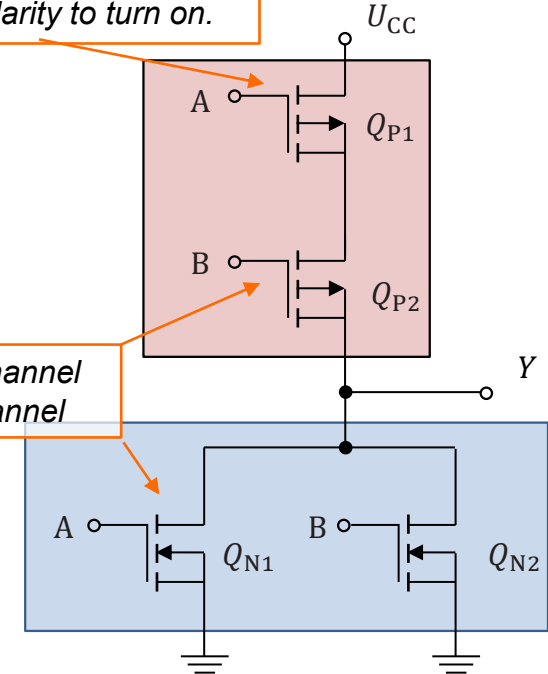
- └ Q_{N1} and Q_{N2} are nMOS: If A OR B is **1** then Y is **0**

$$\bar{Y} = A \vee B \rightarrow Y = A \bar{B}$$

		AND	OR	NAND	NOR
A	B	$A \wedge B$	$A \vee B$	$A \bar{B}$	$A \bar{B}$
0	0	0	0	1	1
0	1	0	1	1	0
1	0	0	1	1	0
1	1	1	1	0	0

Normally-off type requires opposite polarity to turn on.

Arrow out: p-channel
Arrow in: n-channel



NOR gate circuit made of pMOS for the pull-up network and nMOS for the pull-down network.

5.3.4 CMOS

Exercise: Logic gate gate

Analyse the following gate, fill the truth table and name the gate.

— Pull-up network

- Two normally-off pMOS:
Closed when gate voltage is low.

- If A OR B are low then Y is 1

$$Y = \bar{A}\bar{B} \rightarrow \bar{Y} = A\bar{B}$$

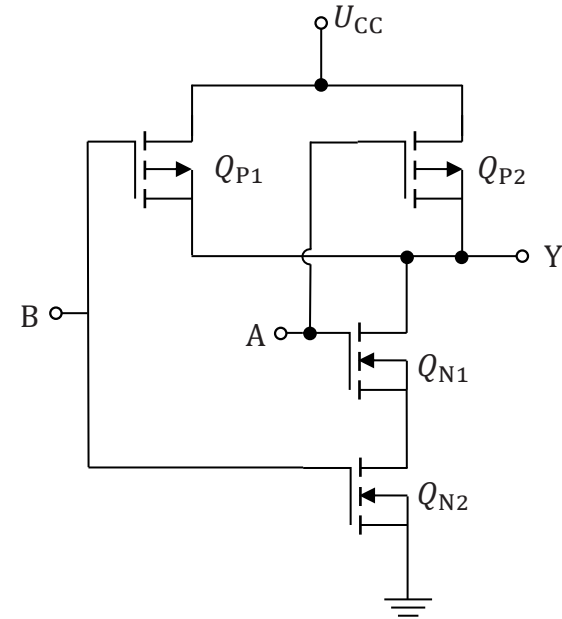
— Pull-down network

- Two normally-off nMOS:
Closed when gate voltage is low.

- If A AND B are high then Y is 0

$$\bar{Y} = A\bar{B}$$

		NAND
A	B	$A\bar{B}$
0	0	1
0	1	1
1	0	1
1	1	0



5.3.5 Constant current circuits

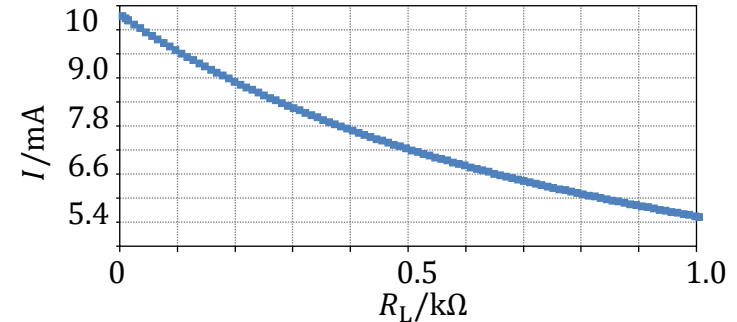
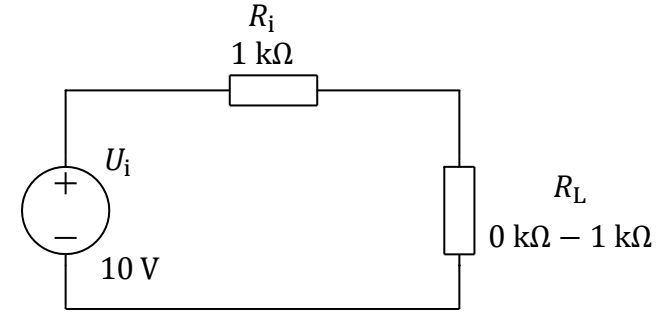
5.3.5 Constant current circuits

Constant current circuits

- **Objective:** Provide a constant current I for varying load resistance R_L
- For a fixed internal resistance, the current drops with increasing load

$$I = \frac{U_i}{R_i + R_L}$$

- **Possible solution:** $R_i \gg R_L$ (i. e. a factor of 100 can be considered „much larger“)
- **Drawback:**
 - ⌈ Works for small currents only.
 - ⌈ Large voltage drop across the internal resistance.
- **Better solution:** automated, load-dependent stabilization



Current in a simple circuit with constant internal resistance R_i and varying load resistance R_L .

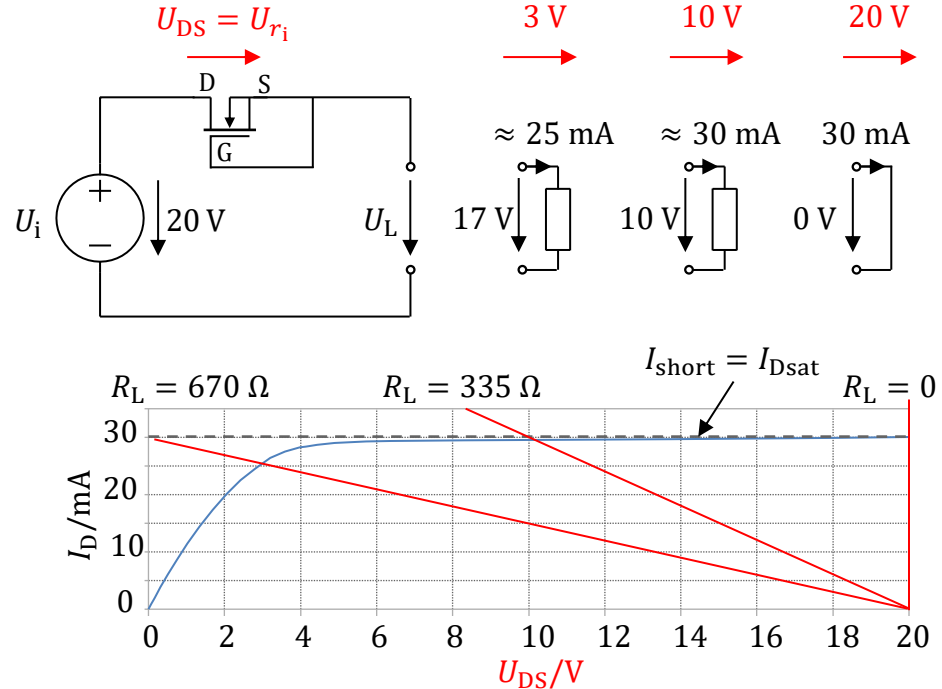
5.3.5 Constant current circuits

Constant current circuits

- Maximum current $I_{\text{short}} = I_{\text{Dsat}}$ when the gate is in short circuit with the source $U_{\text{GS}} = 0 \text{ V}$
- With increasing load resistance R_{L} the voltage U_{DS} is reduced
- Example:
 - The current is almost constant due to the high differential resistance ($r_{\text{DS}} \approx 100 \text{ k}\Omega$). For $R_{\text{L}} = 335 \Omega$

$$\Delta I_{\text{D}} = \frac{\Delta U_{\text{DS}}}{r_{\text{DS}}} = \frac{(U_{\text{i}} - U_{\text{L}})}{r_{\text{DS}}} = \frac{20 \text{ V} - (30 \text{ mA} \cdot 335 \Omega)}{100 \text{ k}\Omega}$$

$$= 0.10 \text{ mA}$$
 - For $R_{\text{L}} = 670 \Omega$ the voltage can not be stabilized (JFET out of working range)
- Note: as the JFET defines the maximum current I_{D} (here 30 mA) no source-resistor is required.

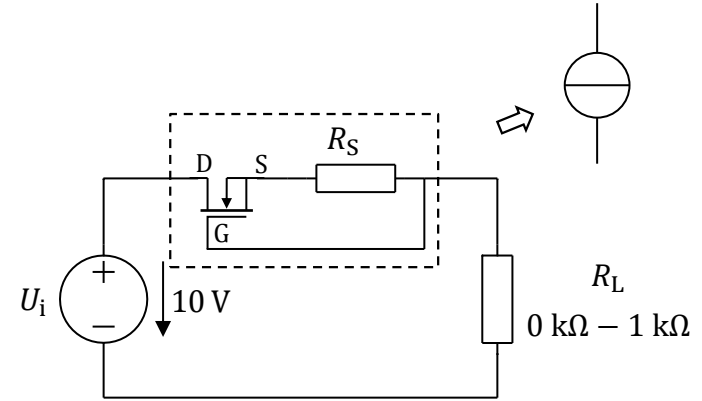


JFET current stabilizer circuit. Voltage stabilization up to $R_{\text{L}} \approx 0.5 \text{ k}\Omega$.

5.3.5 Constant current circuits

Constant current circuits

- By adding a source resistor R_S , the drain current I_D can be adjusted from $0 \leq I_D \leq I_{D\text{sat}}$
- This circuit can be readily used as constant current source
- Circuit calculation
 - ⌈ Step 1: check the manufacturer data sheet or take characteristic curve to **find** U_{GS} for $I_{DS} = 10 \text{ mA}$
 - ⌈ Step 2: **calculate** R_S to yield U_{GS}
 - ⌈ Step 3: Calculate the **minimum** V_i for amplification in the saturation region $U_i = U_{DS} + U_{RS} + U_{RL}$
- Notes
 - ⌈ *Si-JFETs are typically limited to the mA range.*
 - ⌈ *For high power applications SiC-technology is used (U_{GS} up to -20 V , I_D in the 100 A range)*

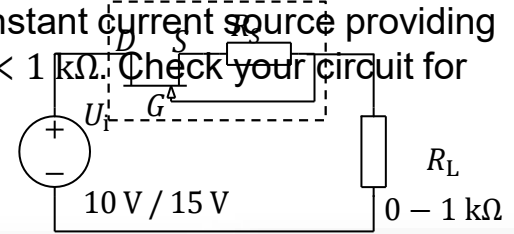


Constant current circuit with symbolic representation as constant current source.

5.3.5 Constant current circuits

Exercise: Design of a constant current source

Utilize the non-linear U-I characteristics of a JFET (LS5912c) to design a constant current source providing constant current of 10 mA for a varying load resistance between $0 \text{ k}\Omega \ll R_L \ll 1 \text{ k}\Omega$. Check your circuit for supply voltages U_i of 15 V and 10 V.



LS5912c parametrization via LTspice LS5912C_characteristic.asc:

$$U_{GS} = -1.6 \text{ V}$$

Required source resistor: $R_S = \frac{|U_{GS}|}{I_D} = \frac{1.6 \text{ V}}{10 \text{ mA}} = 160 \text{ }\Omega$

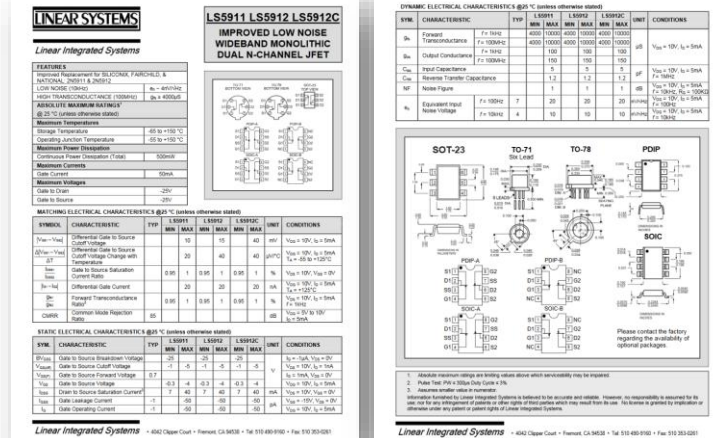
The differential output resistance is low $r_{DS} = 5 \text{ k}\Omega \rightarrow$ better ratios available

For $U_i = 10 \text{ V}$ the amplification is in the linear region ($\rightarrow$ confusingly this means a non-linear behavior for our circuit).

2. Simulation via LTspice LS5912C_current_stabilization.asc

For $U_i = 15 \text{ V}$ the current drops from 10.1 mA to 9.5 mA. The current stabilization is $\frac{I_{\text{no load}} - I_{\text{load}}}{I_{\text{no load}}} = 6.2 \text{ \%}$.

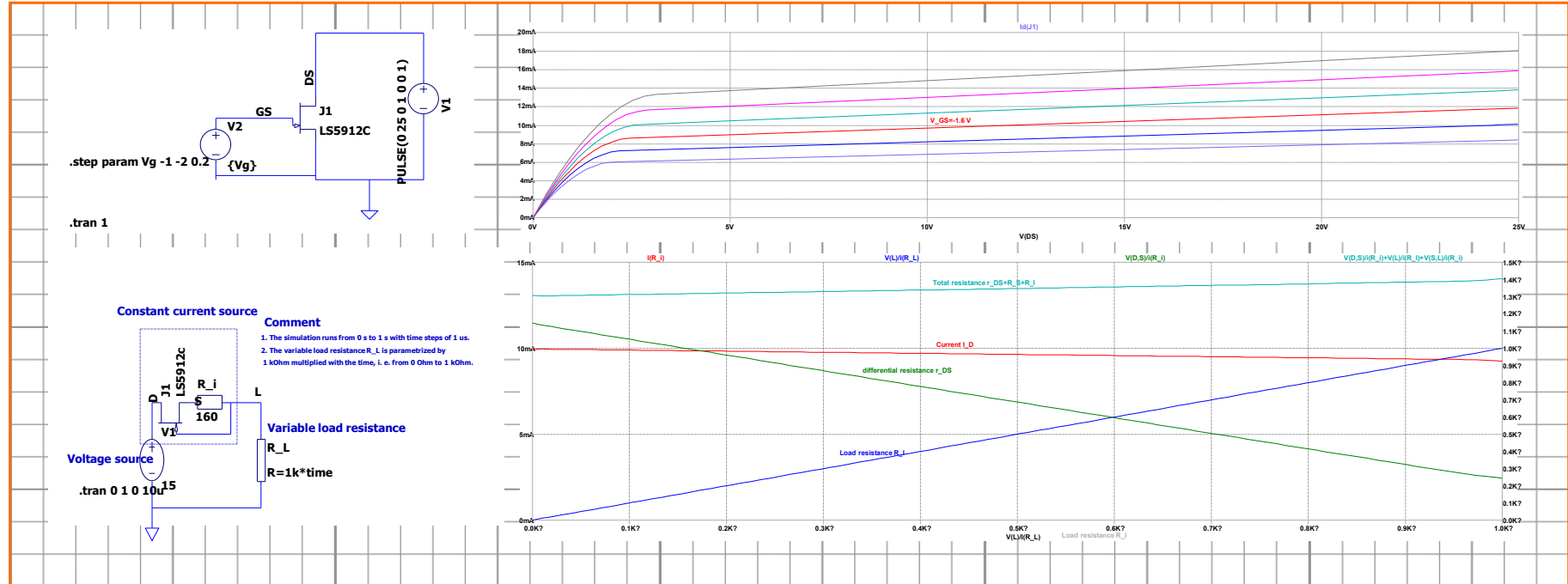
For $U_i = 10 \text{ V}$ the current drops from 9.8 mA to 7.5 mA. The current stabilization is $\frac{I_{\text{no load}} - I_{\text{load}}}{I_{\text{no load}}} = 25 \text{ \%}$.



5.3.5 Constant current circuits

Exercise: Design of a constant current source

Utilize the non-linear U-I characteristics of a JFET (LS5912c) to design a constant current source in LTspice providing constant current of 10 mA for a varying load resistance between $0 \text{ k}\Omega \ll R_L \ll 1 \text{ k}\Omega$.



Version 4.0.0

- Initial version SoSe26

Version 4.1.0

- Minor changes (transistor models, voltage characteristics, transistor basic circuits)

Version 4.2.0

- Editorial changes
- Updates in 5.1.5 – 5.1.7

Version 4.3.0

- DB slide updated
- Approximate BE and BC diode voltages updated for clarification.

Version 4.4.0

- Hint for exam relevant questions added